

Where the Shortcut Sits

Complete research record, from the pilot to the submission-ready state
(standalone input for writing the paper)

Repository: github.com/hossam1999/Where_shortcut_sits Record date: 2026-09-27

Abstract

This document records *everything* done in the project, stage by stage: the question, every dataset and how artifact and ROI labels were obtained, every experiment’s design (traps, environments, seeds, splits, leakage groups, heads, statistics), every result with 95% confidence intervals, the figures (including real images of every dataset), the observations and conclusions drawn at each stage, what failed and why, the external review and the responses, and a plan for the paper. Part I is the narrative; the appendices reproduce every pre-registration and audit document verbatim (converted from the repository), every generated result table, and the pilot-replication ledger. All numbers are produced by scripts in the repository; none are typed from memory. Two paper drafts exist in the repository: `paper/main.tex` (Medical Image Analysis format) and `paper_neurips/main.tex` (NeurIPS 2025 format).

Contents

I Narrative record	4
1 How to use this record	4
2 Executive summary	4
3 Data	5
4 Stage 0 — the pilot and its replication	10
5 Stage 1 — controlled dermoscopy and the mechanism	10
6 Stage 2 — real dermoscopic hair (ISIC 2019)	11
7 Stage 3 — chest radiography, the boundary case	11
8 Stage 4 — new modalities: thyroid, ovarian and capsule traps	12
9 Stage 5 — the confirmatory primary family	12
10 Stage 6 — replication breadth: backbones, scale, foundation models	13
11 Stage 7 — unaltered test distributions and clinical metrics	14
12 Stage 8 — theory	16
13 Stage 9 — remedies for frozen foundation-model features	18
14 Stage 10 — end-to-end fine-tuning and remedies for fine-tuned networks	19
15 Stage 11 — robustness, leakage and integrity	21
16 Stage 12 — datasets considered and not used	22

17 Cross-cutting methods	22
18 External review and responses	22
19 Conclusions and guidance for writing the paper	23
 II Appendices	 23
A All generated paper tables	23
B Result ledgers	34
C Pilot replication: all 88 numbers	34
D Holm-corrected primary claims	35
E Every arm $\times$ cohort $\times$ backbone	36
 III Experiment registry (every design, pre-registration, audit and result document, verbatim)	 39
R1 Replication specification of the pilot (author protocol)	40
R2 Audit of the original thesis claims	42
R3 Data sources and licences	43
R4 Pre-registration: ISIC 2019 real-hair traps	44
R5 Pre-commit: matched traps	45
R6 Pre-registration: chest-radiograph device traps	46
R7 Pre-registration: real chest drains	47
R8 Pre-registration: thyroid ultrasound traps	48
R9 Thyroid caliper detector audit	50
R10 Pre-registration: capsule-endoscopy traps	50
R11 Pre-registration: ovarian ultrasound traps	52
R12 Breast ultrasound feasibility audit (not used)	53
R13 Pre-registration: universal insertion-erasure (U-MtE) and ablations U1-U14	53
R14 Pre-registration: SLAS (few-shot patch suppression)	58
R15 Pre-registration: end-to-end fine-tuning	59
R16 Pre-registration: natural (unaltered) test distributions	60
R17 Sensitivity to artifact-label definitions	61

R1	Artifact-mask visual audit	62
R1	Pre-registration: LaMa inpainting comparison	62
R2	Pre-registration: text-prompted erasure and zero-shot reliance	63
R2	Statistical plan and multiplicity	64
R2	Theory (linear-Gaussian account)	65
R2	Pre-registration: theory versus real encoders	66
R2	Pre-registration: embedding-based leakage groups	67
R2	Pre-registration: backbone scale	68
R2	Pre-registration: U-MtE rank ablation	69
R2	Pre-registration: fine-tune, then erase	70
R2	Pre-registration: erase-while-fine-tuning	70
R2	Related work and novelty assessment	72
R3	Findings overview (living document)	72

Part I

Narrative record

1 How to use this record

- **Scope.** The project asks whether the value of region-of-interest (ROI) masking as a shortcut defence depends on *where* the shortcut-carrying artifact sits, explains the answer, and develops remedies for artifacts inside the ROI.
- **Conventions.** Trap A = artifact mostly inside the ROI (overlap $r \geq 0.5$); Trap B = mostly outside ($r < 0.1$); $r = |\text{artifact} \cap \text{ROI}|/|\text{artifact}|$. Training $P(A=1 | Y=1) = 0.9$, $P(A=1 | Y=0) = 0.1$; test environments: *correlated* (same rates), *reversed* (0.1/0.9; primary metric = reversed-test AUROC, “hard test”), *clean* (artifact independent of label: present in half of each class in real traps, absent in controlled sweeps). Crossover = $[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ (reversed AUROC). Robustness summary: $\min(\text{reversed}, \text{correlated})$; an arm is flagged as *shortcut-flipping* when $\text{correlated} < \text{reversed} - 0.02$.
- **Uncertainty.** Hierarchical paired bootstrap, 10,000 replicates, clusters = training seeds (or folds), paired test images resampled within clusters; AUROC differences computed within clusters. 95% percentile CIs.
- **Status labels:** **[supported]**, **[partial]**, **[not supported]** are used for every claim.

2 Executive summary

1. **[supported] Location law.** In pre-registered traps built from real artifacts in four diseases and three modalities (dermoscopic hair; thyroid and ovarian sonographer calipers; capsule-endoscopy debris), masking helps far more when the artifact is outside the ROI than inside: crossover +0.145 [+0.105, +0.181] (hair), +0.230 [+0.204, +0.255] (thyroid), +0.368 [+0.340, +0.397] (capsule), +0.170 [+0.116, +0.224] (ovary); all four survive Holm correction. Replicated with MedSigLIP, ConvNeXt, DINOv2 ViT-S/L (9/9 cohort $\times$ size), fine-tuned ResNet-50 and ViT-S, and the zero-shot scores of MedSigLIP and DermLIP (4/4).
2. **[supported] Masking can harm.** In-ROI masking *lowers* reversed AUROC when it also removes disease context: real dermoscopic hair (−0.037), controlled sweeps (ovary −0.197, capsule −0.102, dermoscopy rulers −0.129 at full overlap), fine-tuned capsule (−0.284), and on unaltered data (official thyroid split −0.102 on shortcut-conflicting pairs; held-out ISIC hospitals BCN −0.016, MSK −0.056). At a validation-fixed operating point masking halves thyroid sensitivity (0.734 $\rightarrow$ 0.347).
3. **[supported] Mechanism: retention.** Masking keeps an in-ROI artifact and removes competing evidence, so the masked model’s counterfactual reliance on the artifact grows with overlap (dermoscopy $|\Delta p|$ 0.235 $\rightarrow$ 0.491).
4. **[partial] Theory.** A linear-Gaussian closed form reproduces every sign; fitted per model on clean and correlated AUROC only, it matches held-out reversed AUROC (MAE 0.039, 148 models) and the location crossover ($r = 0.997$, MAE 0.014, 13/14 signs). Pre-registered verdict: “qualitative account” (one near-zero sign miss). Fails for chest drains.
5. **[partial] Remedy for frozen features (U-MtE).** Mask-then-erase with a generic overlay library, no artifact annotation: thyroid +0.13 to +0.22 over masking (three backbone families, three model sizes), ovary (MedSigLIP) +0.215; erasure beats augmentation (thyroid, hair); disease protection prevents failure when the artifact resembles the pathology (capsule); text-prompted erasure is a near no-op. Does not help when the shortcut is carried by correlates of the artifact (hair, drains) — label-based reweighting wins there.
6. **[supported] Remedy for fine-tuned networks.** Post-hoc U-MtE on a fine-tuned ResNet-50 is null (+0.004): the shortcut feature forms during fine-tuning. Erasing *during* fine-tuning (running overlay subspace projected out, plus feature invariance and prediction consistency; `umte_cons_ft`) improves on masking for ResNet-50 (+0.157 [+0.114, +0.205]) and ViT-S (+0.143 [+0.103, +0.187]) on thyroid; each

component alone works for only one architecture. Ovary: positive, not significant (+0.094); capsule (pathology-like debris): fails (−0.022).

7. **[supported] Rigour.** Pilot replicated (73/88 numbers MATCH, 12 same sign and significance, 3 mismatch); pre-registration before every experiment; Holm-corrected primary family 12/16 supported; stricter leakage grouping leaves 11/12 unchanged; every CI in the papers traced to a result file (109/109).

Table 1: Timeline of the stages (all dates 2026; UTC). Details in the sections below and the registry (Appendix III).

Stage	Topic	Outcome
0	Pilot + replication	Original thesis experiments E1-E15 re-implemented from the author’s specification; 73/88 MATCH.
1	Controlled dermoscopy	Synthetic ruler swept from outside to inside the lesion: sign reversal; mechanism = retention.
2	Real dermoscopic hair	ISIC 2019 spec traps: harm in-ROI, crossover +0.145; shortcut largely carried by correlates.
3	Chest radiography	Controlled tubes and real drains: boundary case (encoder ignores extra-pulmonary tubes; drains correlate-carried).
4	New modalities	Thyroid (TN3K/TNCD), capsule (SEE-AI), ovary (MMOTU) traps with audited artifact detectors.
5	Confirmatory family	16 primary tests, Holm: 12/16.
6	Replication breadth	MedSigLIP, ConvNeXt, DINOv2 S/B/L, fine-tuning, zero-shot VLMs.
7	Unaltered data	Official thyroid split, ISIC leave-one-hospital-out, capsule frame blocks; clinical metrics.
8	Theory	Closed form, simulation (0.009), real-encoder test ($r = 0.997$).
9	Remedies (frozen)	I2E $\rightarrow$ MtE $\rightarrow$ U-MtE; protection; balancing; U1-U14 ablations; JTT, SPLINCE, LEACE, LaMa, text, SLAS, selectors.
10	Remedies (fine-tuned)	mte_ft invariance; post-hoc erase (null); erase-while-fine-tuning; prediction consistency.
11	Robustness	Label definitions, mask audit, random-vs-grouped leakage, embedding groups, rank ablation, integrity checks.
12	Rejected datasets	CANDID-PTX, CLIP devices, breast US, colonoscopy, pathology pen, ISIC ink, cardiomegaly lines.
13	External review	Seven points; six valid; all addressed (Sec. 18).

3 Data

Table 2: Cohorts. ROI = region of interest. Leakage groups define every split. Full licences: registry §R3.

Cohort	Modality	Images	Artifact (label source)	ROI	Leakage groups
ISIC 2018 (controlled)	dermoscopy	2,437	synthetic ruler (pasted)	lesion masks	lesion ID + pHash
ISIC 2019	dermoscopy	25,331	hair (published masks, Wegley et al. 2026)	U-Net lesion masks (manual for HAM)	lesion ID + pHash
TN3K/TNCD	thyroid US	3,493	calipers (rule-based detector, 95% precision)	nodule masks	pHash chains (≤ 8); no patient IDs
MMOTU OTU_2d	ovarian US	1,202	calipers/overlays (same detector)	tumour masks	pHash (≤ 8)
SEE-AI	capsule endoscopy	5,481	debris/bubbles (patch-token probe, IoU 0.62 vs expert masks)	lesion boxes	frame blocks of 100 + pHash (≤ 2); no video IDs
NIH ChestX-ray14	radiography	29,687	chest drains (image-level; NEATX + detector)	lung masks	patient IDs
NIH pneumothorax (controlled)	radiography	—	synthetic tube	lung masks	patient IDs

Trap cell counts (A0 = artifact-free shared group; A1 per trap; Y0/Y1 = benign/malignant or class). Thyroid: A0 1,052/749; Trap A A1 858/292; Trap B A1 227/99. Ovary: A0 170/178; Trap A 341/276; Trap B 91/103. Capsule: A0 166/338; Trap A 404/412; Trap B 232/1,009. ISIC 2019: 172 reversed-test melanomas per trap and seed. Real example images of every dataset with ROI (green) and artifact (red) are in Figs. 1–5.

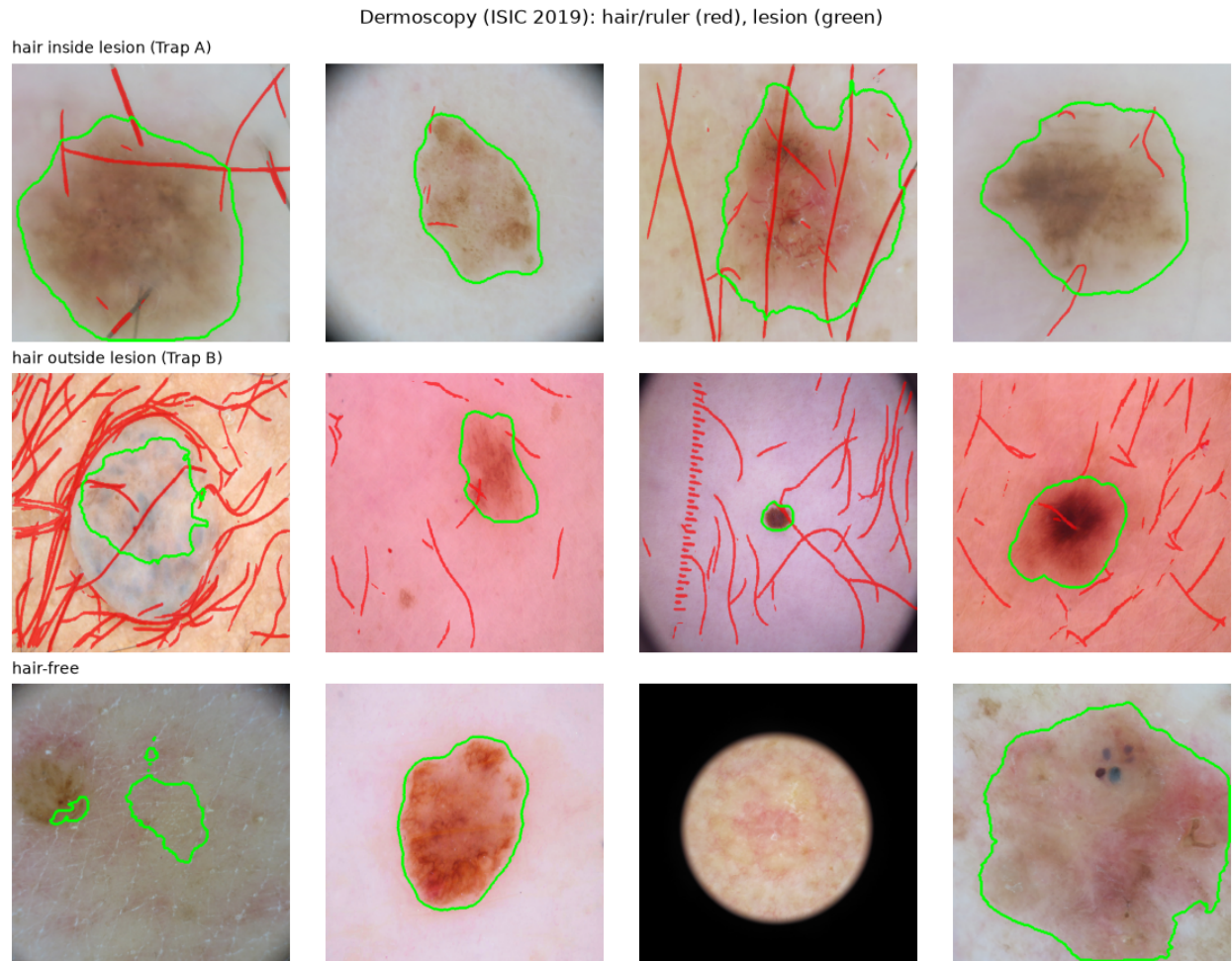
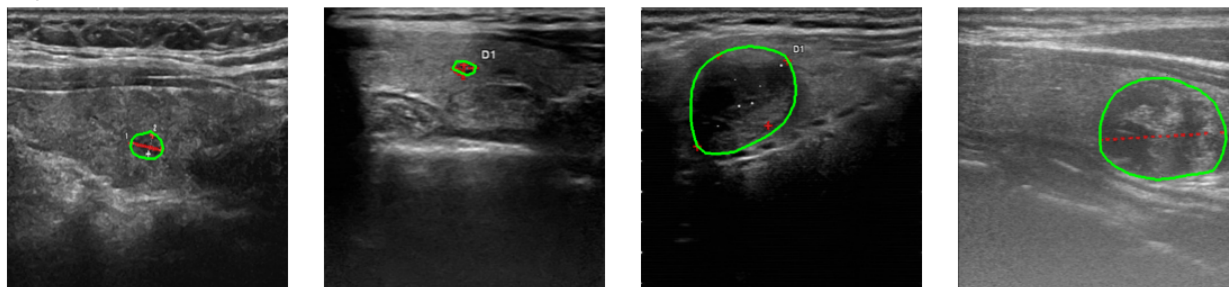


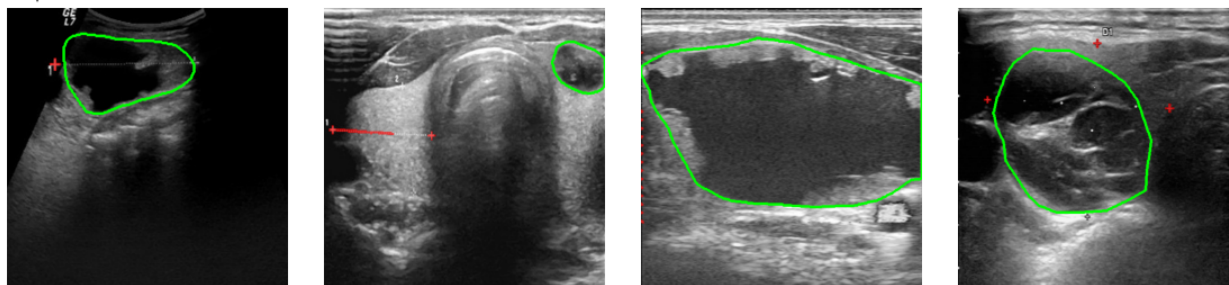
Figure 1: ISIC 2019 dermoscopy: real hair inside (Trap A) and outside (Trap B) the lesion.

Thyroid ultrasound (TN3K): detected calipers (red), nodule (green)

calipers inside nodule



calipers outside nodule



caliper-free

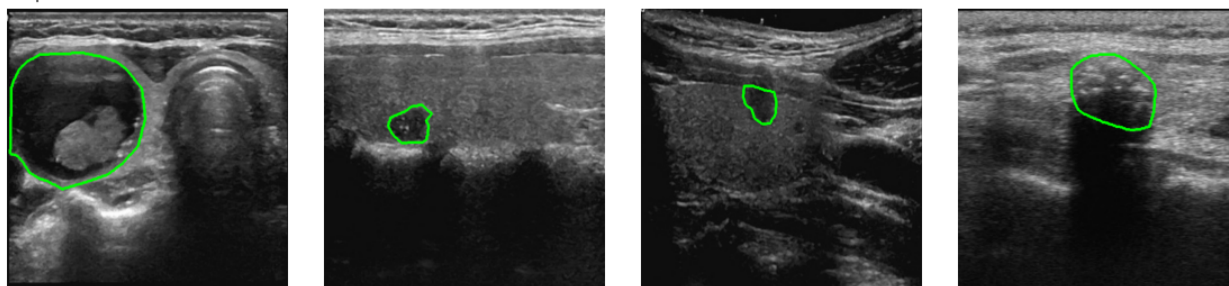
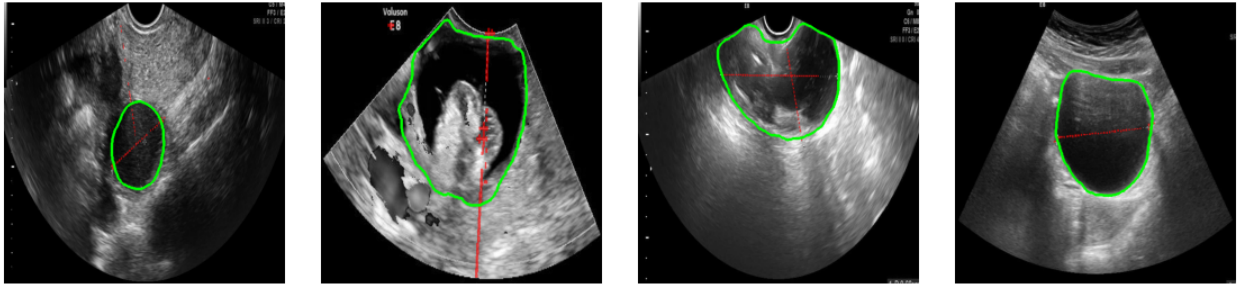


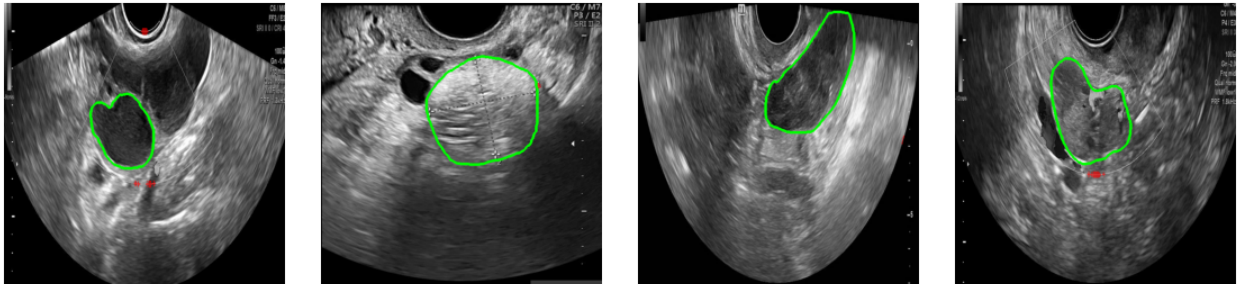
Figure 2: Thyroid ultrasound (TN3K/TNCD): sonographer calipers on and off the nodule.

Ovarian tumour ultrasound (MMOTU): calipers/overlays (red), tumour (green)

calipers inside tumour



overlays outside tumour



caliper-free

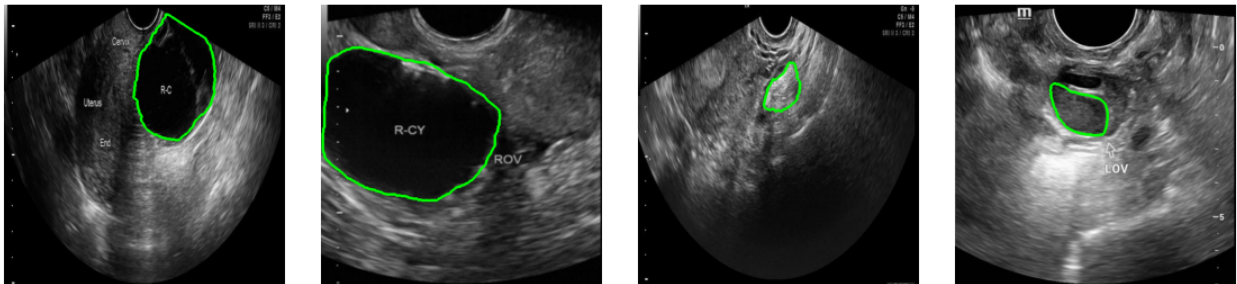


Figure 3: Ovarian ultrasound (MMOTU): calipers and overlays.

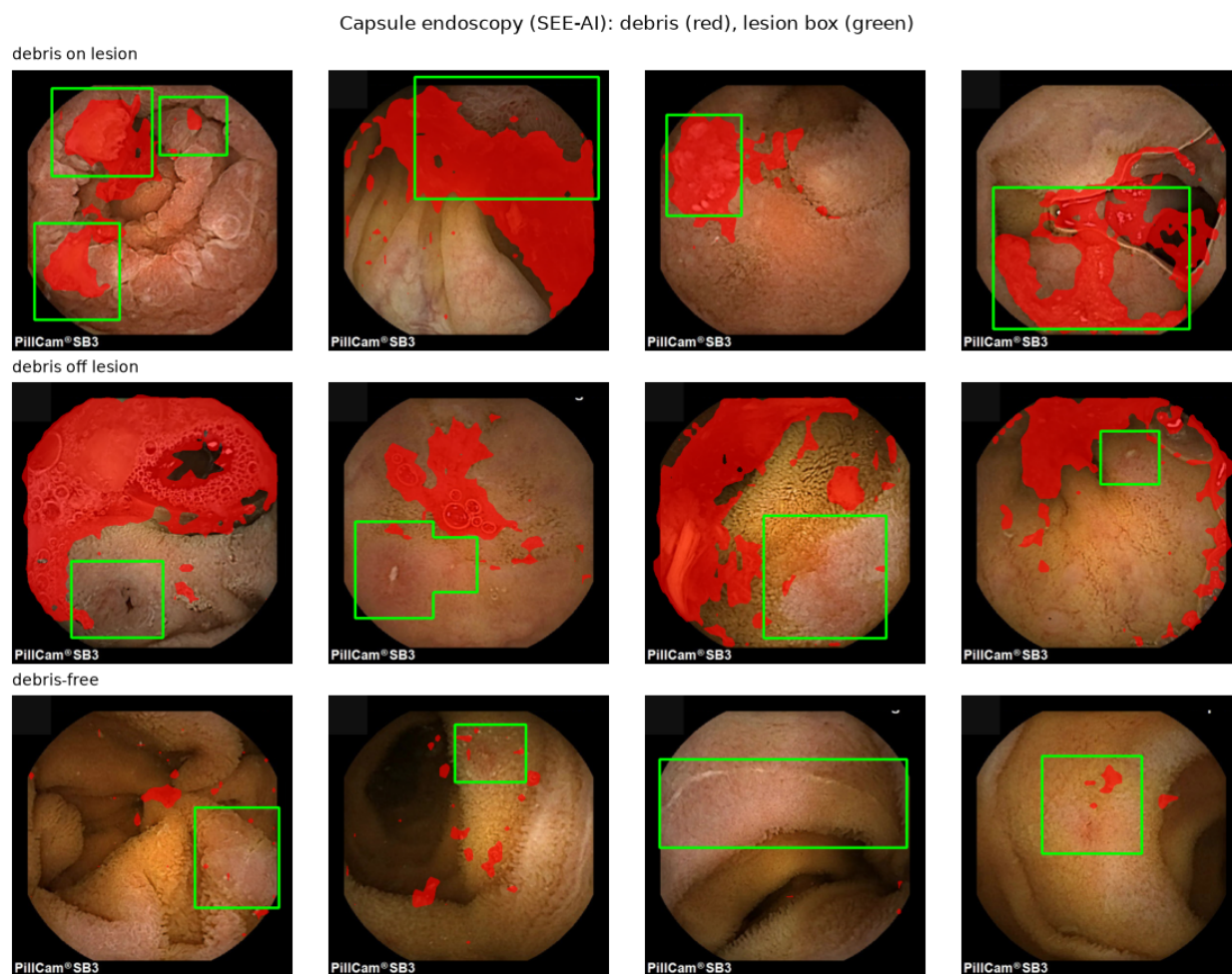


Figure 4: Capsule endoscopy (SEE-AI): debris/bubbles on and off the lesion.

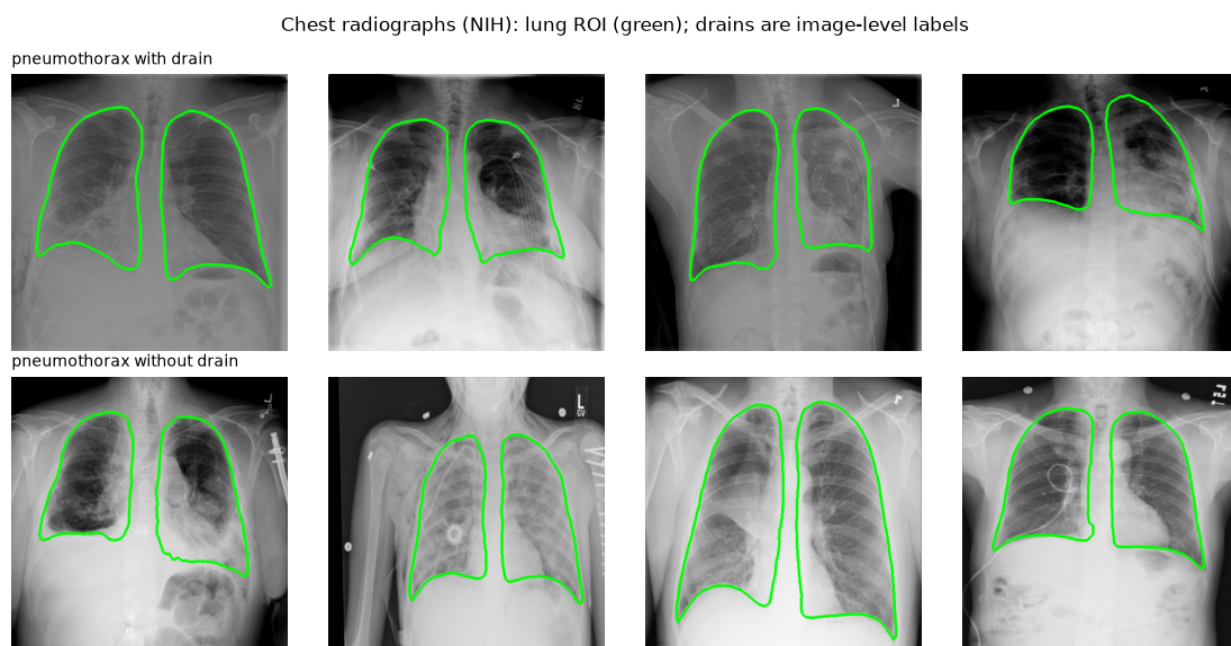


Figure 5: Chest radiography (NIH ChestX-ray14): chest drains and lung masks.

4 Stage 0 — the pilot and its replication

Question. Does the original thesis (“Where the Shortcut Sits”) reproduce? **Design.** The author’s protocol was written down as a specification (registry §R1) and re-implemented from scratch: E1–E7 controlled rulers on ISIC 2018 (DINOv2 ViT-B/14 @518 and @224, DermLIP), E10 hair decodability, E12 contrast trap, E13/E15 real hair traps on ISIC 2019, E14 area ratio. **Result.** 88 numbers compared: 73 MATCH (same sign, same CI verdict, $|\Delta| \leq 0.02$), 12 SIGN+CI (same sign and verdict, $|\Delta| > 0.02$), 3 MISMATCH (E4 occlusion at 50% has a CI that now excludes 0; E12 contrast trap not null; one HAM-only stratum). Full ledger: Appendix B. **Thesis-claims audit:** registry §R2. **Conclusion.** The pilot is reproducible; the three mismatches are documented and do not affect the central claims.

5 Stage 1 — controlled dermoscopy and the mechanism

Design. 2,437 ISIC 2018 images; a synthetic ruler correlated with melanoma (0.9/0.1) pasted at overlap $r \in \{0, 0.25, 0.5, 0.75, 1\}$ with the lesion; DINOv2 ViT-B/14 @518 frozen, logistic heads; seeds 42, 123, 456. **Results.** Masking improves reversed AUROC by +0.184 [+0.146, +0.224] at $r = 0$ and lowers it by $-0.097, -0.112, -0.129$ at 50/75/100% (interaction +0.313 [+0.272, +0.353]); reproduced at 224 px (+0.361 at 0%, -0.095 at 100%), with DermLIP (harm ≈ -0.29 at every in-ROI overlap) and a randomised ruler. At $r = 0$ the masked model’s correlated and reversed AUROC coincide (0.849/0.849); at $r = 1$ the masked model’s correlated AUROC exceeds ERM’s (0.961 vs 0.937) — it uses the ruler more. **Mechanism (three explanations tested separately).** *Occlusion* is not the cause: with the ruler present but uncorrelated (50/50) masking helps at every overlap (+0.036, +0.040, +0.039). *Lost context* alone is not the cause: dilating the mask at 50% overlap increases harm ($-0.097 \rightarrow -0.138 \rightarrow -0.154$) as it restores ruler pixels (retention 0.50 $\rightarrow$ 0.73 $\rightarrow$ 0.95), and at full overlap harm is flat although the ruler’s share of visible pixels falls threefold. *Retention with amplified reliance:* same-head counterfactual sensitivity to the ruler doubles (0.235 $\rightarrow$ 0.491); harm is largest for small lesions (-0.364 vs +0.064 for large lesions at 100%). Fig. 7 shows the same mechanism in all five controlled sweeps.

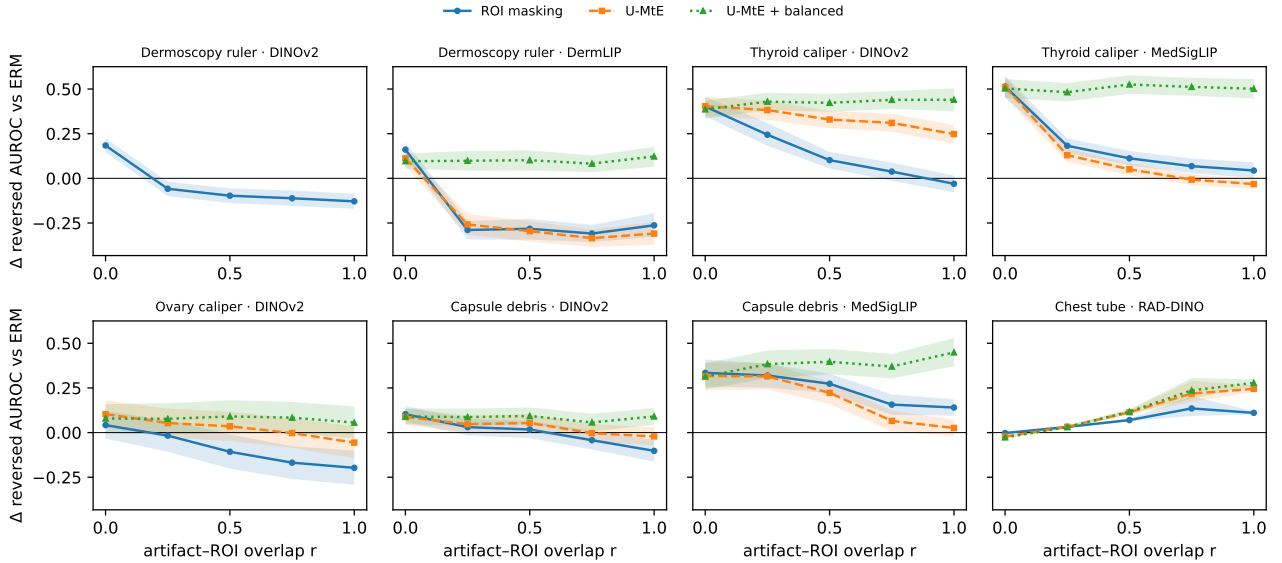


Figure 6: Controlled overlap sweeps: Δ reversed AUROC vs ERM as a synthetic artifact moves into the ROI (95% CIs). Masking (blue) declines with overlap except in chest radiography; U-MtE + balancing (green) is flat.

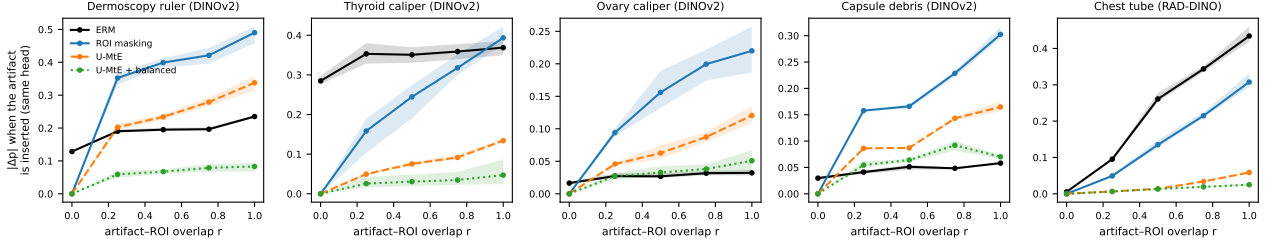


Figure 7: Mechanism: counterfactual sensitivity $|p(x+\text{artifact}) - p(x)|$ vs overlap. Masking increases reliance on an in-ROI artifact; U-MtE and balancing remove it. At full overlap: dermoscopy 0.491 (mask) vs 0.235 (ERM); ovary 0.218 vs 0.032; capsule 0.302 vs 0.058; thyroid 0.393 vs 0.369; chest tube: ERM ignores extra-pulmonary tubes ($|\Delta p| = 0.005$ at $r = 0$).

Leakage assessment (controlled sweep). Five random splits contained 67–124 near-duplicate pairs across the split; the grouped split has none. Random splits change the ERM shortcut gap and the size of the masking effect (Table 3), so every analysis uses grouped splits.

Table 3: Leakage assessment (controlled dermoscopy sweep): grouped (lesion ID + pHash) versus five random splits.

Split	Overlap	Near-dup pairs across split	ERM clean	ERM corr	ERM rev	ERM gap	mask-ERM rev
grouped	0.0	0	0.797	0.906	0.665	0.241	0.184
grouped	1.0	0	0.792	0.937	0.515	0.422	-0.129
random_0	0.0	98	0.812	0.909	0.649	0.260	0.188
random_0	1.0	98	0.810	0.944	0.495	0.450	-0.119
random_1	0.0	101	0.809	0.906	0.658	0.248	0.180
random_1	1.0	101	0.807	0.945	0.523	0.422	-0.109
random_2	0.0	75	0.828	0.924	0.555	0.369	0.297
random_2	1.0	75	0.827	0.945	0.404	0.541	-0.007
random_3	0.0	124	0.837	0.913	0.707	0.206	0.180
random_3	1.0	124	0.821	0.949	0.520	0.429	-0.147
random_4	0.0	67	0.818	0.926	0.591	0.335	0.269
random_4	1.0	67	0.824	0.946	0.463	0.482	-0.059

6 Stage 2 — real dermoscopic hair (ISIC 2019)

Design (registry §R4). Hair-free group A0 = at most 30 native hair pixels; Trap A: in-lesion hair ($r \geq 0.5$); Trap B: $r < 0.1$; cells matched to the A0 source mix; five seeds (42, 123, 456, 789, 2026) $\times$ five group-safe folds (lesion ID + pHash); 172 reversed-test melanomas per trap and seed. **Results.** Masking helps in Trap B (+0.107 [+0.078, +0.135]) and harms in Trap A ($-0.037 [-0.057, -0.019]$, negative in all five seeds); crossover +0.145 [+0.105, +0.181]. Harm holds in BCN (-0.051) but not HAM (-0.009 , n.s.). Oracle hair inpainting helps both traps (+0.103/+0.096). Group-balanced training and DFR (image-level hair label) give the largest gains (+0.225/+0.251 Trap A; +0.209/+0.215 Trap B); paired LEACE recovers less than half. Hair is decodable from the embedding (AUROC 0.943) but removing every hair patch with the oracle mask gains only +0.015: the hair shortcut is mostly carried by correlates (body site, age, sex; registry §R4). **Conclusion.** In-ROI masking harms on real data; hair is the *correlate-carried* regime.

7 Stage 3 — chest radiography, the boundary case

Controlled tube sweep (NIH pneumothorax, RAD-DINO): no sign reversal — ERM does not use a tube outside the lungs ($|\Delta p| = 0.005$ at $r = 0$), so masking has nothing to remove there; inside the lungs masking reduces but does not remove reliance. **Real chest drains** (29,687 images, patient-grouped folds; registry §R7): an extreme shortcut (ERM reversed 0.19 RAD-DINO, 0.28 DINOv2); lung masking +0.05/+0.01; U-MtE +0.03 over masking (RAD-DINO only); DFR and mask+DFR best — a drain comes with a treated lung (correlate-carried). **Pathology traps with real devices** (Atelectasis, Consolidation, Effusion, Infiltration; registry §R6): crossovers near zero (+0.060, -0.002 , -0.004 , -0.032). **Conclusion.** Chest radiography is reported as a boundary case, not as a fifth confirmation.

8 Stage 4 — new modalities: thyroid, ovarian and capsule traps

Artifact labels without annotation. Calipers: a rule-based detector (colour/shape; visual audit: 95% of thyroid detections on real calipers, 92–97% of caliper-free images clean; registry \$R9, \$R18). Capsule debris: a patch-token probe trained on the published expert contamination masks (IoU 0.62). **Traps** as in Stage 2 (five seeds $\times$ five folds). **Results (DINOV2).**

Table 4: Masking gain (reversed AUROC) by artifact location, DINOV2 ViT-B/14 @518.

Cohort	mask-ERM, Trap A (in-ROI)	mask-ERM, Trap B (out)	Crossover
ISIC hair	-0.037 [-0.057, -0.019]	+0.107 [+0.078, +0.135]	+0.145 [+0.105, +0.181]
Thyroid calipers	+0.111 [+0.097, +0.124]	+0.341 [+0.320, +0.363]	+0.230 [+0.204, +0.255]
Ovarian calipers	+0.132 [+0.089, +0.174]	+0.302 [+0.261, +0.343]	+0.170 [+0.116, +0.224]
Capsule debris	+0.096 [+0.081, +0.113]	+0.465 [+0.440, +0.490]	+0.368 [+0.340, +0.397]

Controlled sweeps in the same cohorts (synthetic caliper/debris pasted at controlled overlap): ovary at full overlap -0.197 [-0.292, -0.100], capsule -0.102 [-0.162, -0.040], thyroid -0.030 (n.s.); location interactions capsule +0.205, thyroid +0.432. **Observation.** On real thyroid/ovary/capsule, in-ROI masking still helps a little (it removes clutter) — the universal claim is the *gap* (crossover), and harm appears whenever masking also removes disease context (dermoscopy, sweeps, fine-tuned capsule, unaltered data).

9 Stage 5 — the confirmatory primary family

Each test was pre-registered individually; the family of 16 (four hypotheses $\times$ four cohorts, DINOV2) was defined afterwards for multiplicity control and corrected with Holm (BH for reference; registry \$R21).

Table 5: The 16 pre-registered primary tests (DINOV2 ViT-B/14 @518; reversed-test AUROC; Holm over the family).

Family	Cohort	Estimate [95% CI]	p_{Holm}	Supported
P1 location crossover	ISIC hair (DINOV2)	+0.145 [+0.105, +0.181]	0.0016	yes
P1 location crossover	Thyroid (DINOV2)	+0.230 [+0.204, +0.255]	0.0016	yes
P1 location crossover	Capsule (DINOV2)	+0.368 [+0.340, +0.397]	0.0016	yes
P1 location crossover	Ovary (DINOV2)	+0.170 [+0.116, +0.224]	0.0016	yes
P2 U-MtE (protected) > mask	ISIC hair	+0.041 [+0.030, +0.054]	0.0016	yes
P2 U-MtE (protected) > mask	Thyroid	+0.217 [+0.174, +0.257]	0.0016	yes
P2 U-MtE (protected) > mask	Capsule	-0.010 [-0.021, -0.001]	0.0900	no
P2 U-MtE (protected) > mask	Ovary	+0.043 [+0.026, +0.061]	0.0016	yes
P3 erase > augment	ISIC hair	+0.045 [+0.037, +0.054]	0.0016	yes
P3 erase > augment	Thyroid	+0.207 [+0.182, +0.235]	0.0016	yes
P3 erase > augment	Capsule	-0.130 [-0.147, -0.110]	0.0016	no
P3 erase > augment	Ovary	+0.035 [+0.010, +0.059]	0.0224	yes
P4 U-MtE+balanced > balanced	ISIC hair	-0.013 [-0.041, +0.018]	0.4230	no
P4 U-MtE+balanced > balanced	Thyroid	+0.112 [+0.090, +0.134]	0.0016	yes
P4 U-MtE+balanced > balanced	Capsule	+0.082 [+0.064, +0.100]	0.0016	yes
P4 U-MtE+balanced > balanced	Ovary	+0.028 [+0.000, +0.057]	0.0988	no

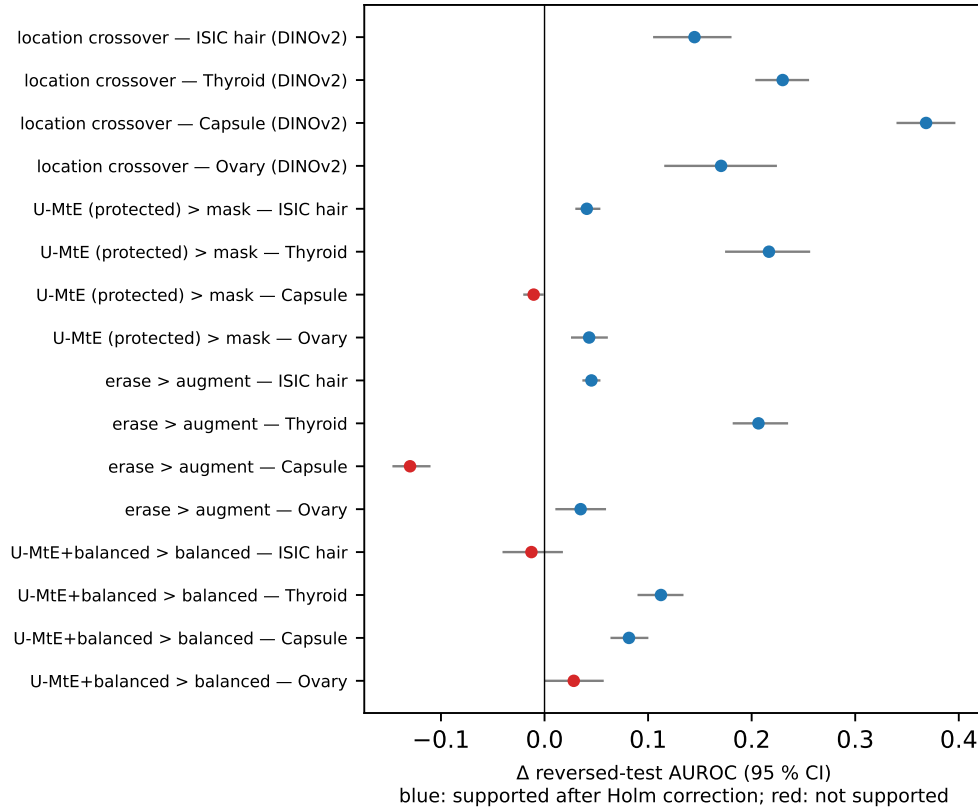


Figure 8: Forest plot of the 16 primary tests (blue: supported after Holm).

10 Stage 6 — replication breadth: backbones, scale, foundation models

Backbone families. Thyroid crossover +0.163 (MedSigLIP), +0.284 (ConvNeXt); capsule +0.332 (MedSigLIP), +0.292 (ConvNeXt); ovary +0.179 (MedSigLIP). **Scale** (DINOv2 ViT-S/B/L; registry \$R25):

Table 6: Backbone scale (Trap A for the U-MtE contrasts; ERM gap = correlated – reversed AUROC, Trap A).

Cohort	DINOv2	Crossover	U-MtE _{prot} –mask	U-MtE–mask	ERM gap
thyroid	ViT-S/14	+0.244 [+0.209, +0.275]	+0.181 [+0.146, +0.218]	+0.182 [+0.144, +0.213]	0.547
thyroid	ViT-B/14	+0.230 [+0.204, +0.255]	+0.217 [+0.174, +0.257]	+0.222 [+0.196, +0.251]	0.590
thyroid	ViT-L/14	+0.271 [+0.234, +0.308]	+0.222 [+0.195, +0.256]	+0.213 [+0.188, +0.235]	0.621
ovary	ViT-S/14	+0.298 [+0.229, +0.365]	+0.103 [+0.078, +0.128]	+0.120 [+0.092, +0.147]	0.319
ovary	ViT-B/14	+0.170 [+0.116, +0.224]	+0.043 [+0.026, +0.061]	+0.019 [-0.007, +0.044]	0.409
ovary	ViT-L/14	+0.202 [+0.153, +0.250]	+0.060 [+0.043, +0.077]	+0.094 [+0.071, +0.118]	0.570
capsule	ViT-S/14	+0.370 [+0.341, +0.401]	-0.008 [-0.014, -0.003]	-0.153 [-0.167, -0.140]	0.353
capsule	ViT-B/14	+0.368 [+0.340, +0.397]	-0.010 [-0.021, -0.001]	-0.115 [-0.128, -0.101]	0.354
capsule	ViT-L/14	+0.408 [+0.378, +0.438]	-0.040 [-0.046, -0.035]	-0.093 [-0.103, -0.083]	0.325

Larger models exploit the in-ROI caliper shortcut *more* (ERM gap thyroid 0.547→0.621, ovary 0.319→0.570).

Zero-shot vision-language models (no training; registry \$R20): zero-shot scores are already swayed by artifacts (capsule correlated–reversed gap 0.40) and masking changes them per the location law: crossovers thyroid +0.103 [+0.079, +0.127], capsule +0.049 [+0.024, +0.076], ovary +0.181 [+0.135, +0.228] (MedSigLIP), hair +0.051 [+0.031, +0.071] (DermLIP). Caveat: MedSigLIP zero-shot thyroid classification is at chance (clean 0.45). **Conclusion.** The effect is a property of the representation, not of the trained head.

11 Stage 7 — unaltered test distributions and clinical metrics

Design (registry §R16). No resampling. Thyroid: official TN3K split; ISIC 2019: each hospital (BCN, HAM, MSK) held out in turn; capsule: held-out frame-block groups (per seed). Natural associations: thyroid calipers 37.6% of benign vs 24.1% of malignant; in-lesion hair 31% of melanomas vs 20% of benign; capsule debris on lesion 23% of polyp-like vs 10% of erosions. “Hard” pairs = shortcut-conflicting pairs.

Table 7: Natural (unaltered) test distributions: every bootstrap contrast (hard = shortcut-conflicting pairs).

Test set	Subset contrast	Δ AUROC [95% CI]
capsule	hard mask-erm	+0.021 [+0.000, +0.044]
capsule	hard mte-mask	+0.000 [-0.010, +0.009]
capsule	hard mte_protect-mask	-0.024 [-0.035, -0.014]
capsule	hard balanced-mask	-0.009 [-0.031, +0.011]
capsule	hard mte_balanced-mask	+0.014 [+0.003, +0.025]
capsule	easy mask-erm	+0.071 [+0.042, +0.104]
capsule	easy mte-mask	-0.001 [-0.009, +0.007]
capsule	easy mte_protect-mask	-0.021 [-0.030, -0.014]
capsule	easy balanced-mask	-0.079 [-0.123, -0.046]
capsule	easy mte_balanced-mask	-0.008 [-0.016, +0.000]
capsule	all mask-erm	+0.077 [+0.062, +0.093]
capsule	all mte-mask	-0.005 [-0.010, +0.000]
capsule	all mte_protect-mask	-0.022 [-0.026, -0.018]
capsule	all balanced-mask	-0.086 [-0.106, -0.067]
capsule	all mte_balanced-mask	-0.008 [-0.013, -0.002]
isic BCN	hard mask-erm	-0.016 [-0.027, -0.005]
isic BCN	hard mte-mask	+0.003 [+0.000, +0.006]
isic BCN	hard mte_protect-mask	-0.002 [-0.007, +0.002]
isic BCN	hard balanced-mask	+0.034 [+0.023, +0.045]
isic BCN	hard mte_balanced-mask	+0.034 [+0.029, +0.039]
isic BCN	easy mask-erm	-0.012 [-0.021, -0.004]
isic BCN	easy mte-mask	+0.003 [-0.000, +0.006]
isic BCN	easy mte_protect-mask	-0.024 [-0.029, -0.019]
isic BCN	easy balanced-mask	-0.008 [-0.017, +0.001]
isic BCN	easy mte_balanced-mask	-0.018 [-0.023, -0.014]
isic BCN	all mask-erm	-0.016 [-0.021, -0.012]
isic BCN	all mte-mask	+0.004 [+0.002, +0.006]
isic BCN	all mte_protect-mask	-0.014 [-0.017, -0.011]
isic BCN	all balanced-mask	+0.011 [+0.006, +0.016]
isic BCN	all mte_balanced-mask	+0.004 [+0.002, +0.006]
isic HAM	hard mask-erm	+0.053 [+0.037, +0.068]
isic HAM	hard mte-mask	+0.023 [+0.017, +0.032]
isic HAM	hard mte_protect-mask	-0.043 [-0.055, -0.031]
isic HAM	hard balanced-mask	-0.022 [-0.038, -0.006]
isic HAM	hard mte_balanced-mask	+0.065 [+0.056, +0.075]
isic HAM	easy mask-erm	+0.015 [+0.002, +0.028]
isic HAM	easy mte-mask	-0.001 [-0.015, +0.010]
isic HAM	easy mte_protect-mask	+0.013 [+0.004, +0.024]
isic HAM	easy balanced-mask	-0.041 [-0.061, -0.021]
isic HAM	easy mte_balanced-mask	-0.024 [-0.030, -0.017]
isic HAM	all mask-erm	+0.040 [+0.033, +0.046]
isic HAM	all mte-mask	+0.010 [+0.004, +0.014]
isic HAM	all mte_protect-mask	-0.011 [-0.018, -0.005]
isic HAM	all balanced-mask	-0.042 [-0.054, -0.026]
isic HAM	all mte_balanced-mask	+0.013 [+0.008, +0.017]
isic MSK	hard mask-erm	-0.056 [-0.084, -0.029]
isic MSK	hard mte-mask	-0.008 [-0.015, -0.002]
isic MSK	hard mte_protect-mask	+0.032 [+0.023, +0.042]
isic MSK	hard balanced-mask	+0.094 [+0.069, +0.119]
isic MSK	hard mte_balanced-mask	+0.041 [+0.029, +0.053]
isic MSK	easy mask-erm	+0.099 [+0.080, +0.118]
isic MSK	easy mte-mask	+0.001 [-0.005, +0.007]
isic MSK	easy mte_protect-mask	-0.027 [-0.034, -0.020]
isic MSK	easy balanced-mask	-0.132 [-0.153, -0.111]
isic MSK	easy mte_balanced-mask	-0.028 [-0.036, -0.021]
isic MSK	all mask-erm	+0.021 [+0.010, +0.031]
isic MSK	all mte-mask	-0.007 [-0.010, -0.004]
isic MSK	all mte_protect-mask	-0.004 [-0.008, +0.001]
isic MSK	all balanced-mask	-0.024 [-0.035, -0.013]
isic MSK	all mte_balanced-mask	-0.016 [-0.021, -0.011]
thyroid	hard mask-erm	-0.102 [-0.139, -0.066]
thyroid	hard mte-mask	+0.051 [+0.033, +0.072]
thyroid	hard mte_protect-mask	-0.058 [-0.079, -0.036]
thyroid	hard balanced-mask	+0.142 [+0.107, +0.178]
thyroid	hard mte_balanced-mask	+0.088 [+0.071, +0.108]
thyroid	easy mask-erm	+0.048 [+0.019, +0.079]
thyroid	easy mte-mask	-0.010 [-0.028, +0.004]
thyroid	easy mte_protect-mask	+0.010 [-0.005, +0.023]
thyroid	easy balanced-mask	-0.115 [-0.146, -0.084]
thyroid	easy mte_balanced-mask	-0.049 [-0.066, -0.035]
thyroid	all mask-erm	-0.005 [-0.027, +0.018]
thyroid	all mte-mask	+0.010 [+0.002, +0.018]
thyroid	all mte_protect-mask	-0.017 [-0.027, -0.007]
thyroid	all balanced-mask	-0.015 [-0.039, +0.008]
thyroid	all mte_balanced-mask	-0.000 [-0.009, +0.008]

Observations. Masking lowers AUROC on shortcut-conflicting pairs for thyroid (-0.102) and two of three held-out hospitals; U-MtE_{bal} improves on masking in every cohort ($+0.014$ to $+0.088$). Pre-registered N1/N2 (protected U-MtE on thyroid) were *not* supported. Integrity note: the thyroid prediction archive failed a gzip CRC check and was regenerated deterministically; every number was identical.

Table 8: Clinical secondary metrics on the unaltered test sets (Sec. 15): mean (SD) over training seeds. Sensitivity and specificity at the operating point fixed on clean validation data; ECE with 10 equal-width bins; AUPRC baseline = prevalence.

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.736 (0.009)	0.606 (0.012)	0.207 (0.005)	0.082 (0.008)	0.734 (0.047)	0.619 (0.061)
	mask	0.731 (0.006)	0.632 (0.007)	0.214 (0.006)	0.120 (0.026)	0.347 (0.095)	0.888 (0.041)
	U-MtE	0.742 (0.003)	0.638 (0.007)	0.203 (0.001)	0.085 (0.007)	0.425 (0.088)	0.854 (0.050)
	balanced	0.716 (0.007)	0.584 (0.012)	0.219 (0.004)	0.110 (0.007)	0.714 (0.052)	0.599 (0.061)
	U-MtE _{bal}	0.731 (0.004)	0.630 (0.007)	0.207 (0.001)	0.084 (0.006)	0.399 (0.069)	0.866 (0.034)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.586 (0.003)	0.239 (0.006)	0.279 (0.011)	0.842 (0.038)	0.519 (0.075)
	mask	0.769 (0.004)	0.571 (0.009)	0.286 (0.007)	0.332 (0.009)	0.860 (0.013)	0.439 (0.023)
	U-MtE	0.773 (0.004)	0.574 (0.008)	0.280 (0.009)	0.328 (0.012)	0.858 (0.039)	0.448 (0.089)
	balanced	0.781 (0.005)	0.585 (0.009)	0.228 (0.006)	0.257 (0.011)	0.811 (0.046)	0.554 (0.087)
	U-MtE _{bal}	0.774 (0.004)	0.568 (0.008)	0.279 (0.011)	0.325 (0.016)	0.860 (0.027)	0.447 (0.051)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.172 (0.007)	0.161 (0.019)	0.608 (0.049)	0.786 (0.039)
	mask	0.790 (0.004)	0.512 (0.009)	0.146 (0.003)	0.115 (0.009)	0.593 (0.052)	0.824 (0.037)
	U-MtE	0.783 (0.004)	0.500 (0.008)	0.150 (0.003)	0.126 (0.009)	0.566 (0.073)	0.831 (0.049)
	balanced	0.766 (0.008)	0.462 (0.014)	0.178 (0.008)	0.170 (0.020)	0.626 (0.047)	0.755 (0.046)
	U-MtE _{bal}	0.774 (0.006)	0.482 (0.009)	0.154 (0.004)	0.131 (0.010)	0.594 (0.061)	0.798 (0.051)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.750 (0.005)	0.353 (0.005)	0.162 (0.006)	0.234 (0.011)	0.535 (0.092)	0.813 (0.068)
	mask	0.789 (0.002)	0.404 (0.006)	0.143 (0.005)	0.217 (0.011)	0.559 (0.112)	0.842 (0.063)
	U-MtE	0.799 (0.008)	0.406 (0.009)	0.145 (0.007)	0.226 (0.016)	0.573 (0.067)	0.846 (0.041)
	balanced	0.747 (0.017)	0.346 (0.019)	0.156 (0.014)	0.228 (0.015)	0.539 (0.078)	0.803 (0.076)
	U-MtE _{bal}	0.802 (0.005)	0.401 (0.009)	0.139 (0.006)	0.215 (0.021)	0.606 (0.054)	0.825 (0.037)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.859 (0.032)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.871 (0.042)	0.897 (0.043)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.758 (0.087)	0.838 (0.059)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.044 (0.012)	0.866 (0.048)	0.897 (0.052)

12 Stage 8 – theory

Closed form (registry §R22): disease block SNR S , artifact SNR A , training rates (p_1, p_0) ; effective artifact SNR $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and $\text{AUROC}_{\text{rev}} = \Phi((S - \tilde{A}) / \sqrt{2(S + \tilde{A})})$. Consequences: out-of-ROI masking sets $\tilde{A} \rightarrow 0$; in-ROI masking keeps $\tilde{A}$ and harms iff it removes disease signal ($S_m < S$); the crossover is positive for every S_m ; erasure scales S by $1 - \rho^2$ (fails for pathology-like artifacts unless protected); group balancing is location-invariant; correlate-carried shortcuts survive masking and erasure. Simulation agreement ≤ 0.009 AUROC over 29 settings (Fig. 10). **Real-encoder test** (pre-registered; registry §R23): the general form for any test composition was fitted per model to clean and correlated AUROC only, and compared with held-out reversed AUROC.

Table 9: Theory-vs-real summary (prediction_summary.json).

Endpoint	Theory	Ref. rev=clean	Ref. rev= 2-clean-corr
T1 reversed AUROC, ERM+mask (148 models): MAE	0.039	0.234	0.108
T1 Pearson r	0.917	0.592	0.763
T1 all ERM-head arms (570): MAE	0.031	0.163	0.069
T2 crossover (14 pairs): r / MAE / signs	0.997 / 0.014 / 13	0.930 / 0.108 / 12	0.989 / 0.036 / 13
T3 sweeps (45 points): r / MAE / signs	0.965 / 0.043 / 42	0.670 / 0.122 / 31	0.869 / 0.081 / 36
T4 in-ROI sign from $S_m - S$	11/14		
Fine-tuned (outside model): rev MAE	0.018		

Pre-registered verdict: **qualitative account**.

Table 10: Theory versus real encoders: location crossover implied by (S, A) fitted to clean and correlated AUROC only.

Cohort	Backbone	Regime	Observed crossover	Theory (fit on clean+corr)
CXR Atelectasis	RAD-DINO	frozen	+0.060	+0.062
CXR Consolidation	RAD-DINO	frozen	-0.002	+0.010
CXR Effusion	RAD-DINO	frozen	-0.004	-0.013
CXR Infiltration	RAD-DINO	frozen	-0.032	-0.040
Capsule	ConvNeXt	frozen	+0.292	+0.288
Capsule	DINOv2	frozen	+0.368	+0.335
Capsule	MedSigLIP	frozen	+0.332	+0.300
Capsule	ResNet-50 FT	end-to-end	+0.584	+0.542
ISIC hair	DINOv2	frozen	+0.145	+0.147
ISIC hair	DermLIP	frozen	-0.002	-0.003
Ovary	DINOv2	frozen	+0.170	+0.157
Ovary	MedSigLIP	frozen	+0.179	+0.177
Ovary	ResNet-50 FT	end-to-end	+0.136	+0.096
Thyroid	ConvNeXt	frozen	+0.284	+0.250
Thyroid	DINOv2	frozen	+0.230	+0.204
Thyroid	MedSigLIP	frozen	+0.163	+0.146
Thyroid	ResNet-50 FT	end-to-end	+0.057	+0.054
Thyroid	ViT-S FT	end-to-end	+0.168	+0.114

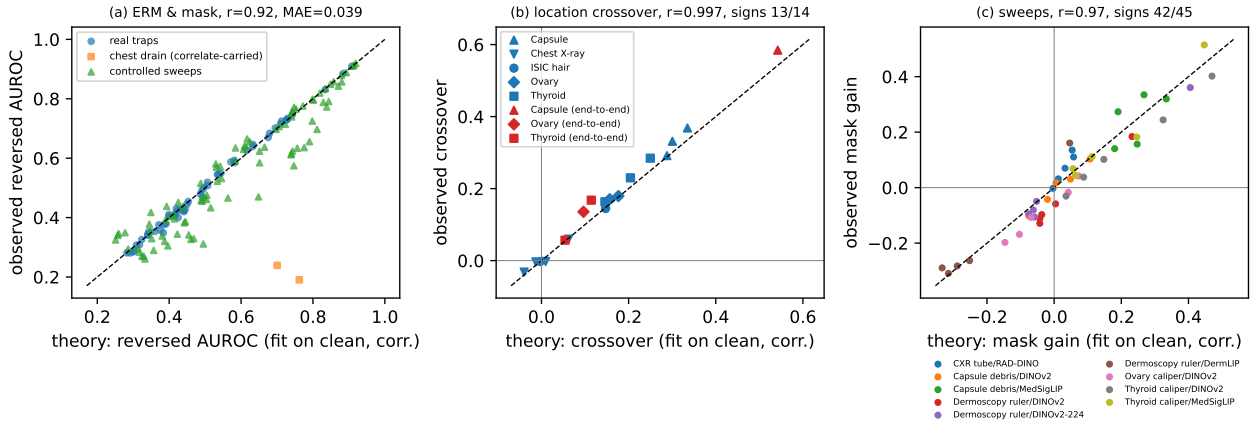


Figure 9: Theory vs real encoders: (a) reversed AUROC; (b) location crossover; (c) sweep mask gains.

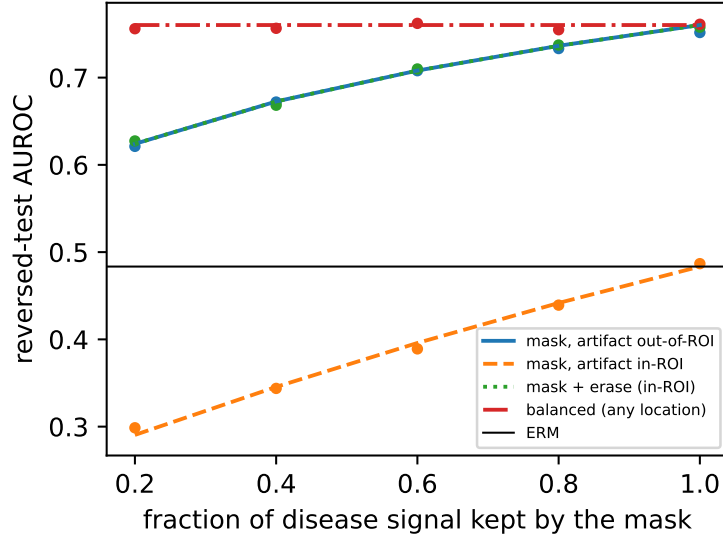


Figure 10: Theory (lines) vs simulation (points).

Conclusion. Quantitatively close agreement, but by the pre-registered rule a “qualitative account” (one of 14 crossover signs disagrees at -0.002). The papers must not say the theory “predicts” magnitudes; they say it matches held-out results. Its failure on chest drains marks its boundary (correlate-carried shortcuts are invisible in the clean and correlated environments).

13 Stage 9 — remedies for frozen foundation-model features

Evolution of the method. (1) *Insert-then-erase (I2E)* in the pilot: insert artifact *templates* (rulers) into images, estimate the subspace of feature changes, erase it — works in controlled sweeps but needs an artifact model. (2) *Mask-then-erase (MtE)*: estimate the subspace on *masked* images so that erasure targets exactly what masking leaves. (3) *U-MtE*: replace templates by one *generic* procedural overlay library (strokes, bands, patches, blobs, dashed lines, glyphs; random colour/opacity/position, half centred in the ROI) used unchanged for every modality; rank k = smallest with 90% held-out difference energy (cap 64); logistic head on erased masked features. Needs no artifact example, artifact mask or artifact label — only the ROI mask. (4) *Disease protection*: erase $\text{orth}((I - WW^\top)U_k)$ with W = label directions on artifact-free images. (5) *U-MtE_{bal}*: group-balanced head when image-level artifact labels exist. Registry \$R13 (U1–U14). **Main results.** Thyroid $+0.222$ [$+0.197, +0.251$] (DINOv2), $+0.204$ [$+0.174, +0.237$] (MedSigLIP), $+0.128$ [$+0.112, +0.145$] (ConvNeXt); ovary (MedSigLIP) $+0.215$ [$+0.182, +0.246$]; hair $+0.052$; controlled sweeps $+0.08$ to $+0.28$. Real capsule: unprotected -0.115 (debris $\approx$ erosion fibrin), protected -0.010 (safety net; costs ≤ 0.012 elsewhere). Erase vs augment with the same overlays (U7): thyroid $+0.207$, hair $+0.045$, ovary $+0.035$ (not robust to embedding groups), capsule -0.130 . With artifact labels, U-MtE_{bal} is the most robust thyroid model (min(rev,corr) 0.788 DINOv2, 0.792 MedSigLIP); mask+DFR best on capsule (0.885 vs 0.857); DFR best on hair (0.741). The full arm $\times$ cohort table is Appendix Table main_inroi. **Rank ablation** (registry \$R26):

Table 11: U-MtE erasure-rank ablation (Trap A reversed AUROC, mean over seeds; e = held-out energy, k = fixed rank).

Cohort	Rank rule	mask	U-MtE	U-MtE _{prot}	U-MtE—mask	prot—mask
capsule	e0.50	0.684	0.630	0.676	-0.054	-0.008
capsule	e0.90	0.684	0.569	0.673	-0.115	-0.010
capsule	e0.99	0.684	0.569	0.673	-0.115	-0.010
capsule	k1	0.684	0.685	0.685	+0.001	+0.001
capsule	k16	0.684	0.634	0.675	-0.050	-0.009
capsule	k4	0.684	0.676	0.681	-0.008	-0.002
capsule	k64	0.684	0.569	0.673	-0.115	-0.010
ovary	e0.50	0.545	0.543	0.578	-0.002	+0.033
ovary	e0.90	0.545	0.564	0.588	+0.019	+0.043
ovary	e0.99	0.545	0.564	0.588	+0.019	+0.043
ovary	k1	0.545	0.558	0.552	+0.012	+0.007
ovary	k16	0.545	0.535	0.565	-0.010	+0.020
ovary	k4	0.545	0.504	0.560	-0.041	+0.015
ovary	k64	0.545	0.564	0.588	+0.019	+0.043
thyroid	e0.50	0.400	0.611	0.611	+0.211	+0.211
thyroid	e0.90	0.400	0.622	0.617	+0.222	+0.217
thyroid	e0.99	0.400	0.622	0.617	+0.222	+0.217
thyroid	k1	0.400	0.471	0.451	+0.071	+0.051
thyroid	k16	0.400	0.580	0.576	+0.180	+0.176
thyroid	k4	0.400	0.463	0.451	+0.063	+0.051
thyroid	k64	0.400	0.622	0.617	+0.222	+0.217

Baselines and alternatives (each pre-registered).

- **JTT** (label-free): U-MtE beats it on thyroid (+0.198) and ovary (+0.137); JTT wins on hair.
- **SPLINCE** (Holstege, Ravfogel & Wouters, NeurIPS 2025; faithful whitened implementation): with artifact labels it removes much disease signal (clean 0.56–0.78) and flips the shortcut on capsule; U-MtE_{bal} more robust on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564); on drains SPLINCE 0.516 vs 0.460.
- **LEACE** (paired/unpaired): unpaired erasure with image-level labels flips the shortcut; paired recovers little.
- **LaMa inpainting** with oracle artifact masks: strong; protected U-MtE matches it on thyroid without masks (−0.009 [−0.045, +0.023]), below it on ovary; U-MtE_{bal} beats it.
- **Text-prompted erasure** (biased prompts; MedSigLIP, DermLIP): near no-op ($\leq +0.008$ over masking).
- **SLAS** (few-shot patch-token suppression): localises hair from 5 annotated images (patch AUROC 0.93), small classification gains (+0.02 hair, +0.08 thyroid).
- **Pseudo-group balancing** (U8; overlay-detector pseudo labels): +0.04 on thyroid, none on hair.
- **Erasure + DFR** (U10): no gain over masking + DFR.
- **Validation-driven selectors** (U13, U14): beat masking in 5/5 cohorts (+0.17 to +0.40), within 0.02 of the best candidate in 5/5, beat DFR in 3/5; split-validation variant fixes drains (0.656); neither uniformly best.

14 Stage 10 — end-to-end fine-tuning and remedies for fine-tuned networks

Design (registry §R15): ResNet-50 (ImageNet, 224 px, 8 epochs, AdamW, lr 10^{-4} , head lr $\times 10$, OneCycle, fp16) and ViT-S/16 (lr $3 \cdot 10^{-5}$); seed 42, folds 0–4 as bootstrap clusters; thresholds on clean validation. **Location law.** Capsule (ResNet-50): masking harms in-ROI (−0.284 [−0.347, −0.226]), helps out (+0.301), crossover +0.584 [+0.469, +0.691]; thyroid ViT-S crossover +0.168 [+0.074, +0.265]; thyroid ResNet-50 +0.057 [−0.021, +0.130]. **Remedies tried for fine-tuned networks.**

- **mte_ft** (invariance penalty between masked images with and without a generic overlay): thyroid ResNet-50 +0.143 [+0.099, +0.188] **[supported]**; ViT-S +0.009 **[not supported]**; capsule −0.021 **[not supported]**.
- **mte_post** (fine-tune like mask, then U-MtE on the penultimate features; registry §R27): +0.004 [+0.000, +0.012] over the same features without erasure **[not supported]**. After unconstrained fine-tuning the caliper is encoded along task-specific directions that generic overlays do not move.

- **umte_ft** (erase-while-fine-tuning: running overlay-shift subspace projected out before the head at every step, plus the invariance penalty; registry \$R28\$): ResNet-50 +0.160 [+0.114, +0.207] over masking and +0.016 over mte_ft; ViT-S +0.061 [−0.005, +0.141] over masking, +0.052 [+0.007, +0.107] over mte_ft **[partial]**. Note: on ResNet-50 correlated AUROC (0.664) is slightly below reversed (0.695) — mild over-correction.
- **cons_ft** (prediction consistency between masked images with and without an overlay): ResNet-50 +0.007 [−0.014, +0.029] **[not supported]**; ViT-S +0.062 [+0.031, +0.087] **[supported]**.
- **umte_cons_ft (all three combined; pre-registered amendment before running)**: ResNet-50 +0.157 [+0.114, +0.205], ViT-S +0.143 [+0.103, +0.187] **[supported]** — the first remedy that works for fully fine-tuned networks of both architecture families without artifact annotation. AUROC clean/corr/rev: ResNet-50 0.688/0.661/0.693 (mask 0.691/0.827/0.536; mild over-correction); ViT-S 0.712/0.798/0.608 (mask 0.688/0.868/0.465).
- Other cohorts (ResNet-50): **umte_ft** ovary +0.107 [−0.003, +0.229]; **umte_cons_ft** (Amendment 2; run after recovering the GPU with a private copy of the identical driver library): ovary +0.094 [−0.011, +0.210] **[partial]** (better than masking on clean, correlated and reversed AUROC, 0.687/0.726/0.650 vs 0.578/0.570/0.556; underpowered cohort); capsule −0.022 [−0.046, +0.001] **[not supported]** (pathology-like debris; fine-tuned masking collapses to 0.211 and only label-based balancing helps, mask+balanced 0.687).

Table 12: Every end-to-end fine-tuning contrast (5 folds of seed 42 as clusters).

Cohort	Architecture	Trap	Contrast	Δ reversed AUROC [95% CI]
capsule	ResNet-50	trapA	mask – erm	-0.284 [-0.347, -0.226]
capsule	ResNet-50	trapA	mte_ft – mask	-0.021 [-0.039, -0.003]
capsule	ResNet-50	trapA	mask_balanced – balanced	+0.017 [-0.070, +0.105]
capsule	ResNet-50	trapA	mte_ft – erm	-0.305 [-0.367, -0.246]
capsule	ResNet-50	trapA	umte_cons_ft – mask	-0.022 [-0.046, +0.001]
capsule	ResNet-50	trapB	mask – erm	+0.301 [+0.225, +0.380]
capsule	ResNet-50	trapB	mte_ft – mask	-0.205 [-0.286, -0.111]
capsule	ResNet-50	trapB	mask_balanced – balanced	-0.024 [-0.117, +0.065]
capsule	ResNet-50	trapB	mte_ft – erm	+0.096 [-0.003, +0.194]
ovary	ResNet-50	trapA	mask – erm	+0.098 [-0.001, +0.212]
ovary	ResNet-50	trapA	mte_ft – mask	+0.094 [-0.025, +0.217]
ovary	ResNet-50	trapA	mask_balanced – balanced	-0.009 [-0.128, +0.109]
ovary	ResNet-50	trapA	mte_ft – erm	+0.192 [+0.103, +0.278]
ovary	ResNet-50	trapA	umte_ft – mask	+0.107 [-0.003, +0.229]
ovary	ResNet-50	trapA	umte_ft – mte_ft	+0.013 [-0.049, +0.074]
ovary	ResNet-50	trapA	umte_cons_ft – mask	+0.094 [-0.011, +0.210]
ovary	ResNet-50	trapB	mask – erm	+0.234 [+0.084, +0.376]
ovary	ResNet-50	trapB	mte_ft – mask	+0.014 [-0.067, +0.104]
ovary	ResNet-50	trapB	mask_balanced – balanced	-0.007 [-0.128, +0.120]
ovary	ResNet-50	trapB	mte_ft – erm	+0.248 [+0.123, +0.366]
thyroid	ResNet-50	trapA	mask – erm	+0.105 [+0.067, +0.141]
thyroid	ResNet-50	trapA	mte_ft – mask	+0.143 [+0.099, +0.188]
thyroid	ResNet-50	trapA	mask_balanced – balanced	+0.147 [+0.104, +0.192]
thyroid	ResNet-50	trapA	mte_ft – erm	+0.248 [+0.209, +0.288]
thyroid	ResNet-50	trapA	mte_post – mask	-0.027 [-0.053, -0.001]
thyroid	ResNet-50	trapA	mte_post – mask_post	+0.004 [+0.000, +0.012]
thyroid	ResNet-50	trapA	umte_ft – mask	+0.160 [+0.114, +0.207]
thyroid	ResNet-50	trapA	umte_ft – mte_ft	+0.016 [+0.002, +0.030]
thyroid	ResNet-50	trapA	cons_ft – mask	+0.007 [-0.014, +0.029]
thyroid	ResNet-50	trapA	umte_cons_ft – mask	+0.157 [+0.114, +0.205]
thyroid	ResNet-50	trapB	mask – erm	+0.161 [+0.096, +0.226]
thyroid	ResNet-50	trapB	mte_ft – mask	-0.034 [-0.062, -0.008]
thyroid	ResNet-50	trapB	mask_balanced – balanced	+0.123 [+0.061, +0.185]
thyroid	ResNet-50	trapB	mte_ft – erm	+0.128 [+0.066, +0.192]
thyroid	ViT-S/16	trapA	mask – erm	+0.088 [+0.051, +0.126]
thyroid	ViT-S/16	trapA	mte_ft – mask	+0.009 [-0.021, +0.040]
thyroid	ViT-S/16	trapA	mask_balanced – balanced	+0.017 [-0.023, +0.055]
thyroid	ViT-S/16	trapA	mte_ft – erm	+0.098 [+0.066, +0.129]
thyroid	ViT-S/16	trapA	umte_ft – mask	+0.061 [-0.005, +0.141]
thyroid	ViT-S/16	trapA	umte_ft – mte_ft	+0.052 [+0.007, +0.107]
thyroid	ViT-S/16	trapA	cons_ft – mask	+0.062 [+0.031, +0.087]
thyroid	ViT-S/16	trapA	umte_cons_ft – mask	+0.143 [+0.103, +0.187]
thyroid	ViT-S/16	trapB	mask – erm	+0.256 [+0.164, +0.354]
thyroid	ViT-S/16	trapB	mte_ft – mask	-0.000 [-0.020, +0.019]
thyroid	ViT-S/16	trapB	mask_balanced – balanced	+0.069 [+0.009, +0.141]
thyroid	ViT-S/16	trapB	mte_ft – erm	+0.256 [+0.161, +0.358]

Table 13: End-to-end fine-tuning: AUROC per arm (mean over folds).

Cohort	Arch.	Trap	Arm	Clean	Correlated	Reversed
capsule	ResNet-50	trapA	balanced	0.742	0.799	0.670
capsule	ResNet-50	trapA	erm	0.719	0.900	0.495
capsule	ResNet-50	trapA	mask	0.596	0.915	0.211
capsule	ResNet-50	trapA	mask_balanced	0.754	0.780	0.687
capsule	ResNet-50	trapA	mte_ft	0.567	0.904	0.190
capsule	ResNet-50	trapA	umte_cons_ft	0.555	0.879	0.190
capsule	ResNet-50	trapB	balanced	0.787	0.839	0.724
capsule	ResNet-50	trapB	erm	0.711	0.915	0.409
capsule	ResNet-50	trapB	mask	0.786	0.851	0.710
capsule	ResNet-50	trapB	mask_balanced	0.752	0.787	0.700
capsule	ResNet-50	trapB	mte_ft	0.569	0.613	0.505
ovary	ResNet-50	trapA	balanced	0.575	0.554	0.602
ovary	ResNet-50	trapA	erm	0.576	0.671	0.457
ovary	ResNet-50	trapA	mask	0.578	0.570	0.556
ovary	ResNet-50	trapA	mask_balanced	0.569	0.551	0.593
ovary	ResNet-50	trapA	mte_ft	0.687	0.743	0.649
ovary	ResNet-50	trapA	umte_cons_ft	0.687	0.726	0.650
ovary	ResNet-50	trapA	umte_ft	0.694	0.725	0.663
ovary	ResNet-50	trapB	balanced	0.595	0.559	0.595
ovary	ResNet-50	trapB	erm	0.560	0.716	0.366
ovary	ResNet-50	trapB	mask	0.600	0.585	0.600
ovary	ResNet-50	trapB	mask_balanced	0.528	0.466	0.589
ovary	ResNet-50	trapB	mte_ft	0.667	0.704	0.614
thyroid	ResNet-50	trapA	balanced	0.623	0.631	0.599
thyroid	ResNet-50	trapA	cons_ft	0.690	0.823	0.543
thyroid	ResNet-50	trapA	erm	0.587	0.744	0.431
thyroid	ResNet-50	trapA	mask	0.691	0.827	0.536
thyroid	ResNet-50	trapA	mask_balanced	0.721	0.677	0.747
thyroid	ResNet-50	trapA	mask_post	0.673	0.831	0.504
thyroid	ResNet-50	trapA	mte_ft	0.698	0.703	0.679
thyroid	ResNet-50	trapA	mte_post	0.674	0.829	0.508
thyroid	ResNet-50	trapA	umte_cons_ft	0.685	0.661	0.693
thyroid	ResNet-50	trapA	umte_ft	0.688	0.664	0.695
thyroid	ResNet-50	trapB	balanced	0.583	0.595	0.592
thyroid	ResNet-50	trapB	erm	0.580	0.646	0.553
thyroid	ResNet-50	trapB	mask	0.700	0.714	0.714
thyroid	ResNet-50	trapB	mask_balanced	0.692	0.693	0.715
thyroid	ResNet-50	trapB	mte_ft	0.668	0.717	0.681
thyroid	ViT-S/16	trapA	balanced	0.701	0.700	0.693
thyroid	ViT-S/16	trapA	cons_ft	0.712	0.866	0.528
thyroid	ViT-S/16	trapA	erm	0.640	0.860	0.377
thyroid	ViT-S/16	trapA	mask	0.688	0.868	0.465
thyroid	ViT-S/16	trapA	mask_balanced	0.760	0.788	0.710
thyroid	ViT-S/16	trapA	mte_ft	0.695	0.878	0.475
thyroid	ViT-S/16	trapA	umte_cons_ft	0.712	0.798	0.608
thyroid	ViT-S/16	trapA	umte_ft	0.660	0.772	0.527
thyroid	ViT-S/16	trapB	balanced	0.710	0.714	0.712
thyroid	ViT-S/16	trapB	erm	0.642	0.774	0.504
thyroid	ViT-S/16	trapB	mask	0.758	0.808	0.760
thyroid	ViT-S/16	trapB	mask_balanced	0.767	0.773	0.780
thyroid	ViT-S/16	trapB	mte_ft	0.748	0.783	0.760

15 Stage 11 – robustness, leakage and integrity

Artifact-label definitions (registry §R17): large-marker (≥ 50 px) and strict-location ($r \geq 0.7 / r < 0.05$) variants keep the crossover (thyroid +0.215 to +0.262; ovary +0.170 to +0.195) and the protected-U-MtE gain (thyroid +0.217 to +0.282; ovary +0.043 to +0.067) significant; strict capsule Trap B +0.259. **Mask audit** (registry §R18): 40 images per cell inspected. **Leakage.** (i) Random vs grouped splits (Table 3). (ii) Stricter embedding groups for the three cohorts without patient IDs (registry §R24; amendment: embeddings mean-centred because raw cosine was ≈ 0.99 for all ovary pairs):

Table 14: Embedding-based near-duplicate groups (mean-centred DINOv2 CLS, cosine $\geq \tau$).

Cohort	Images	Groups (pHash/frames)	Groups (+embedding)	τ	Largest group share
thyroid	3493	3455	3081	0.8	0.045805897509304326
ovary	1202	1059	969	0.8	0.020798668885191347
capsule	5481	137	135	0.93	0.03667214012041598

Table 15: Primary contrasts under stricter embedding-based leakage groups.

Cohort	Test	Main groups	Embedding groups	Robust
Thyroid	P1 location crossover	+0.230 [+0.204, +0.255]	+0.235 [+0.211, +0.259]	yes
Thyroid	P2 U-MtE (protected) > mask	+0.217 [+0.174, +0.257]	+0.251 [+0.219, +0.279]	yes
Thyroid	P3 erase > augment	+0.207 [+0.182, +0.235]	+0.192 [+0.162, +0.218]	yes
Thyroid	P4 U-MtE+balanced > balanced	+0.112 [+0.090, +0.134]	+0.152 [+0.138, +0.167]	yes
Ovary	P1 location crossover	+0.170 [+0.116, +0.224]	+0.157 [+0.080, +0.235]	yes
Ovary	P2 U-MtE (protected) > mask	+0.043 [+0.026, +0.061]	+0.047 [+0.020, +0.072]	yes
Ovary	P3 erase > augment	+0.035 [+0.010, +0.059]	+0.020 [-0.011, +0.053]	no
Ovary	P4 U-MtE+balanced > balanced	+0.028 [+0.000, +0.057]	+0.039 [+0.015, +0.064]	yes
Capsule	P1 location crossover	+0.368 [+0.340, +0.397]	+0.394 [+0.359, +0.430]	yes
Capsule	P2 U-MtE (protected) > mask	-0.010 [-0.021, -0.001]	-0.007 [-0.013, -0.001]	yes
Capsule	P3 erase > augment	-0.130 [-0.147, -0.110]	-0.136 [-0.153, -0.122]	yes
Capsule	P4 U-MtE+balanced > balanced	+0.082 [+0.064, +0.100]	+0.068 [+0.050, +0.085]	yes

Integrity. Every CI in both papers is traced automatically to a result file (`scripts/verify/audit_paper_numbers.py`; 109/109). All 146 result archives pass a gzip check after one regeneration (identical numbers). Deterministic arms were verified identical across historical run folders. Deviations from pre-registrations are recorded in each document (e.g. grid trimming in the rank ablation, fp32 features in fine-tune-then-erase, one skipped sweep in the theory test).

16 Stage 12 — datasets considered and not used

CANDID-PTX (no access); CXR CLiP real devices (masking helps at both locations — no in-ROI regime); CXR cardiomegaly lines (too few labelled); TCGA/GrandQC pathology pen marks (unreliable masks); ISIC ink (70 in-lesion images); breast ultrasound BUSI/BUS-BRA (two attempts: caliper labels $\approx 70\%$ pure; only 19/2,522 overlay-free images — registry \$R12); EAD endoscopy (no disease labels); BKAI-IGH NeoPolyp colonoscopy (no specular-free images); DINOv3 (weights gated, access refused).

17 Cross-cutting methods

- **Backbones.** DINOv2 ViT-S/B/L-14 @518 (CLS), MedSigLIP-448 (image features, joint space), DermLIP/PanDerm (512-d joint space), RAD-DINO @518, ConvNeXt-Base @384; fine-tuned ResNet-50 and ViT-S/16. Views rendered at 518px; features cached per view (`erm`, `mask`, `mask_insert_generic`, ...).
- **Heads.** Logistic regression; C and the decision threshold (balanced accuracy) chosen on clean validation only.
- **Seeds.** Real traps: 42, 123, 456, 789, 2026×5 folds; controlled sweeps: 42, 123, 456; fine-tuning: seed 42, folds 0-4; natural tests: 5 seeds; bootstrap seeds deterministic per comparison (string hash).
- **Statistics.** Hierarchical paired bootstrap (10,000); two-sided bootstrap p ; Holm over the 16 primary tests.
- **Compute.** One RTX 3090 (24 GB), 8 CPU cores, 31 GB RAM; about two days end to end.
- **Reproduction.** `make data data_extra verify synthetic isic2019 thyroid ovary capsule cxr drain natural finetune sensitivity scale lama baselines analysis paper`; tool: `python -m wtss.umte, python -m wtss.slas`.

18 External review and responses

1. “U-MtE needs no mask” is misleading (it uses the ROI mask) — *valid*; wording fixed everywhere.
2. Table 1 calls the method I2E — *valid*; renamed “insert-then-erase (artifact templates)” and explained.
3. GroupDRO claimed as benchmarked — *partly valid*; now scoped to the controlled dermoscopy sweeps.
4. Title omits X-ray — *valid, minor*; chest radiography named as a boundary case.
5. Patient/video leakage in TN3K/SEE-AI — *valid*; embedding-group sensitivity added (11/12 robust).
6. Theory only checked against its own simulation — *valid*; real-encoder test added ($r = 0.997$).

7. End-to-end evidence weaker — *valid*; U-MtE framed for frozen features; post-hoc and erase-while-fine-tuning tested.
8. Clinical metrics missing — *valid*; AUPRC, Brier, ECE, sensitivity/specificity added.
9. Related work thin — *valid*; six verified references added (Pewton 2024, Wang 2024, Germani 2026 MedIA, Lin 2024 MICCAI, Zech 2018, Brown 2023).

19 Conclusions and guidance for writing the paper

Claims the evidence supports (safe to make).

1. ROI masking helps much less for in-ROI than out-of-ROI artifacts (4/4 real cohorts, Holm; all replications).
2. In-ROI masking can be harmful, including on unaltered data and in clinical metrics (thyroid sensitivity).
3. The mechanism is retention with amplified reliance (counterfactual sensitivity, dilation, lesion-size strata).
4. A two-parameter linear-Gaussian account reproduces all signs and closely matches held-out results of real encoders; call it a qualitative account.
5. For frozen foundation-model features, U-MtE removes in-ROI shortcuts carried by distinct overlays (thyroid across 3 families $\times$ 3 sizes; ovary with MedSigLIP) without artifact annotation; erasure beats augmentation (thyroid, hair); protection makes it safe for pathology-like artifacts; text-prompted erasure does not work.
6. What carries the shortcut decides the method: pixels (U-MtE), pathology-like appearance (protected U-MtE or mask+DFR), correlates (DFR/balancing).

Claims to avoid. “Masking always harms in-ROI” (it helps a little on real thyroid/ovary/capsule); “the theory predicts magnitudes”; “U-MtE works post hoc on fine-tuned networks” (null); “erase-while-fine-tuning works for any architecture and disease” (shown for ResNet-50 and ViT-S on thyroid; see Stage 10 for other cohorts); “erase beats augment in ovary” (not robust); “four diseases and four modalities” (chest radiography is a boundary case); any claim of patient-level independence for TN3K, MMOTU, SEE-AI. **Limitations.** Detector-based artifact labels (audited); no patient IDs for three cohorts; reversed environments are constructed by subsampling; disease protection needs diagnosis labels of artifact-free images; balanced variants need image-level artifact labels; the theory is fitted per model and blind to correlate-carried shortcuts. **Suggested paper structure** (as in `paper_neurips/main.tex`): teaser with real images; location-law table; theory with the real-encoder figure; natural-data table; compact remedies table; forest plot; mechanism and dose-response figures in the appendix; scale, leakage, clinical metrics and fine-tuning in the robustness appendix.

Part II

Appendices

A All generated paper tables

main inroi

Table 16: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by scripts/make_main_table.py.

Method	Needs	Hair/DINOv2	Hair/DermatLP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConvNext	Ovary/DINOv2	Ovary/MedSigLIP	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConvNext	Drain/RAD-DINO	Drain/DINOv2
ERM	-	0.49 (0.49)	0.67 (0.67)	0.29 (0.29)	0.28 (0.28)	0.35 (0.35)	0.41 (0.41)	0.42 (0.42)	0.59 (0.59)	0.59 (0.59)	0.55 (0.55)	0.19 (0.19)	0.28 (0.28)
ROI masking	ROI	0.45 (0.45)	0.58 (0.58)	0.40 (0.40)	0.35 (0.35)	0.36 (0.36)	0.55 (0.55)	0.43 (0.43)	0.68 (0.68)	0.70 (0.70)	0.64 (0.64)	0.24 (0.24)	0.29 (0.29)
Mask + overlay augmentation	ROI	0.46 (0.46)	-	0.42 (0.42)	0.38 (0.38)	0.37 (0.37)	0.53 (0.53)	-	0.70 (0.70)	0.72 (0.72)	0.65 (0.65)	0.24 (0.24)	0.29 (0.29)
U-MIE (ours)	ROI	0.51 (0.51)	0.51 (0.51)	0.62 (0.62)	0.55 (0.55)	0.49 (0.49)	0.56 (0.56)	0.64 (0.64)	0.57 (0.57)	0.47 (0.47)	0.56 (0.56)	0.27 (0.27)	0.28 (0.28)
U-MIE, protected (ours)	ROI + Y of A=0	0.49 (0.49)	0.59 (0.59)	0.62 (0.62)	0.61 (0.61)	0.55 (0.55)	0.59 (0.59)	0.57 (0.57)	0.67 (0.67)	0.67 (0.67)	0.64 (0.64)	0.27 (0.27)	0.28 (0.28)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.66 (0.66)	0.71 (0.71)	0.73 (0.73)	0.78 (0.78)	0.81 (0.81)	0.72 (0.72)	0.54 (0.54)	0.40 (0.40)
DFR	A	0.74 (0.74)	0.78 (0.78)	0.74 (0.66)	0.76 (0.72)	0.71 (0.67)	0.72 (0.69)	-	0.80 (0.80)	0.80 (0.80)	0.77 (0.77)	0.68 (0.68)	0.55 (0.55)
Mask + DFR	ROI + A	0.69 (0.69)	-	0.83 (0.74)	0.81 (0.76)	0.77 (0.72)	0.75 (0.75)	-	0.88 (0.88)	0.90 (0.90)	0.86 (0.86)	0.65 (0.65)	0.57 (0.57)
U-MIE + balanced (ours)	ROI + A	0.70 (0.70)	0.65 (0.65)	0.79 (0.79)	0.79 (0.79)	0.74 (0.74)	0.74 (0.74)	0.79 (0.79)	0.86 (0.86)	0.85 (0.85)	0.80 (0.80)	0.46 (0.46)	0.44 (0.44)

Table 17: In-ROI artifacts (Trap A): robustness = min(reversed, correlated) AUROC, which penalises both shortcut use and shortcut flipping. Bold: best per column. Full reversed/correlated/clean values for all cohorts and backbones: Table 16.

Method	Needs	Hair DINOv2	Thyroid DINOv2	Thyroid MedSigLIP	Ovary DINOv2	Capsule DINOv2	Drain RAD-DINO
ROI masking	ROI	0.45	0.40	0.35	0.55	0.68	0.24
Mask + overlay augmentation	ROI	0.46	0.42	0.38	0.53	0.70	0.24
U-MtE (ours)	ROI	0.51	0.62	0.55	0.56	0.57	0.27
U-MtE, protected (ours)	ROI	0.49	0.62	0.61	0.59	0.67	0.27
Group-balanced	A	0.72	0.68	0.70	0.71	0.78	0.54
DFR	A	0.74	0.66	0.72	0.69	0.80	0.68
Mask + DFR	ROI + A	0.69	0.74	0.76	0.75	0.88	0.65
U-MtE + balanced (ours)	ROI + A	0.70	0.79	0.79	0.74	0.86	0.46

main inroi compact

Table 18: Clinical secondary metrics on the unaltered test sets (Sec. 15): mean (SD) over training seeds. Sensitivity and specificity at the operating point fixed on clean validation data; ECE with 10 equal-width bins; AUPRC baseline = prevalence.

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.736 (0.009)	0.606 (0.012)	0.207 (0.005)	0.082 (0.008)	0.734 (0.047)	0.619 (0.061)
	mask	0.731 (0.006)	0.632 (0.007)	0.214 (0.006)	0.120 (0.026)	0.347 (0.095)	0.888 (0.041)
	U-MtE	0.742 (0.003)	0.638 (0.007)	0.203 (0.001)	0.085 (0.007)	0.425 (0.088)	0.854 (0.050)
	balanced	0.716 (0.007)	0.584 (0.012)	0.219 (0.004)	0.110 (0.007)	0.714 (0.052)	0.599 (0.061)
	U-MtE _{bal}	0.731 (0.004)	0.630 (0.007)	0.207 (0.001)	0.084 (0.006)	0.399 (0.069)	0.866 (0.034)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.586 (0.003)	0.239 (0.006)	0.279 (0.011)	0.842 (0.038)	0.519 (0.075)
	mask	0.769 (0.004)	0.571 (0.009)	0.286 (0.007)	0.332 (0.009)	0.860 (0.013)	0.439 (0.023)
	U-MtE	0.773 (0.004)	0.574 (0.008)	0.280 (0.009)	0.328 (0.012)	0.858 (0.039)	0.448 (0.089)
	balanced	0.781 (0.005)	0.585 (0.009)	0.228 (0.006)	0.257 (0.011)	0.811 (0.046)	0.554 (0.087)
	U-MtE _{bal}	0.774 (0.004)	0.568 (0.008)	0.279 (0.011)	0.325 (0.016)	0.860 (0.027)	0.447 (0.051)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.172 (0.007)	0.161 (0.019)	0.608 (0.049)	0.786 (0.039)
	mask	0.790 (0.004)	0.512 (0.009)	0.146 (0.003)	0.115 (0.009)	0.593 (0.052)	0.824 (0.037)
	U-MtE	0.783 (0.004)	0.500 (0.008)	0.150 (0.003)	0.126 (0.009)	0.566 (0.073)	0.831 (0.049)
	balanced	0.766 (0.008)	0.462 (0.014)	0.178 (0.008)	0.170 (0.020)	0.626 (0.047)	0.755 (0.046)
	U-MtE _{bal}	0.774 (0.006)	0.482 (0.009)	0.154 (0.004)	0.131 (0.010)	0.594 (0.061)	0.798 (0.051)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.750 (0.005)	0.353 (0.005)	0.162 (0.006)	0.234 (0.011)	0.535 (0.092)	0.813 (0.068)
	mask	0.789 (0.002)	0.404 (0.006)	0.143 (0.005)	0.217 (0.011)	0.559 (0.112)	0.842 (0.063)
	U-MtE	0.799 (0.008)	0.406 (0.009)	0.145 (0.007)	0.226 (0.016)	0.573 (0.067)	0.846 (0.041)
	balanced	0.747 (0.017)	0.346 (0.019)	0.156 (0.014)	0.228 (0.015)	0.539 (0.078)	0.803 (0.076)
	U-MtE _{bal}	0.802 (0.005)	0.401 (0.009)	0.139 (0.006)	0.215 (0.021)	0.606 (0.054)	0.825 (0.037)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.859 (0.032)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.871 (0.042)	0.897 (0.043)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.758 (0.087)	0.838 (0.059)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.044 (0.012)	0.866 (0.048)	0.897 (0.052)

natural clinical

Table 19: Synthetic tube, NIH ChestX-ray14 pneumothorax, RAD-DINO @518

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.915	0.913	0.911	0.772	0.907	0.614			
ROI masking	ROI mask (train+test)	0.910	0.910	0.908	0.843	0.905	0.725	-0.002 [-0.007, +0.002]	+0.070 [+0.062, +0.078]	+0.110 [+0.101, +0.120]
Inpainting (oracle)	artifact pixels (train+test)	0.915	0.915	0.916	0.913	0.916	0.914	+0.003 [+0.002, +0.003]	+0.140 [+0.127, +0.153]	+0.299 [+0.280, +0.321]
Group-balanced	image-level A (train)	0.914	0.913	0.913	0.920	0.912	0.924	+0.000 [-0.003, +0.003]	+0.147 [+0.134, +0.161]	+0.309 [+0.289, +0.330]
DFR	image-level A (val)	0.900	0.898	0.899	0.889	0.897	0.894	-0.014 [-0.019, -0.010]	+0.116 [+0.105, +0.128]	+0.280 [+0.256, +0.306]
LEACE (paired, oracle renders)	paired renders (train)	0.916	0.915	0.914	0.912	0.908	0.901	+0.003 [+0.002, +0.003]	+0.139 [+0.127, +0.153]	+0.287 [+0.266, +0.309]
Insert-then-erase (artifact templates)	artifact templates	0.911	0.910	0.908	0.907	0.906	0.905	-0.002 [-0.006, +0.001]	+0.135 [+0.122, +0.149]	+0.291 [+0.271, +0.312]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.907	0.907	0.907	0.907	0.907	0.907	-0.006 [-0.010, -0.001]	+0.134 [+0.120, +0.150]	+0.292 [+0.270, +0.316]
Insert-then-erase rank-1 (ablation)	artifact templates	0.915	0.915	0.914	0.910	0.910	0.867	+0.002 [+0.001, +0.003]	+0.138 [+0.125, +0.151]	+0.253 [+0.235, +0.271]
Insertion augmentation	artifact templates	0.915	0.915	0.913	0.856	0.911	0.726	+0.002 [+0.001, +0.003]	+0.084 [+0.074, +0.093]	+0.112 [+0.098, +0.125]

synthetic cxr raddino518

Table 20: Location interaction $[\Delta \text{ vs ERM}]_{0\%} - [\Delta \text{ vs ERM}]_{100\%}$ (DINOv2 @518)

Method	Interaction (0% – 100%)
ROI masking	+0.313 [+0.272, +0.353]
Inpainting (oracle)	-0.103 [-0.118, -0.088]
Group-balanced	-0.143 [-0.163, -0.127]
GroupDRO	-0.116 [-0.130, -0.103]
DFR	-0.155 [-0.171, -0.140]
LEACE (paired, oracle renders)	-0.127 [-0.150, -0.104]
Repair consistency	-0.000 [-0.019, +0.021]
Repair consistency, $\lambda=0$	-0.011 [-0.029, +0.009]

synthetic interaction

Table 21: Synthetic ruler, ISIC 2018, DermLIP/PanDerm @224

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.932	0.697	0.930	0.624	0.932	0.574			
ROI masking	ROI mask (train+test)	0.858	0.858	0.818	0.342	0.823	0.311	+0.161 [+0.121, +0.202]	-0.282 [-0.343, -0.227]	-0.263 [-0.328, -0.195]
Group-balanced	image-level A (train)	0.948	0.920	0.947	0.915	0.948	0.914	+0.223 [+0.193, +0.252]	+0.291 [+0.243, +0.342]	+0.339 [+0.294, +0.391]
DFR	image-level A (val)	0.946	0.934	0.944	0.922	0.943	0.923	+0.237 [+0.200, +0.278]	+0.298 [+0.247, +0.362]	+0.348 [+0.310, +0.386]
LEACE (paired, oracle renders)	paired renders (train)	0.951	0.946	0.950	0.942	0.954	0.938	+0.249 [+0.214, +0.288]	+0.318 [+0.262, +0.378]	+0.364 [+0.318, +0.415]
Insert-then-erase (artifact templates)	artifact templates	0.915	0.879	0.917	0.847	0.921	0.826	+0.183 [+0.147, +0.220]	+0.223 [+0.171, +0.282]	+0.251 [+0.205, +0.300]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.918	0.911	0.918	0.905	0.922	0.900	+0.214 [+0.181, +0.248]	+0.281 [+0.233, +0.332]	+0.325 [+0.269, +0.384]

synthetic isic dermlip224

Table 22: Synthetic ruler, ISIC 2018, DINOv2 @224

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.794	0.438	0.797	0.410	0.800	0.391			
ROI masking	ROI mask (train+test)	0.798	0.798	0.771	0.329	0.762	0.295	+0.361 [+0.308, +0.413]	-0.081 [-0.119, -0.043]	-0.095 [-0.131, -0.059]
Group-balanced	image-level A (train)	0.803	0.732	0.805	0.709	0.808	0.708	+0.294 [+0.264, +0.328]	+0.299 [+0.271, +0.327]	+0.318 [+0.288, +0.347]
DFR	image-level A (val)	0.737	0.729	0.741	0.734	0.740	0.726	+0.291 [+0.229, +0.357]	+0.324 [+0.258, +0.396]	+0.335 [+0.269, +0.405]
LEACE (paired, oracle renders)	paired renders (train)	0.830	0.830	0.829	0.824	0.826	0.816	+0.392 [+0.351, +0.434]	+0.415 [+0.373, +0.456]	+0.426 [+0.383, +0.469]
Insert-then-erase (artifact templates)	artifact templates	0.813	0.795	0.810	0.785	0.805	0.769	+0.357 [+0.315, +0.400]	+0.375 [+0.334, +0.418]	+0.378 [+0.336, +0.419]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.811	0.811	0.809	0.808	0.807	0.805	+0.373 [+0.337, +0.410]	+0.398 [+0.360, +0.437]	+0.414 [+0.376, +0.452]

synthetic isic dino224

Table 23: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test Δ vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.476	0.825	0.386	+0.184 [+0.146, +0.224]	-0.097 [-0.138, -0.056]	-0.129 [-0.170, -0.087]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.759	+0.141 [+0.120, +0.162]	+0.211 [+0.187, +0.234]	+0.243 [+0.218, +0.269]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.792	0.806	0.796	+0.137 [+0.109, +0.168]	+0.219 [+0.188, +0.252]	+0.281 [+0.248, +0.315]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.091, +0.160]	+0.221 [+0.184, +0.261]	+0.279 [+0.240, +0.320]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.832	0.810	0.810	0.809	0.809	+0.167 [+0.139, +0.197]	+0.237 [+0.212, +0.263]	+0.294 [+0.265, +0.325]
Repair consistency	artifact pixels (train)	0.828	0.707	0.822	0.616	0.814	0.558	+0.042 [+0.015, +0.069]	+0.043 [+0.014, +0.072]	+0.042 [+0.012, +0.073]
Insert-then-erase (artifact templates)	artifact templates	0.848	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.152, +0.206]	+0.260 [+0.229, +0.291]	+0.303 [+0.271, +0.336]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.941	0.942	0.943	0.940	0.944	0.942	+0.177 [+0.151, +0.205]	+0.267 [+0.238, +0.298]	+0.327 [+0.293, +0.362]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.730	+0.134 [+0.115, +0.154]	+0.197 [+0.174, +0.220]	+0.215 [+0.190, +0.240]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.654	0.810	0.603	+0.061 [+0.049, +0.074]	+0.081 [+0.068, +0.094]	+0.088 [+0.077, +0.099]

synthetic isic dino518

Table 24: ISIC 2019 real hair, DINOv2 @518: Trap A (in-lesion) and Trap B (extra-lesional)

Method	Needs	A clean	A corr	A rev	B clean	B corr	B rev	A Δrev	A Δclean	B Δrev	B Δclean
ERM	—	0.753	0.925	0.389	0.760	0.925	0.359				
ROI masking	ROI mask (train+test)	0.755	0.925	0.401	0.746	0.846	0.537	+0.012 [-0.042, +0.062]	+0.002	+0.178 [+0.142, +0.214]	-0.014
Inpainting (oracle)	artifact pixels (train+test)	0.753	0.896	0.559	0.750	0.882	0.478	+0.171 [+0.132, +0.210]	-0.000	+0.119 [+0.094, +0.141]	-0.010
Group-balanced	image-level A (train)	0.771	0.850	0.723	0.768	0.781	0.734	+0.335 [+0.269, +0.400]	+0.018	+0.375 [+0.332, +0.423]	+0.008
DFR	image-level A (val)	0.777	0.782	0.794	0.782	0.719	0.815	+0.405 [+0.322, +0.484]	+0.024	+0.456 [+0.396, +0.513]	+0.022
LEACE (paired, removal)	artifact pixels (train)	0.754	0.918	0.504	0.751	0.907	0.453	+0.116 [+0.079, +0.156]	+0.001	+0.094 [+0.074, +0.113]	-0.009
LEACE (unpaired labels)	image-level A (train)	0.761	0.639	0.869	0.754	0.615	0.833	+0.480 [+0.414, +0.534]	+0.007	+0.474 [+0.430, +0.516]	-0.006
Insert-then-erase (artifact templates)	artifact templates	0.745	0.926	0.456	0.751	0.919	0.406	+0.067 [+0.035, +0.100]	-0.008	+0.047 [+0.027, +0.068]	-0.009
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.755	0.855	0.703	0.788	0.832	0.706	+0.314 [+0.257, +0.366]	+0.002	+0.347 [+0.308, +0.391]	+0.028
Insert-then-erase rank-1 (ablation)	artifact templates	0.753	0.926	0.389	0.757	0.925	0.358	+0.001 [-0.002, +0.003]	-0.000	-0.001 [-0.005, +0.002]	-0.003
Insertion augmentation	artifact templates	0.754	0.926	0.415	0.760	0.924	0.373	+0.026 [+0.016, +0.037]	+0.001	+0.014 [+0.005, +0.022]	+0.000
Prevalence calibration	image-level A (train+test)	0.753	0.468	0.900	0.760	0.488	0.887	+0.511 [+0.456, +0.572]	+0.000	+0.528 [+0.495, +0.560]	+0.000

traps isic2019 dino518

Table 25: ISIC 2019 real hair, DINOv2 @518: Trap A (in-lesion) and Trap B (extra-lesional) — location crossover $[\Delta \text{ vs ERM}]_{\text{TrapB}} - [\Delta \text{ vs ERM}]_{\text{TrapA}}$

Method	Crossover (B – A)
Prevalence calibration	+0.017 [-0.046, +0.072]
ROI masking	+0.166 [+0.116, +0.215]
Inpainting (oracle)	-0.052 [-0.096, -0.016]
Group-balanced	+0.040 [-0.024, +0.109]
DFR	+0.051 [-0.008, +0.115]
LEACE (paired, removal)	-0.022 [-0.069, +0.013]
LEACE (unpaired labels)	-0.007 [-0.053, +0.042]
Insert-then-erase (artifact templates)	-0.020 [-0.051, +0.011]
Insert-then-erase + balanced (artifact templates)	+0.033 [-0.028, +0.090]
Insert-then-erase rank-1 (ablation)	-0.002 [-0.006, +0.003]
Insertion augmentation	-0.012 [-0.029, +0.004]

traps isic2019 dino518 crossover

B Result ledgers

C Pilot replication: all 88 numbers

Source: results/SUMMARY.md

Rule: MATCH = same sign, same CI-excludes-0 verdict, $|\Delta| \leq 0.02$; SIGN+CI = sign and CI verdict match, $|\Delta| > 0.02$; MISMATCH otherwise.

Verdict counts: {'MATCH': 73, 'SIGN+CI': 12, 'MISMATCH': 3}

E	quantity	expected	obtained	verdict
E1	mask – ERM rev @0%	+0.184 [+0.146, +0.224]	+0.184 [+0.146, +0.224]	MATCH
E1	mask – ERM rev @25%	-0.058 [-0.100, -0.018]	-0.059 [-0.100, -0.019]	MATCH
E1	mask – ERM rev @50%	-0.097 [-0.138, -0.056]	-0.097 [-0.138, -0.056]	MATCH
E1	mask – ERM rev @75%	-0.111 [-0.153, -0.068]	-0.112 [-0.155, -0.068]	MATCH
E1	mask – ERM rev @100%	-0.128 [-0.170, -0.086]	-0.129 [-0.170, -0.087]	MATCH
E1	interaction 0% – 100% (mask)	+0.312 [+0.272, +0.354]	+0.313 [+0.272, +0.353]	MATCH
E2	dino224 mask – ERM @0%	+0.364	+0.361 [+0.308, +0.413]	MATCH
E2	dino224 mask – ERM @25%	-0.040	-0.050 [-0.090, -0.008]	MATCH
E2	dino224 mask – ERM @50%	-0.076	-0.081 [-0.119, -0.043]	MATCH
E2	dino224 mask – ERM @75%	-0.102	-0.106 [-0.145, -0.069]	MATCH
E2	dino224 mask – ERM @100%	-0.092	-0.095 [-0.131, -0.059]	MATCH
E2	dermlip224 mask – ERM @0%	+0.160	+0.161 [+0.121, +0.202]	MATCH
E2	dermlip224 mask – ERM @25%	-0.291	-0.289 [-0.343, -0.239]	MATCH
E2	dermlip224 mask – ERM @50%	-0.300	-0.282 [-0.343, -0.227]	MATCH
E2	dermlip224 mask – ERM @75%	-0.313	-0.309 [-0.356, -0.261]	MATCH
E2	dermlip224 mask – ERM @100%	-0.268	-0.263 [-0.328, -0.195]	MATCH
E3	variable ruler mask – ERM @0%	+0.147 [+0.109, +0.185]	+0.147 [+0.110, +0.185]	MATCH
E3	variable ruler mask – ERM @50%	-0.094 [-0.140, -0.048]	-0.094 [-0.141, -0.047]	MATCH
E3	variable ruler mask – ERM @100%	-0.155 [-0.199, -0.112]	-0.155 [-0.199, -0.111]	MATCH
E3	variable ruler balanced – ERM @0%	+0.103	+0.103 [+0.077, +0.130]	MATCH
E3	variable ruler balanced – ERM @50%	+0.171	+0.171 [+0.143, +0.201]	MATCH
E3	variable ruler balanced – ERM @100%	+0.222	+0.222 [+0.193, +0.252]	MATCH
E4	occlusion (50/50) mask – ERM @0%	+0.035 [+0.006, +0.065]	+0.036 [+0.007, +0.065]	MATCH
E4	occlusion (50/50) mask – ERM @50%	+0.033 [-0.000, +0.066]	+0.040 [+0.011, +0.070]	MISMATCH
E4	occlusion (50/50) mask – ERM @100%	+0.039 [+0.010, +0.067]	+0.039 [+0.011, +0.068]	MATCH
E5	dilate0 – ERM @50%	-0.097	-0.097 [-0.138, -0.057]	MATCH
E5	dilate10 – ERM @50%	-0.138	-0.138 [-0.178, -0.099]	MATCH
E5	dilate25 – ERM @50%	-0.153	-0.154 [-0.194, -0.115]	MATCH
E5	dilate50 – ERM @50%	-0.142	-0.142 [-0.182, -0.105]	MATCH
E5	dilate0 – ERM @100%	-0.128	-0.129 [-0.170, -0.086]	MATCH
E5	dilate10 – ERM @100%	-0.143	-0.144 [-0.184, -0.103]	MATCH
E5	dilate25 – ERM @100%	-0.144	-0.144 [-0.184, -0.106]	MATCH
E5	dilate50 – ERM @100%	-0.138	-0.139 [-0.177, -0.099]	MATCH
E6	Δp		erm @100%	+0.235
E6	Δp		mask @100%	+0.490
E6	Δp		mask @0%	+0.000
E6	Δp		erm @0%	+0.128
E6	Δp		erm @50%	+0.195
E7	mask – ERM, large lesions @0%	+0.249	+0.248	MATCH
E7	mask – ERM, medium lesions @0%	+0.214	+0.215	MATCH
E7	mask – ERM, small lesions @0%	+0.150	+0.150	MATCH
E7	mask – ERM, large lesions @50%	+0.087	+0.087	MATCH
E7	mask – ERM, medium lesions @50%	-0.129	-0.129	MATCH
E7	mask – ERM, small lesions @50%	-0.325	-0.326	MATCH
E7	mask – ERM, large lesions @100%	+0.064	+0.064	MATCH
E7	mask – ERM, medium lesions @100%	-0.158	-0.158	MATCH
E7	mask – ERM, small lesions @100%	-0.364	-0.364	MATCH
E14	in-lesion hair median area ratio	+0.033	+0.033	MATCH
E14	Trap A mask – ERM, area-ratio tertile T1	-0.064 [-1.000, -0.001]	-0.043 [-0.066, -0.021]	SIGN+CI
E14	Trap A mask – ERM, area-ratio tertile T2	-0.057 [-1.000, -0.001]	-0.033 [-0.055, -0.012]	SIGN+CI
E14	Trap A mask – ERM, area-ratio tertile T3	-0.059 [-1.000, -0.001]	-0.026 [-0.044, -0.007]	SIGN+CI
E10	hair decodability test AUROC	+0.943 [+0.919, +0.964]	+0.943 [+0.919, +0.964]	MATCH
E10	detector mean IoU	+0.220	+0.222	MATCH

E	quantity	expected	obtained	verdict
E10	detector median IoU	+0.180	+0.184	MATCH
E10	LEACE A-decoder after erasure	+0.500	+0.500	MATCH
E10	clean melanoma AUROC after erasure	+0.814	+0.814	MATCH
E10	melanoma-head	Δp	after erasure	+0.058
E10	LEACE projection rank	+1.000	+1.000	MATCH
E12	overlap-contrast trap mask – ERM (spec: null)	+0.047 [-0.050, +0.150]	-0.083 [-0.097, -0.072]	MISMATCH
§3.1	shortcut-gap inflation by leakage @0% (ratio – 1)	+0.270	+0.179	SIGN+C
§3.1	shortcut-gap inflation by leakage @100% (ratio – 1)	+0.270	+0.103	SIGN+C
E8/E9	inpaint – ERM @50%	+0.211 [+0.187, +0.234]	+0.211 [+0.187, +0.234]	MATCH
E8/E9	inpaint – ERM @100%	+0.244 [+0.218, +0.269]	+0.243 [+0.218, +0.269]	MATCH
E8/E9	inpaint_consistency – ERM @50%	+0.046	+0.043 [+0.014, +0.072]	MATCH
E8/E9	inpaint_consistency_lam0 – ERM @50%	+0.036	+0.035 [+0.005, +0.066]	MATCH
E8/E9	leace – ERM @100%	+0.294 [+0.265, +0.325]	+0.294 [+0.265, +0.325]	MATCH
E8/E9	balanced – ERM @100%	+0.281 [+0.249, +0.315]	+0.281 [+0.248, +0.315]	MATCH
E8/E9	dfr – ERM @100%	+0.279 [+0.240, +0.319]	+0.279 [+0.240, +0.320]	MATCH
E8/E9	groupdro – ERM @100%	+0.197 [+0.168, +0.226]	+0.197 [+0.168, +0.226]	MATCH
E13	trapA mask – ERM (rev)	-0.064 [-0.079, -0.049]	-0.037 [-0.057, -0.019]	SIGN+C
E13	trapB mask – ERM (rev)	+0.124 [+0.105, +0.143]	+0.107 [+0.078, +0.135]	MATCH
E13	trapA inpaint – ERM (rev)	+0.101 [+0.094, +0.107]	+0.103 [+0.093, +0.114]	MATCH
E13	trapB inpaint – ERM (rev)	+0.108 [+0.098, +0.117]	+0.096 [+0.084, +0.107]	MATCH
E13	trapA balanced – ERM (rev)	+0.213 [+0.190, +0.235]	+0.225 [+0.199, +0.248]	MATCH
E13	trapB balanced – ERM (rev)	+0.234 [+0.217, +0.252]	+0.209 [+0.192, +0.226]	SIGN+C
E13	trapA dfr – ERM (rev)	+0.202 [+0.176, +0.228]	+0.251 [+0.223, +0.277]	SIGN+C
E13	trapB dfr – ERM (rev)	+0.192 [+0.168, +0.216]	+0.215 [+0.178, +0.252]	SIGN+C
E13	trapA leace_paired – ERM (rev)	+0.058 [+0.053, +0.063]	+0.086 [+0.078, +0.095]	SIGN+C
E13	trapB leace_paired – ERM (rev)	+0.078 [+0.070, +0.085]	+0.081 [+0.069, +0.094]	MATCH
E13	trapA leace_unpaired – ERM (rev)	+0.316	+0.326 [+0.301, +0.350]	MATCH
E13	C3 crossover B – A (mask)	+0.188 [+0.179, +0.194]	+0.145 [+0.105, +0.181]	SIGN+C
E13	trapA mask – ERM, HAM only	-0.030 [-0.059, -0.003]	-0.009 [-0.036, +0.017]	MISMATCH
E13	trapA mask – ERM, BCN only	-0.078 [-0.099, -0.058]	-0.051 [-0.075, -0.027]	SIGN+C
E15	trapA mask – ERM (rev)	-0.069 [-0.088, -0.051]	-0.085 [-0.107, -0.063]	MATCH
E15	trapA balanced – ERM (rev)	+0.126 [+0.115, +0.138]	+0.143 [+0.130, +0.157]	MATCH
E15	trapA dfr – ERM (rev)	+0.114 [+0.091, +0.138]	+0.110 [+0.078, +0.138]	MATCH
E15	trapA leace_paired – ERM (rev)	+0.031 [+0.028, +0.034]	+0.041 [+0.034, +0.048]	MATCH
E15	trapA leace_unpaired – ERM (rev)	+0.192	+0.198 [+0.178, +0.217]	MATCH

D Holm-corrected primary claims

Source: results/PRIMARY_CLAIMS.md

Supported after Holm (correct direction): 12/16

family	cohort	estimate [95 % CI]	p	p (Holm)	p
P1 location crossover	ISIC hair (DINOv2)	+0.145 [+0.105, +0.181]	0.0001	0.0016 ☐	0.0001
P1 location crossover	Thyroid (DINOv2)	+0.230 [+0.204, +0.255]	0.0001	0.0016 ☐	0.0001
P1 location crossover	Capsule (DINOv2)	+0.368 [+0.340, +0.397]	0.0001	0.0016 ☐	0.0001
P1 location crossover	Ovary (DINOv2)	+0.170 [+0.116, +0.224]	0.0001	0.0016 ☐	0.0001
P2 U-MtE (protected) > mask	ISIC hair	+0.041 [+0.030, +0.054]	0.0001	0.0016 ☐	0.0001
P2 U-MtE (protected) > mask	Thyroid	+0.217 [+0.174, +0.257]	0.0001	0.0016 ☐	0.0001
P2 U-MtE (protected) > mask	Capsule	-0.010 [-0.021, -0.001]	0.0300	0.0900	0.0300
P2 U-MtE (protected) > mask	Ovary	+0.043 [+0.026, +0.061]	0.0001	0.0016 ☐	0.0001
P3 erase > augment	ISIC hair	+0.045 [+0.037, +0.054]	0.0001	0.0016 ☐	0.0001
P3 erase > augment	Thyroid	+0.207 [+0.182, +0.235]	0.0001	0.0016 ☐	0.0001
P3 erase > augment	Capsule	-0.130 [-0.147, -0.110]	0.0001	0.0016 ☐ (opposite direction)	0.0001
P3 erase > augment	Ovary	+0.035 [+0.010, +0.059]	0.0056	0.0224 ☐	0.0056
P4 U-MtE+balanced > balanced	ISIC hair	-0.013 [-0.041, +0.018]	0.4230	0.4230	0.4230
P4 U-MtE+balanced > balanced	Thyroid	+0.112 [+0.090, +0.134]	0.0001	0.0016 ☐	0.0001
P4 U-MtE+balanced > balanced	Capsule	+0.082 [+0.064, +0.100]	0.0001	0.0016 ☐	0.0001
P4 U-MtE+balanced > balanced	Ovary	+0.028 [+0.000, +0.057]	0.0494	0.0988	0.0494

E Every arm $\times$ cohort $\times$ backbone

Source: *results/CROSS_COHORT.md*

Arms prefixed U- use the generic, artifact-agnostic overlay library. rev (clean). † = gaming (correlated-test AUROC < reversed – 0.02: the shortcut is flipped, not removed; excluded by pre-registration).

min(rev, corr) is a post-hoc secondary summary that penalises both shortcut use and shortcut flipping.

Label needs: erm/mask/U-mte/U-mte_protect: none; balanced/dfr/leace/_balanced/_dfr: image-level artifact labels.

E.0.1 ISIC 2019 hair — DINOv2

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.491 / 0.908 (0.723)	0.491	0.519 / 0.898 (0.722)	0.519
mask	0.454 / 0.916 (0.712)	0.454	0.626 / 0.818 (0.722)	0.626
inpaint	0.594 / 0.899 (0.762)	0.594	0.614 / 0.864 (0.743)	0.614
balanced	0.716 / 0.836 (0.775)	0.716	0.728 / 0.802 (0.763)	0.728
dfr	0.743 / 0.741 (0.740)	0.741	0.734 / 0.686 (0.711) †	0.686
leace_paired	0.577 / 0.905 (0.758)	0.577	0.600 / 0.874 (0.740)	0.600
leace_unpaired	0.817 / 0.618 (0.717) †	0.618	0.802 / 0.619 (0.713) †	0.619
U-mte	0.506 / 0.913 (0.735)	0.506	0.651 / 0.805 (0.727)	0.651
U-mte_balanced	0.704 / 0.852 (0.779)	0.704	0.773 / 0.751 (0.759) †	0.751
U-mte_aug	0.461 / 0.919 (0.718)	0.461	0.636 / 0.812 (0.722)	0.636
U-mte_protect	0.495 / 0.895 (0.715)	0.495	0.650 / 0.775 (0.706)	0.650
U-mte_protect_balanced	0.684 / 0.845 (0.768)	0.684	0.735 / 0.745 (0.735)	0.735
U-mte_dfr	0.704 / 0.787 (0.748)	0.704	0.760 / 0.655 (0.707) †	0.655
U-mask_dfr	0.690 / 0.754 (0.722)	0.690	0.725 / 0.655 (0.689) †	0.655
U-umte_pbal	0.510 / 0.913 (0.736)	0.510	0.657 / 0.797 (0.726)	0.657

E.0.2 ISIC 2019 hair — DermLIP

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.670 / 0.921 (0.812)	0.670	0.734 / 0.828 (0.780)	0.734
mask	0.585 / 0.893 (0.752)	0.585	0.646 / 0.762 (0.699)	0.646
inpaint	0.714 / 0.921 (0.829)	0.714	0.778 / 0.826 (0.804)	0.778
balanced	0.813 / 0.879 (0.847)	0.813	0.853 / 0.792 (0.820) †	0.792
dfr	0.780 / 0.840 (0.814)	0.780	0.812 / 0.712 (0.766) †	0.712
leace_paired	0.711 / 0.920 (0.827)	0.711	0.783 / 0.821 (0.802)	0.783
leace_unpaired	0.867 / 0.650 (0.765) †	0.650	0.878 / 0.611 (0.750) †	0.611
mte	0.524 / 0.868 (0.711)	0.524	0.613 / 0.730 (0.673)	0.613
mte_balanced	0.631 / 0.811 (0.721)	0.631	0.743 / 0.631 (0.687) †	0.631
U-mte	0.509 / 0.868 (0.704)	0.509	0.604 / 0.740 (0.668)	0.604
U-mte_balanced	0.648 / 0.811 (0.733)	0.648	0.756 / 0.655 (0.704) †	0.655
U-mte_protect	0.589 / 0.872 (0.741)	0.589	0.677 / 0.694 (0.678)	0.677
U-mte_protect_balanced	0.693 / 0.816 (0.753)	0.693	0.765 / 0.652 (0.707) †	0.652

E.0.3 Thyroid US markers — DINOv2

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.289 / 0.879 (0.610)	0.289	0.406 / 0.796 (0.612)	0.406
mask	0.400 / 0.902 (0.687)	0.400	0.747 / 0.784 (0.763)	0.747
inpaint	0.390 / 0.849 (0.636)	0.390	0.442 / 0.765 (0.611)	0.442
balanced	0.675 / 0.760 (0.721)	0.675	0.675 / 0.661 (0.673)	0.661
dfr	0.738 / 0.659 (0.702) †	0.659	0.723 / 0.556 (0.639) †	0.556
leace_paired	0.339 / 0.873 (0.629)	0.339	0.434 / 0.789 (0.621)	0.434
leace_unpaired	0.793 / 0.616 (0.713) †	0.616	0.757 / 0.604 (0.687) †	0.604
mte	0.703 / 0.841 (0.782)	0.703	0.756 / 0.782 (0.774)	0.756
mte_balanced	0.818 / 0.795 (0.812) †	0.795	0.777 / 0.766 (0.774)	0.766
U-mte	0.622 / 0.822 (0.728)	0.622	0.740 / 0.782 (0.761)	0.740

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
U-mte_balanced	0.788 / 0.818 (0.807)	0.788	0.771 / 0.765 (0.767)	0.765
U-mte_aug	0.415 / 0.898 (0.690)	0.415	0.743 / 0.769 (0.756)	0.743
U-mte_protect	0.617 / 0.780 (0.703)	0.617	0.742 / 0.755 (0.750)	0.742
U-mte_protect_balanced	0.804 / 0.760 (0.787) †	0.760	0.766 / 0.735 (0.753) †	0.735
U-mte_dfr	0.839 / 0.743 (0.793) †	0.743	0.761 / 0.742 (0.747)	0.742
U-mask_dfr	0.832 / 0.736 (0.785) †	0.736	0.764 / 0.726 (0.742) †	0.726
U-umte_pbal	0.663 / 0.840 (0.759)	0.663	0.743 / 0.786 (0.764)	0.743

E.0.4 Thyroid US markers — MedSigLIP

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.284 / 0.905 (0.628)	0.284	0.500 / 0.773 (0.638)	0.500
mask	0.348 / 0.914 (0.677)	0.348	0.727 / 0.791 (0.758)	0.727
inpaint	0.379 / 0.868 (0.640)	0.379	0.564 / 0.748 (0.662)	0.564
balanced	0.695 / 0.781 (0.744)	0.695	0.691 / 0.654 (0.673) †	0.654
dfr	0.757 / 0.719 (0.742) †	0.719	0.727 / 0.620 (0.677) †	0.620
leace_paired	0.352 / 0.868 (0.623)	0.352	0.534 / 0.760 (0.652)	0.534
leace_unpaired	0.825 / 0.639 (0.739) †	0.639	0.768 / 0.602 (0.690) †	0.602
mte	0.658 / 0.842 (0.758)	0.658	0.763 / 0.797 (0.783)	0.763
mte_balanced	0.814 / 0.794 (0.805) †	0.794	0.782 / 0.774 (0.779)	0.774
U-mte	0.553 / 0.811 (0.687)	0.553	0.753 / 0.803 (0.781)	0.753
U-mte_balanced	0.792 / 0.800 (0.798)	0.792	0.785 / 0.780 (0.781)	0.780
U-mte_aug	0.375 / 0.904 (0.678)	0.375	0.725 / 0.784 (0.756)	0.725
U-mte_protect	0.611 / 0.791 (0.705)	0.611	0.745 / 0.783 (0.767)	0.745
U-mte_protect_balanced	0.780 / 0.763 (0.777)	0.763	0.758 / 0.733 (0.747) †	0.733
U-mte_dfr	0.816 / 0.735 (0.777) †	0.735	0.752 / 0.731 (0.738) †	0.731
U-mask_dfr	0.806 / 0.757 (0.781) †	0.757	0.757 / 0.717 (0.735) †	0.717

E.0.5 Thyroid US markers — ConvNeXt

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.354 / 0.867 (0.630)	0.354	0.428 / 0.740 (0.595)	0.428
mask	0.357 / 0.882 (0.649)	0.357	0.716 / 0.742 (0.726)	0.716
balanced	0.661 / 0.745 (0.705)	0.661	0.668 / 0.643 (0.657) †	0.643
dfr	0.710 / 0.669 (0.694) †	0.669	0.710 / 0.562 (0.633) †	0.562
U-mte	0.486 / 0.826 (0.667)	0.486	0.715 / 0.757 (0.732)	0.715
U-mte_balanced	0.745 / 0.764 (0.759)	0.745	0.763 / 0.751 (0.751)	0.751
U-mte_aug	0.365 / 0.880 (0.652)	0.365	0.717 / 0.741 (0.726)	0.717
U-mte_protect	0.552 / 0.790 (0.680)	0.552	0.717 / 0.740 (0.726)	0.717
U-mte_protect_balanced	0.733 / 0.732 (0.737)	0.732	0.745 / 0.715 (0.726) †	0.715
U-mte_dfr	0.783 / 0.714 (0.750) †	0.714	0.743 / 0.705 (0.713) †	0.705
U-mask_dfr	0.767 / 0.722 (0.743) †	0.722	0.730 / 0.676 (0.685) †	0.676

E.0.6 Capsule debris — ConvNeXt

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.545 / 0.947 (0.775)	0.545	0.445 / 0.908 (0.705)	0.445
mask	0.639 / 0.964 (0.830)	0.639	0.831 / 0.945 (0.895)	0.831
balanced	0.716 / 0.884 (0.801)	0.716	0.660 / 0.865 (0.769)	0.660
dfr	0.775 / 0.862 (0.814)	0.775	0.730 / 0.845 (0.788)	0.730
U-mte	0.559 / 0.963 (0.803)	0.559	0.809 / 0.939 (0.883)	0.809
U-mte_balanced	0.803 / 0.892 (0.846)	0.803	0.857 / 0.915 (0.891)	0.857
U-mte_aug	0.655 / 0.964 (0.835)	0.655	0.828 / 0.943 (0.892)	0.828
U-mte_protect	0.643 / 0.953 (0.816)	0.643	0.815 / 0.935 (0.881)	0.815
U-mte_protect_balanced	0.789 / 0.916 (0.851)	0.789	0.862 / 0.927 (0.896)	0.862
U-mte_dfr	0.869 / 0.848 (0.854) †	0.848	0.871 / 0.889 (0.881)	0.871

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
U-mask_dfr	0.864 / 0.878 (0.866)	0.864	0.884 / 0.913 (0.899)	0.884

E.0.7 NIH chest drains (in-ROI only) — RAD-DINO

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.191 / 0.948 (0.886)	0.191	–	–
mask	0.239 / 0.949 (0.870)	0.239	–	–
balanced	0.538 / 0.870 (0.853)	0.538	–	–
dfr	0.683 / 0.738 (0.800)	0.683	–	–
U-mte	0.268 / 0.940 (0.847)	0.268	–	–
U-mte_balanced	0.460 / 0.857 (0.826)	0.460	–	–
U-mte_aug	0.237 / 0.948 (0.866)	0.237	–	–
U-mte_protect	0.269 / 0.946 (0.867)	0.269	–	–
U-mte_protect_balanced	0.401 / 0.913 (0.851)	0.401	–	–
U-mask_dfr	0.646 / 0.770 (0.801)	0.646	–	–

E.0.8 NIH chest drains (in-ROI only) — DINOv2

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.280 / 0.922 (0.823)	0.280	–	–
mask	0.292 / 0.932 (0.834)	0.292	–	–
balanced	0.399 / 0.841 (0.781)	0.399	–	–
dfr	0.549 / 0.808 (0.770)	0.549	–	–
U-mte	0.285 / 0.931 (0.834)	0.285	–	–
U-mte_balanced	0.443 / 0.849 (0.804)	0.443	–	–
U-mte_aug	0.294 / 0.933 (0.835)	0.294	–	–
U-mte_protect	0.281 / 0.926 (0.818)	0.281	–	–
U-mte_protect_balanced	0.377 / 0.856 (0.789)	0.377	–	–
U-mask_dfr	0.566 / 0.827 (0.788)	0.566	–	–

E.0.9 Ovary US markers — DINOv2

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.414 / 0.823 (0.632)	0.414	0.423 / 0.815 (0.636)	0.423
mask	0.545 / 0.828 (0.690)	0.545	0.725 / 0.752 (0.732)	0.725
inpaint	0.570 / 0.785 (0.682)	0.570	0.469 / 0.808 (0.651)	0.469
balanced	0.707 / 0.732 (0.716)	0.707	0.679 / 0.710 (0.694)	0.679
dfr	0.723 / 0.689 (0.708) †	0.689	0.701 / 0.632 (0.671) †	0.632
leace_paired	0.555 / 0.801 (0.685)	0.555	0.454 / 0.806 (0.643)	0.454
leace_unpaired	0.782 / 0.611 (0.698) †	0.611	0.728 / 0.588 (0.657) †	0.588
mte	0.615 / 0.846 (0.739)	0.615	0.745 / 0.788 (0.766)	0.745
mte_balanced	0.729 / 0.790 (0.763)	0.729	0.770 / 0.760 (0.766)	0.760
U-mte	0.564 / 0.855 (0.719)	0.564	0.736 / 0.787 (0.762)	0.736
U-mte_balanced	0.736 / 0.811 (0.776)	0.736	0.765 / 0.754 (0.762)	0.754
U-mte_aug	0.529 / 0.830 (0.684)	0.529	0.728 / 0.758 (0.739)	0.728
U-mte_protect	0.588 / 0.832 (0.714)	0.588	0.745 / 0.757 (0.748)	0.745
U-mte_protect_balanced	0.677 / 0.820 (0.749)	0.677	0.775 / 0.707 (0.743) †	0.707
U-mask_dfr	0.752 / 0.757 (0.754)	0.752	0.770 / 0.673 (0.726) †	0.673

E.0.10 Ovary US markers — MedSigLIP

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.425 / 0.855 (0.656)	0.425	0.549 / 0.835 (0.698)	0.549
mask	0.428 / 0.871 (0.670)	0.428	0.732 / 0.836 (0.785)	0.732

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
balanced	0.729 / 0.793 (0.762)	0.729	0.703 / 0.763 (0.725)	0.703
U-mte	0.643 / 0.871 (0.767)	0.643	0.744 / 0.853 (0.800)	0.744
U-mte_balanced	0.787 / 0.833 (0.809)	0.787	0.771 / 0.811 (0.791)	0.771
U-mte_protect	0.573 / 0.837 (0.714)	0.573	0.713 / 0.811 (0.760)	0.713

E.0.11 Capsule debris — DINOv2

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.587 / 0.941 (0.784)	0.587	0.420 / 0.933 (0.723)	0.420
mask	0.684 / 0.975 (0.859)	0.684	0.885 / 0.964 (0.929)	0.885
inpaint	0.418 / 0.948 (0.726)	0.418	0.307 / 0.937 (0.675)	0.307
balanced	0.775 / 0.913 (0.844)	0.775	0.659 / 0.875 (0.777)	0.659
dfr	0.805 / 0.880 (0.839)	0.805	0.733 / 0.826 (0.779)	0.733
leace_paired	0.598 / 0.934 (0.780)	0.598	0.431 / 0.923 (0.712)	0.431
leace_unpaired	0.879 / 0.696 (0.791) †	0.696	0.851 / 0.720 (0.788) †	0.720
mte	0.289 / 0.944 (0.672)	0.289	0.762 / 0.894 (0.834)	0.762
mte_balanced	0.711 / 0.873 (0.791)	0.711	0.790 / 0.864 (0.829)	0.790
U-mte	0.569 / 0.975 (0.819)	0.569	0.873 / 0.959 (0.921)	0.873
U-mte_balanced	0.857 / 0.928 (0.893)	0.857	0.901 / 0.941 (0.923)	0.901
U-mte_aug	0.699 / 0.976 (0.865)	0.699	0.887 / 0.963 (0.929)	0.887
U-mte_protect	0.673 / 0.973 (0.851)	0.673	0.860 / 0.951 (0.910)	0.860
U-mte_protect_balanced	0.848 / 0.941 (0.893)	0.848	0.892 / 0.941 (0.918)	0.892
U-mte_dfr	0.885 / 0.862 (0.873) †	0.862	0.880 / 0.896 (0.889)	0.880
U-mask_dfr	0.885 / 0.895 (0.888)	0.885	0.880 / 0.914 (0.900)	0.880

E.0.12 Capsule debris — MedSigLIP

arm	Trap A rev / corr (clean)	Trap A min(rev,corr)	Trap B rev / corr (clean)	Trap B min(rev,corr)
erm	0.593 / 0.965 (0.811)	0.593	0.469 / 0.952 (0.763)	0.469
mask	0.702 / 0.979 (0.869)	0.702	0.910 / 0.966 (0.940)	0.910
inpaint	0.350 / 0.958 (0.718)	0.350	0.327 / 0.956 (0.715)	0.327
balanced	0.810 / 0.932 (0.872)	0.810	0.718 / 0.901 (0.820)	0.718
dfr	0.801 / 0.906 (0.853)	0.801	0.751 / 0.886 (0.820)	0.751
leace_paired	0.595 / 0.965 (0.811)	0.595	0.473 / 0.953 (0.765)	0.473
leace_unpaired	0.901 / 0.692 (0.804) †	0.692	0.871 / 0.751 (0.814) †	0.751
mte	0.222 / 0.910 (0.602)	0.222	0.697 / 0.844 (0.777)	0.697
mte_balanced	0.675 / 0.850 (0.766)	0.675	0.756 / 0.793 (0.774)	0.756
U-mte	0.465 / 0.970 (0.776)	0.465	0.860 / 0.956 (0.914)	0.860
U-mte_balanced	0.854 / 0.946 (0.896)	0.854	0.904 / 0.940 (0.924)	0.904
U-mte_aug	0.721 / 0.979 (0.876)	0.721	0.911 / 0.967 (0.941)	0.911
U-mte_protect	0.667 / 0.973 (0.852)	0.667	0.903 / 0.952 (0.931)	0.903
U-mte_protect_balanced	0.871 / 0.942 (0.904)	0.871	0.918 / 0.937 (0.929)	0.918
U-mte_dfr	0.871 / 0.867 (0.867)	0.867	0.877 / 0.889 (0.885)	0.877
U-mask_dfr	0.901 / 0.920 (0.908)	0.901	0.909 / 0.934 (0.922)	0.909

Part III

Experiment registry (every design, pre-registration, audit and result document, verbatim)

R1 Replication specification of the pilot (author protocol)

Source: docs/REPLICATION_SPEC.md

Verbatim copy of the author's README "Replicating the 'Where the Shortcut Sits' pilot". Deviations are recorded in CHANGES.md; match status in results/SUMMARY.md.

R1.0.1 0. What the pilot claims

1. **Location decides the method.** Masking an image to the lesion (the region of interest, ROI) removes a shortcut artifact when the artifact lies *outside* the lesion, and makes the model *worse than doing nothing* when it lies *inside*.
2. **Mechanism.** The harm is governed by how much of the artifact survives the mask (retention). It is not caused by occluding lesion tissue, and not by losing surrounding context.
3. **Supervision.** Methods that need only an *image-level* artifact label (balanced training, DFR) fix both regimes with nothing required at test time. Pixel-level artifact localisation fails in practice.
4. **Linear erasure is insufficient** for real artifacts under the sound (paired) protocol.
5. All of the above is shown on a synthetic artifact under full control, and **confirmed on real hair** with pre-registered criteria, on two backbones.

R1.0.2 1. Environment

- One GPU; original on a V100 32 GB; frozen backbones + linear heads.
- Python 3.10+, PyTorch ≥ 2.1 , DINOv2 hub loader, open_clip_torch (DermLIP), scikit-learn, opencv, concept-erasure $\geq 0.2.4$, imagehash, pandas, numpy.
- Performance pitfall: cache resized images/masks; use AMP.
- Never edit a running script.

R1.0.3 2. Data sources

Source	Content	Use
ISIC 2018 Task 1	2,594 images + manual lesion masks	syn
ISIC 2016 + 2017 ground truth	melanoma labels for Task 1/2 images	lab
ISIC 2019	25,331 images, MEL vs rest, sources HAM / BCN / MSK	rea
HAM10000 lesion segmentations (Tschandl)	10,015 manual masks	lesi
ISIC 2019 artifact mask archive (Missouri S&T)	hair+ruler (one channel), ink, vignetting; 255 = artifact	rea
Mendeley hair masks (DOI 10.17632/j5ywpd2p27.2, CC BY 4.0)	500 manual hair masks (478 join the Task 1 cohort)	rea
Bissoto et al. artifact annotations	image-level presence of 7 artifact types	hai

R1.0.4 3. Protocol rules

3.1 Splits — leakage-safe Group by lesion_id, merge groups with pHash Hamming ≤ 8 ; split by group; verify zero cross-split group edges. Near-duplicate leakage inflated the shortcut gap by 27 % (+0.0896 [+0.014, +0.166]) in an earlier round. Report threshold counts at 6 / 8 / 10.

3.2 Backbones and heads Frozen DINOv2 ViT-B/14 CLS @518 (primary) and @224; DermLIP/PanDerm (hf-hub:redlessone/encode_image, native preprocessing @224). L2-normalised features. Head: LogisticRegression(liblinear, class_weight="balanced" max_iter=3000), C $\in \{0.01, 0.1, 1, 10\}$ on clean validation only.

3.3 Environments train_corr 0.90/0.10; test_corr 0.90/0.10; test_rev 0.10/0.90 (primary); clean — synthetic: no artifact; **real: artifact uncorrelated with label**. Presence from a stable hash of (image_id, label, seed, env), independent of the overlap level.

3.4 Selection and thresholds All hyperparameters on clean validation only; thresholds = max balanced accuracy on clean validation, frozen.

3.5 Statistics — hierarchical paired bootstrap Resample seed clusters, then paired test images within cluster; paired AUROC delta within each sampled cluster; average per replicate; never pool heads. 2.5/97.5 percentiles; seed SD separately. **For k-fold designs, aggregate fold test predictions per seed before computing AUROC.**

3.6 Pre-registration Decision rules committed to git before each verdict-producing experiment; outcomes reported either way; amendments only if stricter.

R1.0.5 4. Known pitfalls

Pitfall	Symptom	Fix
Pooled bootstrap across heads	CIs subtly wrong	average seed-specific d
Artifact placement identical across seeds	seeds less independent	choose uniformly among
Placement cache key missing tolerance / n_candidates	stale placements	include them in the cach
Evaluating arms on different image sets	inpaint "+0.53" was really +0.003/−0.008	inner-join every paired
Near-constant probe for $ \Delta p $	$ \Delta p \approx 1e-4$	report probability SD; fl
Overlap-contrast trap	null everywhere	comparison group must
Unpaired LEACE	huge reversed gains, clean drops, corr < rev	paired orig↔inpaint era
Masks for placement $\neq$ masks for masking	arms see different stimuli	ground-truth lesion mas
Threshold tie-break on a shared cell	decided nothing	tie-break on cells the th
Source imbalance across cells	comparing sites	match cells to a commo

R1.0.6 5. Experiments and expected numbers

Tolerance: same versions ± 0.01 ; across versions every sign and CI-excludes-0 verdict matches, points ± 0.02 . Seeds: synthetic {42, 123, 456}; real-data traps {42, 123, 456, 789, 2026}.

- **E1** synthetic overlap sweep, DINOv2@518, 2,437 images (154 excluded: median lesion area 0.012 vs 0.152, melanoma 4.5 % vs 20.9 %). Mask − ERM: +0.184 [+0.146,+0.224], −0.058 [−0.100,−0.018], −0.097 [−0.138,−0.056], −0.111 [−0.153,−0.068], −0.128 [−0.170,−0.086]; interaction +0.312 [+0.272,+0.354]; mask corr=rev=0.849 at 0 %; masked corr 0.961 vs ERM 0.937 at 100 %.
- **E2** mask − ERM by backbone (0/25/50/75/100 %): DINOv2@518 +0.184/−0.058/−0.098/−0.111/−0.129; DINOv2@224 +0.364/−0.040/−0.076/−0.102/−0.092; DermLIP@224 +0.160/−0.291/−0.300/−0.313/−0.268. One of 39 comparisons changed CI status after the bootstrap fix (DINOv2@224, 25 %).
- **E3** appearance randomisation: ERM rev 0.702/0.625/0.576; mask − ERM +0.147/−0.094/−0.155; balanced +0.103/+0.171/+0.222; LEACE |corr − rev| 0.002/0.002/0.000; interaction +0.302 [+0.264,+0.340].
- **E4** occlusion (50/50): mask − ERM +0.035 [+0.006,+0.065] (0 %), +0.033 [−0.000,+0.066] (50 %), +0.039 [+0.010,+0.067] (100 %); occlusion cost vs clean $\approx$ 0.000−0.004.
- **E5** dilation: 50 %: margins 0/10/25/50 → retention 0.50/0.73/0.95/1.00, harm −0.097/−0.138/−0.153/−0.142; 100 %: −0.128/−0.143/−0.144/−0.138; pixel share 0.055 → 0.018.
- **E6** $|\Delta p|$ at 100 %: ERM 0.235, mask 0.490; at 0 %: mask 0.000, ERM 0.128; ERM 0.195 at 50 %.
- **E7** lesion-size tertiles (large/medium/small): 0 %: +0.249/+0.214/+0.150; 50 %: +0.087/−0.129/−0.325; 100 %: +0.064/−0.158/−0.364. Artifact-to-lesion ratio $\sim$ 0.012 (large) vs $\sim$ 0.106−0.123 (small).
- **E8** inpaint rev 0.806/0.783/0.779/0.759; Δ +0.211/+0.209/+0.244; consistency +0.038/+0.046/+0.031/+0.043, $|\Delta p|$ higher (+0.073/+0.086/+0.085), PARTIAL_SUPPORT; $\lambda=0$ +0.030/+0.036/+0.021/+0.040.
- **E9** bridge at 100 %: LEACE +0.294, balanced +0.281, DFR +0.279, inpaint +0.244, GroupDRO +0.197, ERM 0.515, mask −0.128.
- **E10** hair decodability (Bissoto) test AUROC 0.943 [0.919,0.964], n=356, base rate 0.567, P(hair|mel)=0.411 vs 0.608; paired LEACE on 478 Mendeley images (340 train pairs): rank 1/768, A-decoder 0.968→0.500, clean 0.812→0.814, $|\Delta p|$ 0.072→0.058; DullRazor detector IoU mean 0.22, median 0.18, P 0.32, R 0.46.

- **E11** ISIC 2019 $\cap$ HAM10000 atlas (10,015; 5,850 groups; split 6,529/1,310/2,176): hair+ruler n 9,760 r mean 0.280 median 0.219 share $\geq$ 0.5 0.199; ink 65, 0.045/0.000/0.031; vignetting 1,134, 0.003/0.000/0.002. Vignetting trap (5 seeds): ERM rev 0.305; mask +0.190 [+0.160,+0.219]; balanced +0.347; DFR +0.490; LEACE +0.004; inpaint -0.008 [-0.017,+0.001].
- **E12** overlap-contrast trap (A=1 r $\geq$ 0.5 vs A=0 r<0.1): C1 +0.019 [-0.085,+0.122], C2 +0.047 — null.
- **E13** full ISIC 2019; lesion masks HAM manual, BCN/MSK from a U-Net trained on ISIC 2018 Task 1 (Ronneberger, widths 64–1024; held-out Dice 0.876, median 0.915; vs HAM manual 0.895 / 0.947); manual $\leftrightarrow$ U-Net swap: r MAE 0.042, Pearson 0.929, 10.7 % change stratum. Two traps sharing one hair-free A=0 group; Trap A r $\geq$ 0.5, Trap B r < 0.1 (strict 0.6/0.05 sensitivity; chosen on min A=1 cell 1,131 vs 884). Source matching: subsample cells to the A=0 source mix within each label (benign HAM 30.2/BCN 63.3/MSK 6.5 %; melanoma 25.5/68.2/6.4 %). 5-fold group-safe CV, 90/10 $\rightarrow$ 10/90; gate $\geq$ 40 reversed melanomas, every matched cell $\geq$ 25. Matched cells Trap A: A0_Y0 708, A0_Y1 157, A1_Y0 2,903, A1_Y1 1,144 (4,912); Trap B: 708, 157, 4,093, 809 (5,767). Reversed-test melanomas 172 per trap ($\sim$ 34/fold). Cross-fold groups 0. DINOv2@518: ERM clean 0.782 rev 0.516 (Trap B rev 0.510); mask 0.769/0.452, -0.064 [-0.079,-0.049] / B +0.124 [+0.105,+0.143]; inpaint 0.811/0.616, +0.101 [+0.094,+0.107] / +0.108 [+0.098,+0.117]; balanced 0.812/0.729, +0.213 [+0.190,+0.235] / +0.234 [+0.217,+0.252]; DFR 0.724/0.718, +0.202 [+0.176,+0.228] / +0.192 [+0.168,+0.216]; LEACE paired 0.800/0.574, +0.058 [+0.053,+0.063] / +0.078 [+0.070,+0.085]. C1–C5 supported; C3 +0.188 [+0.179,+0.194], per seed 0.171–0.195; C5 +0.068 vs +0.112. Source-stratified Trap A mask – ERM: HAM -0.030 [-0.059,-0.003], BCN -0.078 [-0.099,-0.058]; Trap B without BCN +0.276; HAM paired mask-source check: -0.006 [-0.029,+0.016].
- **E14** area ratio: in-lesion hair median 0.033, ruler 0.030; Trap A tertiles (0.005/0.032/0.116): -0.064/-0.057/-0.059; ruler vs hair at matched ratio -0.364 vs -0.059.
- **E15** DermLIP@224 Trap A: ERM 0.845/0.926/0.677; mask 0.800/0.893/0.608, -0.069 [-0.088,-0.051]; balanced 0.858/0.882/0.803, +0.126 [+0.115,+0.138]; DFR 0.827/0.840/0.792, +0.114 [+0.091,+0.138]; LEACE paired 0.856/0.924/0.708, +0.031 [+0.028,+0.034]. Hair decodability 0.883 [0.872,0.894]; LEACE rank 1/512 $\rightarrow$ 0.492. LEACE 2 $\times$ 2: DINOv2 paired 0.800/0.920/0.574 (+0.058), A-erasure 0.736/0.624/0.831 (+0.316); DermLIP paired 0.856/0.924/0.708 (+0.031), A-erasure 0.764/0.643/0.869 (+0.192).

R1.0.7 6–7. Run order and deliverables

Data/joins/dedup counts $\rightarrow$ features $\rightarrow$ E1–E7 $\rightarrow$ E8–E9 $\rightarrow$ E10–E11 $\rightarrow$ U-Net, E12, E13 $\rightarrow$ E14–E15. Deliverables: results/SUMMARY.md (expected vs obtained, MATCH/MISMATCH), results/PRECOMMITTS/, COMPARABILITY.md, CHANGES.md, unit tests (placement, bootstrap, LEACE rank, zero cross-split edges, presence independent of overlap).

R2 Audit of the original thesis claims

Source: docs/THESIS_CLAIMS_AUDIT.md

Status legend: **VERIFIED-ARCHIVE** = recomputed exactly from archived predictions/tables; **VERIFIED-RERUN** = reproduced by rerunning from raw images with this repo; **NO CODE** = not produced by any code in the archived repository; **NOT SUPPORTED** = archived evidence contradicts the claim. Filled in as verification proceeds (see results/verification/).

§	Claim in proposal	Status
7	Masking Δ vs ERM (DINOv2@518): +0.184 / -0.097 / -0.111 / -0.128 at 0/50/75/100%	VERIFIED-
7	Interaction 0% vs 100%: +0.312 [+0.272, +0.354]	VERIFIED-
7	Variable-ruler stress +0.147 / -0.155; interaction +0.302	VERIFIED-
8	Occlusion control +0.035/+0.033/+0.039; cost 0.000–0.004	pending
8	Dilation at 50%: -0.097 $\rightarrow$ -0.153 as retention 0.50 $\rightarrow$ 0.95; flat at 100%	VERIFIED-
8	Same-head $ \Delta p $ 0.235 (ERM) $\rightarrow$ 0.490 (mask) at 100%	VERIFIED-
6.1	“Before the [leakage] correction, the measured shortcut gap was inflated by 27%”	NOT SUPP
6.1	U-Net held-out Dice 0.876; 0.895 vs manual masks	NO CODE -
9	ISIC 2019 Traps A/B (25,331 images), C1–C5, DermLIP replication, source stratification, five-seed crossover	NO CODE
10	Hair detector IoU 0.22	VERIFIED-
10	Training-time repair +0.043; $\lambda=0$ control	VERIFIED-
10	Hair linearly decodable AUROC 0.943 (DINOv2)	VERIFIED-
10	... 0.883 (DermLIP)	NO CODE
10	Rank-1 projection 0.812 $\rightarrow$ 0.814	VERIFIED-
10	Unpaired erasure +0.316 (DINOv2), +0.192 (DermLIP)	NO CODE
11	Synthetic ruler vs real hair at matched area ratio (-0.364 vs -0.059)	NO CODE
11	$\approx$ 11% of images change overlap stratum if mask source swapped	NO CODE -

R2.0.1 Rerun from raw images (DINOv2@518, all nine pilot arms)

results/verification/SYNTHETIC_RERUN_VS_ARCHIVE.md: 252 per-seed AUROCs reproduced within ≤ 0.0005 (fp16 non-determinism) for every sklearn-head arm; the GPU-trained consistency head within ≤ 0.008 . All 21 reversed-test Δ -vs-ERM estimates and CIs agree to 3 decimals; all 21 CI decisions identical. Location interaction (mask): +0.312 [+0.272, +0.353].

R2.0.2 §9 / E13 under the author's protocol (docs/REPLICATION_SPEC.md), DINOv2 @518, 5 seeds $\times$ 5 folds

Outcome-free anchors: hair-free group ≤ 30 native px reproduces A0 cells and all six source proportions; reversed-test melanomas 172 per trap (spec 172). A1 cells differ by 5–15 % (lesion masks from a re-trained spec U-Net, held-out Dice 0.890 vs 0.876).

Claim	Spec	Replication	Ver
C1 mask – ERM, Trap B	+0.124 [+0.105, +0.143]	+0.107 [+0.078, +0.135]	MAT
C2 mask – ERM, Trap A	−0.064 [−0.079, −0.049]	−0.037 [−0.057, −0.019]	SIG
C3 crossover B – A	+0.188 [+0.179, +0.194]	+0.145 [+0.105, +0.181]; seeds 0.087–0.175	SIG
C4 balanced / DFR both traps	all CIs > 0	+0.225 / +0.251 (A), +0.209 / +0.215 (B), all CIs > 0	MAT
C5 paired LEACE < ½ best label-only	+0.068 vs +0.112	+0.084 vs +0.117	MAT
Inpaint (oracle) A / B	+0.101 / +0.108	+0.103 / +0.096	MAT
Source-stratified Trap A, BCN	−0.078 [−0.099, −0.058]	−0.051 [−0.075, −0.027]	SIG
Source-stratified Trap A, HAM	−0.030 [−0.059, −0.003]	−0.009 [−0.036, +0.017]	sign
Unpaired A-erasure Trap A	+0.316, corr < rev	+0.326, corr 0.618 < rev 0.817	MAT

Sensitivity: the author-independent reconstruction (results/real/isic2019/dino518_main: hair-free < 0.1 % of pixels, hair ≥ 0.5 %, artifact-free clean test) gives C1/C3/C4/C5 but a null C2 (+0.012 [−0.042, +0.062]): the in-ROI harm depends on including lightly-haired images and on the strictly hair-free comparison group.

R2.0.3 Leakage (§3.1) — controlled experiment (scripts/run_leakage_experiment.py)

Same 2,437 images, same features and protocol; grouped split vs 5 random image-level splits (93 near-duplicate pHash ≤ 8 pairs cross train/test). ERM shortcut gap: 0.241 $\rightarrow$ 0.284 at 0 % (+18 %), 0.422 $\rightarrow$ 0.465 at 100 % (+10 %); mask – ERM: +0.184 $\rightarrow$ +0.223 (0 %), −0.129 $\rightarrow$ −0.088 (100 %). **Direction of the “27 %” claim supported; supported magnitude 10–18 %.**

R2.0.4 E12 — overlap-contrast trap

Author protocol, full ISIC 2019 spec cohort: A=1 hair $r \geq 0.5$ vs A=0 hair $r < 0.1$. ERM corr 0.884 / rev 0.479; mask – ERM −0.083 [−0.097, −0.072]; balanced +0.169, DFR +0.214. **Not null** (spec: null). The spec's E12 predates E13 and likely used a different cohort; here the contrast is learnable and masking harms, consistent with the retention mechanism.

R2.0.5 DermLIP availability

redlessone/DermLIP_PanDerm-base-w-PubMed-256 became gated (gated="auto") on 2026-09-21, after the pilot ran; access is granted automatically after accepting the terms on the model page.

R3 Data sources and licences

Source: docs/DATA.md

Dataset	Use	Source
ISIC 2018 Task 1/2 images + Task 1 masks	synthetic-ruler pilot cohort; U-Net training	ISIC challenge
ISIC 2016/2017 Part 3 labels	pilot melanoma labels (union; 3 conflicts dropped)	ISIC
ISIC 2019 training (25,331)	real-hair traps	ISIC challenge
HAM10000 lesion segmentations	manual lesion masks (10,015)	Tschandl et al.
ISIC 2019 artifact masks (hair+ruler, ink, vignetting)	artifact geometry, oracle inpainting, I2E donors	Wegley et al. 2
NIH ChestX-ray14 (112,120)	CXR cohorts	NIH CC (HF m

Dataset	Use	Source
NEATX torchxrayvision PSPNet	chest-drain labels for 3,543 NIH pneumothorax images lung ROI masks	Jiménez-Sánchez Cohen et al.

R3.0.1 Derived objects (all rebuilt by make data)

- `isic2019/prepared/cohort_groups.csv` — pHash (64-bit) and leakage groups ($\text{lesion_id} \cup \text{pHash} \leq 2$).
- `isic2019/unet/UNET_REPORT.json` — lesion U-Net held-out Dice.
- `isic2019/prepared/cohort_518.csv` — per image: source, lesion-mask source, hair fraction, $\text{hair} \cap \text{lesion}$ overlap r , ink, vignetting.
- `cxr/prepared/DRAIN_DETECTOR.json` — drain detector CV AUROC and thresholds; `nih_drain_scores.csv`.
- `cache/features/**` — per-view frozen features (npz); `cache/images/**` — memmapped resized images.

R4 Pre-registration: ISIC 2019 real-hair traps

Source: `docs/PREREGISTRATION_ISIC2019_TRAPS.md`

Committed to version control **before any trap AUROC is computed** (check `git log` for this file's timestamp vs. `results/real/`). The code that produced thesis §9 was not present in the archived repository (`isic_pcm_code_results/`), so this is a fresh, documented re-implementation. Thesis numbers are *targets for qualitative agreement*, not for exact reproduction.

R4.0.1 Cohort

- ISIC 2019 training set, 25,331 images; melanoma label = MEL column.
- Source from `lesion_id` prefix: HAM, BCN, MSK (incl. `_downsampled`), else ISIC_legacy.
- Leakage groups: union-find over `lesion_id` and 64-bit pHash with Hamming ≤ 8 (pilot rule).
- Lesion masks: HAM10000 manual (Tschandl 2020) or ISIC 2018 Task 1 manual where available, otherwise U-Net prediction (ResNet34 U-Net, 384 px, trained on those manual masks).
- Artifact masks: Wegley et al. 2026 archive (Scholars' Mine, doi:10.71674/man1-qa33): hair+ruler (one channel), ink, vignetting.
- All geometry at 518×518 (nearest-neighbour masks).
- Quality control: exclude images with lesion area fraction < 0.005 or > 0.95 (failed masks) and images with any ink mark (ink is a different artifact).

R4.0.2 Artifact groups

- **Hair-free (A=0)**: hair/ruler fraction < 0.001 of the image.
- **Hair-present**: hair/ruler fraction ≥ 0.005 .
- Overlap $r = |\text{hair} \cap \text{lesion}| / |\text{hair}|$.
- **Trap A (in-ROI)**: hair-present with $r \geq 0.5$. **Trap B (out-of-ROI)**: hair-present with $r < 0.1$.
- Hair-present images with $0.1 \leq r < 0.5$ are in neither trap. They are the **template-donor pool** for Insert-to-Erase, so no donor is ever a train/test image of a trap.

R4.0.3 Splits and environments

- 5-fold StratifiedGroupKFold on (group, y), seed 20260926. Fold k is test; fold $(k+1) \bmod 5$ is validation; the other three are training. Folds are the clusters of the hierarchical bootstrap.
- Environments per trap, built inside each source (HAM/BCN/MSK/ISIC_legacy) by maximum-size subsampling: train 90/10 ($P(A=1|Y=1)=0.9$, $P(A=1|Y=0)=0.1$), correlated test 90/10, reversed test 10/90. **Source matching**: per source $\times (A, Y)$ cell the counts are the minimum over the two traps, so both traps have identical cell counts per source; the $A=0$ images drawn are the same for both traps.
- Clean test: every hair-free image of the test fold. Clean validation: every hair-free image of the validation fold (used for C , λ and the balanced-accuracy threshold — never artifact data).
- Power: ≥ 40 reversed-test melanomas per trap summed over folds (thesis minimum). Counts are reported before any model is fitted.

R4.0.4 Arms (frozen backbone + linear head)

Backbones: DINOv2 ViT-B/14 @518 (full sweep), DermLIP/PanDerm @224 (replication). | arm | input / supervision
| | --- | --- | | erm | original image | | mask | lesion-masked image at train and test (lesion mask) | | inpaint | Telea
inpaint of the true hair mask at train and test (oracle; upper bound) | | balanced | four (A×Y) groups equally weighted
(image-level A at train) | | dfr | last layer on a group-balanced subset of the validation fold (image-level A on val)
| | leace_paired | LEACE on (original, hair-inpainted) pairs of training hair images | | leace_unpaired | LEACE on
image-level A of the training set (the variant expected to *game* the test) | | i2e | **proposed**: subspace erasure from
(original, hair-inserted) pairs; no artifact label, no mask | | i2e_balanced | i2e + balanced head (image-level A at train)
| | i2e_rank1 | ablation: LEACE on insertion pairs | | insert_aug | ablation: label-independent insertion augmentation
(WP3 route 2) |

Hair insertion: donor from the donor pool (deterministic hash), random flip/rotation, alpha-composited with a $\sigma=0.7$
px Gaussian-softened donor hair mask. I2E energy threshold 0.90, max rank 64 — fixed now.

R4.0.5 Claims and decision rules (95% hierarchical paired bootstrap CI, 10,000 replicates)

- **C1** mask – ERM on Trap B reversed AUROC > 0 (CI excludes 0).
- **C2** mask – ERM on Trap A reversed AUROC < 0 (CI excludes 0).
- **C3** crossover [mask – ERM]_B – [mask – ERM]_A > 0 (CI excludes 0).
- **C4** balanced – ERM and DFR – ERM > 0 on both traps (CIs exclude 0).
- **C5** leace_paired gain on Trap A < ½ of the balanced gain (linear paired erasure insufficient).
- **P1** i2e – ERM on Trap A > 0 (CI excludes 0) **and** clean AUROC loss ≤ 0.02 vs ERM **and** correlated AUROC ≥
reversed AUROC – 0.02 (no inversion, i.e. not gamed).
- **P2** i2e recovers ≥ ½ of the balanced gain on Trap A without any artifact label.
- **P3** i2e_balanced ≥ balanced on Trap A reversed AUROC (point estimate) with clean loss ≤ 0.02. Outcomes
are reported whether positive or negative.

R4.0.6 Deviations recorded after commit (before any AUROC)

- Leakage groups use pHash ≤ 2 (not the pilot's ≤ 8): at ISIC 2019 scale ≤ 8 chains unrelated lesions into one
component of 10,974 images. Recorded in `scripts/data/prepare_isic2019.py`.
- Artifact-mask file names in the archive are irregular; masks are matched on the zero-padded numeric id. 24
ink files carry truncated ids (e.g. ISIC_002431); all ten candidate images of each are flagged `ink_uncertain`
and excluded by quality control, like confirmed ink images.

R5 Pre-commit: matched traps

Source: `docs/PRECOMMIT_MATCHED_TRAPS.md`

Committed **after** the pre-registered ISIC 2019 trap results (DINOv2) and the post-hoc diagnostics (`scripts/analysis/real_trap_di`
and **before** any matched-trap AUROC exists. It is a follow-up, not a replacement: the pre-registered verdicts (C2
not supported on DINOv2) stand as reported.

R5.0.1 Motivation (from diagnostics, no outcome of this follow-up known)

- Removing hair pixels at test only closes 47% (Trap A) / 44% (Trap B) of the ERM clean-reversed gap: most of
what the ERM head exploits in the real trap is not the hair pixels.
- Hair-present and hair-free images differ in sex (HAM: 42% vs 57% female), age (52 vs 58 y) and anatomical
site (HAM lower extremity: 16% vs 42%), i.e. 'hair present' is also a patient/site proxy.
- The in-ROI harm mechanism (retention of a used artifact) presupposes that the shortcut is the artifact.

R5.0.2 Design change (only this)

Within each source, artifact-present and hair-free images are drawn from the same strata of (anatomical site group
× sex × age band [<40 , 40-59, ≥ 60]); in each stratum the environments are built by the same maximum-size
subsampling with $P(A|Y) = 0.9/0.1$ (train, correlated test) and $0.1/0.9$ (reversed). Everything else (folds, QC, groups,
arms, thresholds, bootstrap, source matching across traps) is unchanged.

R5.0.3 Questions (descriptive, 95% hierarchical CI)

- M1: pixel share of the ERM shortcut (hair removed at test only) is larger than in the unmatched trap.
- M2: mask – ERM on Trap A reversed AUROC (sign and CI); M3: mask – ERM on Trap B; M4: crossover.
- M5: I2E – ERM on Trap A, relative to the pixel-removal upper bound (oracle inpaint – ERM). Outcomes are reported whichever way they go.

R6 Pre-registration: chest-radiograph device traps

Source: docs/PREREGISTRATION_CXR_DEVICE_TRAPS.md

Committed before any model is fitted on this cohort (check git log against results/cxr_traps/). Protocol = the author's E13 protocol (docs/REPLICATION_SPEC.md) transferred to radiographs; only the items below differ.

R6.0.1 Cohort

- RANZCR-CLiP training images (30,083; human image-level labels for ETT / NGT / CVC / Swan-Ganz; human polylines on 9,095 images) linked to NIH ChestX-ray14 by 64-bit pHash (nearest neighbour ≤ 6 bits, runner-up ≥ 4 bits further): 28,786 images, 3,227 patients. Disease labels = NIH findings.
- ROI = both lungs (TorchXRyVision PSPNet, $p > 0.5$) at 518 px.
- Device masks = CLiP polylines rasterised at 518 px (ETT 12 px, NGT 7 px, CVC / Swan-Ganz 4 px wide).
- Overlap $r = |\text{device} \cap \text{lungs}| / |\text{device}|$ per device type (observed, outcome-free: median r ETT 0.00, NGT 0.00, CVC 0.52).

R6.0.2 Traps (artifact-free comparison group per device, as in E13)

- **Trap A (in-ROI): CVC.** $A = 1$: CVC present and annotated with $r \geq 0.5$. $A = 0$: no CVC (human label).
- **Trap B (out-of-ROI): ETT.** $A = 1$: ETT present and annotated with $r < 0.1$. $A = 0$: no ETT.
- Source matching: within each label, $A = 1$ cells are subsampled to the $A = 0$ **view-position** (AP/PA) mix.
- Groups: NIH patient id; 5-fold group-safe CV per seed; seeds 42, 123, 456, 789, 2026; fold predictions pooled per seed (seed = bootstrap cluster). Train 90/10, correlated test 90/10, reversed 10/90, clean 50/50, inner validation = 20 % of training patients (clean 50/50 for C and thresholds; all groups for DFR).

R6.0.3 Diseases (co-primary; chosen on counts only)

Every NIH finding with ≥ 40 estimated reversed-test positives in **both** traps: **Infiltration, Effusion, Atelectasis, Consolidation**. Holm correction over the four diseases for each claim.

R6.0.4 Arms

erm, mask (lung ROI), inpaint (Telea of the trap's device mask; oracle), balanced, dfr, leace_paired (orig $\leftrightarrow$ device-inpainted training $A = 1$ images), leace_unpaired (A-erasure), prevcal (Kina & Petersen 2026), i2e / i2e_balanced (device pixels transplanted from donor images of the same device type that are in neither trap: CVC with $0.1 \leq r < 0.5$; ETT with $r \geq 0.1$). Backbones: RAD-DINO @518 (primary; chest-X-ray foundation model), MedSigLIP-448, DINOv2 ViT-B/14 @518.

R6.0.5 Claims (hierarchical paired bootstrap, 10,000 replicates, 95 % CI; per disease, Holm across diseases)

- **X1** mask – ERM on Trap B (ETT, outside the lungs) reversed AUROC > 0 .
- **X2** mask – ERM on Trap A (CVC, inside the lungs) reversed AUROC < 0 .
- **X3** crossover [mask – ERM]_B – [mask – ERM]_A > 0 .
- **X4** balanced and DFR – ERM > 0 in both traps.
- **X5** paired LEACE gain $< \frac{1}{2}$ best label-only gain.
- **X6** (proposed) i2e – ERM > 0 in Trap A with clean AUROC loss ≤ 0.02 and correlated $\geq$ reversed – 0.02. Reported whatever the outcome; the primary backbone is RAD-DINO, the others are replications.

R6.0.6 Amendment 1 (committed before any model fit; stricter)

View matching to the A = 0 view mix is infeasible for Trap B: intubated patients are imaged supine (AP), so ETT images have almost no PA counterpart (matched cells collapse to 7–35 images). Both traps are therefore **restricted to AP radiographs**, which removes the view confound instead of re-weighting it. Counts after the amendment (outcome-free): every matched cell ≥ 25 for all four pre-registered diseases; estimated reversed-test positives Trap A / Trap B: Infiltration 218 / 3,318, Effusion 125 / 2,316, Atelectasis 140 / 1,918, Consolidation 66 / 915. Disease selection unchanged.

R6.0.7 Follow-up 1 (committed after the pre-registered Infiltration/RAD-DINO result, before any follow-up fit)

Observed confound (descriptive): in the Trap A pool the CVC-free group carries far more out-of-lung tubes than the CVC group (ETT 77 % vs 39 %, NGT 83 % vs 42 %), because CLiP's CVC-free images were selected for other tubes. The A contrast is then partly "fewer tubes outside the lungs", which lung masking removes. The pre-registered results stand as reported. Follow-up design (stricter, same everything else): cells are matched within label on **view × presence of the other devices** — Trap A: view × ETT × NGT; Trap B: view × CVC × NGT — so the only device difference between A = 1 and A = 0 is the trap's own device. Questions: X1–X3 and X6 as above; reported beside, never instead of, the pre-registered results.

R6.0.8 Results — pre-registered primary (RAD-DINO @518, AP films, 5 seeds × 5 folds)

Disease	X1 mask–ERM, ETT (out)	X2 mask–ERM, CVC (in)	X3 crossover	balanced A / B	I2E+ba
Infiltration	+0.063 [+0.058, +0.068]	+0.094 [+0.075, +0.113]	−0.032 [−0.051, −0.013]	+0.319 / +0.234	+0.329
Effusion	+0.100 [+0.092, +0.109]	+0.104 [+0.078, +0.129]	−0.004 [−0.027, +0.021]	+0.302 / +0.315	+0.316
Atelectasis	+0.105 [+0.094, +0.114]	+0.045 [+0.013, +0.077]	+0.060 [+0.028, +0.089]	+0.266 / +0.302	+0.256
Consolidation	+0.068 [+0.049, +0.086]	+0.070 [+0.044, +0.096]	−0.002 [−0.030, +0.025]	+0.286 / +0.190	+0.298

X1 supported (4/4). **X2 not supported (0/4): lung masking helps even for the in-lung CVC.** X3 supported 1/4 (Atelectasis). X4 supported (4/4). Interpretation (descriptive, see Follow-up 1): the CVC contrast is confounded by out-of-lung tubes that masking removes; the device-matched follow-up tests this.

R6.0.9 Controlled synthetic tube, RAD-DINO@518, NIH pneumothorax (results/synthetic/ni_h_ptx/raddino)

Reversed AUROC (ERM / mask / U-MtE): r=0: 0.913 / 0.910 / 0.891; r=0.5: 0.772 / 0.843 / 0.886; r=1: 0.614 / 0.725 / 0.859.

- The thesis sign reversal does **not** replicate here: ERM does not use an out-of-lung tube at all (reversed $\approx$ clean at r=0), and lung masking helps more as overlap grows (location interaction [r=0] – [r=1] = −0.113 [−0.123, −0.103]); masking cannot remove the in-lung tube (r=1: 0.725 vs clean 0.905).
- U-MtE – mask = +0.044 [+0.036, +0.053] (r=0.5), **+0.134 [+0.120, +0.151] (r=1)**; U-MtE – ERM (r=1) = +0.245.
- With artifact labels, balanced alone is better than U-MtE_balanced (−0.031 [−0.038, −0.025]). Interpretation: with a strong disease signal (clean AUROC 0.91) RAD-DINO ignores extra-pulmonary overlays; the in-ROI residual shortcut after masking is still present and is what U-MtE removes.

R7 Pre-registration: real chest drains

Source: docs/PREREGISTRATION_CXR_DRAIN.md

Committed before any model is fitted with this runner (the drain trap was built earlier but never run).

- Cohort (wtss.data.cxr_drain, frozen): 29,687 NIH ChestX-ray14 images; y = pneumothorax; A = chest drain (NEATX human labels for PTX+; RAD-DINO drain detector at precision 0.95 / below the median negative score for PTX−); view (AP/PA) as the matching source; patient-grouped 5 folds; train 90/10, correlated 90/10, reversed 10/90, clean = drain-free. Drains lie inside the lung field: this is an in-ROI trap only (ROI = lungs, PSPNet). No pixel drain masks exist, so no inpainting arm.
- Folds act as the bootstrap clusters (5), as in the original run_traps design.
- Backbone: RAD-DINO @518 (chest-X-ray foundation model). Generic overlay library (unchanged).

- Arms: erm, mask, balanced, dfr, U-MtE (mte), mte_balanced, mte_aug, mte_protect(+balanced), mask_dfr, jtt, mask_jtt.
- Claims (reversed AUROC, 95 % CI): **D1** mask – ERM $\leq$ small (in-ROI; reported with CI, no direction claimed); **D2** U-MtE – mask > 0 ; **D3** U-MtE – mte_aug > 0 ; **D4** U-MtE_protect – mask > 0 ; **D5** U-MtE – JTT > 0 (label-free comparison). Reported whichever way they go.

R7.0.1 Results — RAD-DINO@518 (results/cxr_drain/raddino518_universal)

Reversed (clean) AUROC: ERM 0.191 (0.886) — an extreme shortcut; mask 0.239; U-MtE 0.268; U-MtE_protect 0.269 (0.867); JTT 0.175; balanced 0.538; DFR 0.683 (corr 0.738); mask+DFR 0.646.

- D1: mask – ERM = +0.049 [+0.035, +0.061] (lung masking barely helps; drains lie in the lungs).
- D2 supported (small)**: U-MtE – mask = +0.029 [+0.021, +0.036]. **D3 supported**: U-MtE – mte_aug = +0.031 [+0.022, +0.042]. **D4 supported**: U-MtE_protect – mask = +0.030 [+0.022, +0.037] at no clean cost (–0.003 [–0.009, +0.002]). **D5 supported**: U-MtE – JTT = +0.094 [+0.077, +0.109].
- With artifact labels, DFR is best (0.683); U-MtE_balanced is worse than balanced (–0.079). Interpretation: a real drain co-occurs with a treated (re-expanded) pneumothorax, so the shortcut is not only the tube's pixels; pixel/insertion-based repair recovers little (+0.03), label-based group methods much more.

R7.0.2 Amendment (before fitting): DINOv2@518 replication of the drain trap — same arms and claims D1-D5.

R7.0.3 Results — DINOv2@518 replication (results/cxr_drain/dino518_universal)

Reversed: ERM 0.280, mask 0.292, U-MtE 0.285, U-MtE_protect 0.281, JTT 0.294, DFR 0.549, mask+DFR 0.566 (best), U-MtE_balanced 0.443 (vs balanced 0.399: +0.044 [+0.025, +0.064]). D1 mask – ERM +0.012 [+0.002, +0.023]; **D2 not supported** (U-MtE – mask –0.008 [–0.015, +0.000]); **D3 not supported** (–0.010); **D4 not supported** (–0.011); **D5 not supported** (–0.010). With RAD-DINO D2-D5 held but with small effects (+0.03): on real drains, insertion-based erasure does not transfer across backbones; group methods with drain labels are required.

R8 Pre-registration: thyroid ultrasound traps

Source: docs/PREREGISTRATION_THYROID_TRAPS.md

Committed before any model is fitted on this cohort. Protocol = the author's E13 protocol (docs/REPLICATION_SPEC.md) transferred to ultrasound; only the items below differ.

R8.0.1 Cohort (counts are outcome-free; /root/data/us/tncd/markers_stats.csv)

- TN3K images with TNCD benign(0)/malignant(1) labels: 3,493 (1,210 malignant). Nodule masks: TN3K manual.
- Marker masks: audited rule-based detector wtss.data.us_markers.marker_mask (docs/THYROID_MARKER_AUDIT.md; image-level precision/recall $\approx$ 0.86–0.89 on 20 unseen images).
- A = 0 (marker-free, shared by both traps): no detected marker pixel. A = 1: ≥ 15 detected marker pixels with overlap $r = |\text{marker} \cap \text{nodule}| / |\text{marker}|$: **Trap A (in-ROI) $r \geq 0.5$** (1,150 images); **Trap B (out-of-ROI) $r < 0.1$** (326 images). $0.1 \leq r < 0.5$ images are the donor pool for template insertion (I2E).
- Single source (no site metadata): cells are matched on label only (source matching is vacuous).
- Leakage groups: pHash Hamming ≤ 8 (largest group 3 images). Patient ids are not published (3,493 images from 2,421 patients): same-patient different-view leakage cannot be excluded — reported as a limitation.
- Images resized to 518 px (nearest for masks).

R8.0.2 Design (as E13)

5 seeds (42, 123, 456, 789, 2026) $\times$ 5-fold group-safe CV; train 90/10, correlated test 90/10, reversed 10/90, clean 50/50; inner validation 20 % of training groups; fold predictions pooled per seed; hierarchical paired bootstrap with seeds as clusters; gate: every matched cell ≥ 25 and ≥ 40 reversed-test positives.

R8.0.3 Arms

erm, mask (nodule ROI), inpaint (Telea of the detected marker mask; oracle w.r.t. the detector), balanced, dfr, leace_paired, leace_unpaired, prevcal, i2e / i2e_balanced (real-marker templates from the donor pool), U-I2E / U-MtE (+ balanced) with the generic artifact library (unchanged from docs/PREREGISTRATION_UNIVERSAL_I2E.md). Backbones: DINOv2 ViT-B/14 @518 (primary), MedSigLIP-448 (medical foundation model, replication).

R8.0.4 Claims (95 % hierarchical CI)

- **T1** mask – ERM on Trap B (markers outside the nodule) reversed AUROC > 0.
- **T2** mask – ERM on Trap A (markers inside the nodule) reversed AUROC < 0.
- **T3** crossover [mask – ERM]_B – [mask – ERM]_A > 0.
- **T4** balanced and DFR – ERM > 0 in both traps.
- **T5** paired LEACE gain < ½ best label-only gain.
- **T6** U-MtE – mask > 0 in Trap A; U-I2E – ERM > 0 in Trap A; no gaming (clean loss ≤ 0.02, corr ≥ rev – 0.02). Outcomes reported whichever way they go.

R8.0.5 Amendment 1 (2026-09-27, before any synthetic-thyroid model is fitted) — controlled synthetic calipers

Real-marker results (DINOv2) are known at this point: mask – ERM is positive in both traps, much smaller in Trap A. To test the location effect under control, the thesis Result-1 design is transferred unchanged:

- Cohort: marker-free images only (1,663 after common support; wtss.data.thyroid); split TN3K test -> test, trainval -> train/val 80/20 by pHash group.
- Artifact: synthetic sonographer caliper (wtss.synthetic.draw_caliper: dotted line + '+' ends), 65x19 box at 518 px (same geometry as the ISIC ruler); overlap with the nodule $r \in \{0, 0.25, 0.5, 0.75, 1\}$.
- Seeds 42/123/456; arms erm, mask, inpaint, balanced, dfr, leace + proposed arms (I2E, MtE, U-I2E, U-MtE).
- Claims: **TS1** mask – ERM (reversed AUROC) decreases with overlap ($r=1$ minus $r=0$ interaction < 0); **TS2** mask – ERM < 0 at $r = 1$; **TS3** mask – ERM > 0 at $r = 0$; **TS4** U-MtE – mask > 0 at $r = 1$.

R8.0.6 Results — DINOv2@518, real markers (results/thyroid/dino518_main, dino518_universal)

Reversed-test AUROC, delta vs ERM [95 % CI]:

- **T1 supported:** mask – ERM, Trap B = +0.341 [+0.320, +0.363].
- **T2 not supported:** mask – ERM, Trap A = +0.111 [+0.097, +0.124] (helps, does not harm). After masking the model remains marker-driven in Trap A (test_corr 0.902 vs test_rev 0.400).
- **T3 supported:** crossover [mask – ERM]_B – [mask – ERM]_A = +0.230 [+0.204, +0.255].
- **T4 supported:** balanced +0.387 / +0.269, DFR +0.449 / +0.317 (A / B).
- **T5 supported:** paired LEACE +0.050 (A) vs best label-only +0.449 (DFR; prevcal excluded — it needs A at test time and collapses test_corr to 0.258).
- **T6 supported:** MtE – mask (Trap A) = +0.303 [+0.284, ...] with real-marker templates and **U-MtE – mask = +0.222 [+0.197, +0.251] with the generic library (no marker example used)**. U-MtE_balanced: reversed 0.788, clean 0.807 (template MtE_balanced 0.818 / 0.812); Trap B U-MtE 0.740 vs mask 0.747. No gaming: U-MtE corr 0.822 ≥ rev 0.622.

R8.0.7 Replication — MedSigLIP-448 (results/thyroid/medsiglip448_main)

T1 mask – ERM (B) +0.228 [+0.203, +0.252] supported; T3 crossover +0.163 [+0.132, +0.194] supported; T2 not supported (Trap A mask 0.348 vs ERM 0.284: small gain); T6 MtE – mask (A) +0.310 supported; MtE_balanced reversed 0.814 / clean 0.805 (best). Same qualitative pattern as DINOv2.

R8.0.8 Results — Amendment 1, controlled synthetic calipers, DINOv2@518 (results/synthetic/thyroid/dino518_universal)

Reversed AUROC vs ERM [95 % CI]:

- **TS3 supported:** mask – ERM at $r=0$ = +0.402 [+0.347, +0.458].

- **TS2 not supported:** mask – ERM at $r=1 = -0.030 [-0.079, +0.016]$ (the entire masking benefit is lost, no significant harm).
- **TS1 supported:** location interaction ($[r=0] - [r=1] = +0.432 [+0.377, +0.492]$).
- **TS4 supported:** U-MtE – mask at $r=1 = +0.278 [+0.232, +0.325]$; U-MtE_balanced – balanced = $+0.091 [+0.042, +0.142]$; U-MtE_balanced – mask = $+0.471 [+0.402, +0.534]$. U-MtE's location interaction is $+0.156$ (vs $+0.432$ for masking); U-MtE_balanced's is $-0.054 [-0.098, -0.004]$ (location-invariant or better inside).

R8.0.9 Amendment 2 (before fitting): controlled synthetic-caliper replication with MedSigLIP-448 — same design and claims TS1-TS4.

Results — Amendment 2, MedSigLIP controlled calipers (results/synthetic/thyroid/medsiglip448_caliper_c

- TS3 supported: mask – ERM at $r=0 = +0.514 [+0.454, +0.573]$. TS1 supported: interaction $+0.470 [+0.405, +0.544]$.
- TS2 not supported: mask – ERM at $r=1 = +0.044 [+0.001, +0.089]$ (benefit almost entirely lost, no harm).
- TS4 **not supported:** U-MtE – mask at $r=1 = -0.076 [-0.110, -0.043]$. U-MtE_balanced – balanced = $+0.068 [+0.022, +0.115]$; U-MtE_balanced – mask = $+0.458 [+0.409, +0.508]$; U-MtE_balanced location interaction $+0.002 [-0.038, +0.041]$ (location-invariant). Across the four controlled sweeps (capsule/thyroid $\times$ DINOv2/MedSigLIP) unbalanced U-MtE beats masking with DINOv2 only; U-MtE_balanced beats masking in all four and is location-invariant in all four.

R9 Thyroid caliper detector audit

Source: docs/THYROID_MARKER_AUDIT.md

Detector: `wtss.data.us_markers.marker_mask` (dotted measurement lines ≥ 7 evenly spaced isolated bright dots, spacing CV ≤ 0.20 ; '+' calipers by template matching on the top-hat image; top 8 % / bottom 6 px excluded). Tuned on one random sample (seed 3, 16 images) and a first audit sheet (seed 2026, images 0-19); **frozen** before the final audit on images 20-39 of the seed-2026 sample, which had not been viewed.

Final audit (20 unseen images, image-level presence judged by eye from the raw image):

- true positives 6 (+2 probable: 33, 37), true negatives 10, false negatives 1 (thin solid-line Doppler box, #22), false positives 1 (single tiny dot, #28) $\rightarrow$ precision ≈ 0.86 -0.89, recall ≈ 0.86 -0.89.
- Known weakness: thin *solid* box outlines (Doppler/ROI boxes) are not detected; dotted boxes are. Pixel masks under-cover faint dotted lines (the detector marks dots, not the full segment), which biases the overlap r mildly toward the caliper ends.

R10 Pre-registration: capsule-endoscopy traps

Source: docs/PREREGISTRATION_CAPSULE_TRAPS.md

Committed before any classifier is fitted on this cohort. Protocol = E13 (docs/REPLICATION_SPEC.md) as transferred in docs/PREREGISTRATION_THYROID_TRAPS.md; only the items below differ.

R10.0.1 Cohort (outcome-free; /root/data/capsule/capsule_cohort.csv, scripts/data/prepare_capsule.py)

- SEE-AI (Kaggle capsuleyolo/kyucapsule, CC BY 4.0) single-class frames: erosion ($y=1$, 3,550) vs polyp-like ($y=0$, 1,931). ROI = union of the expert lesion boxes.
- Artifact = contamination (debris, bubbles, bile). Masks: expert masks (figshare 27645021) where a frame is in the expert set (22 frames), else a DINOv2 patch-token probe (wtss.slas) trained on 2,160 expert-masked frames; held-out (540 frames) pixel accuracy 0.92, IoU 0.62 (SEE-AI subset 0.61).
- $A = 0$: contamination $< 3\%$ of the frame (504). $A = 1$: contamination $\geq 10\%$ with $r = |\text{contam} \cap \text{ROI}| / |\text{contam}|$: **Trap A $r \geq 0.5$** (816), **Trap B $r < 0.1$** (1,241). $0.1 \leq r < 0.5 \rightarrow$ donor pool (template I2E / MtE).
- Leakage groups: contiguous blocks of 100 frame numbers $\cup$ pHash ≤ 2 (137 groups, largest 173). No video or patient ids are published: limitation.
- Cells (A0_Y0, A0_Y1, A1_Y0, A1_Y1): Trap A 166/338/404/412, Trap B 166/338/232/1009; reversed positives 373.

R10.0.2 Arms / backbones

As thyroid: erm, mask, inpaint (Telea of the probe mask; mask-quality-limited), balanced, dfr, leace_paired, leace_unpaired, prevcal, i2e/i2e_balanced/i2e_rank1/insert_aug, mte/mte_balanced (real-contamination templates), and the generic library run (U-I2E / U-MtE / mte_aug). DINOv2@518 primary; MedSigLIP-448 replication.

R10.0.3 Claims (reversed-test AUROC, 95 % hierarchical CI)

C1 mask – ERM > 0 in Trap B · C2 mask – ERM < 0 in Trap A · C3 crossover [B] – [A] > 0 · C4 balanced, DFR – ERM > 0 · C5 U-MtE – mask > 0 in Trap A · C6 U-MtE – mte_aug > 0 in Trap A · no gaming (clean loss <= 0.02; corr >= rev – 0.02). Reported whichever way they go. Then (user rule): controlled synthetic-artifact test on the same cohort's low-contamination frames.

R10.0.4 Controlled synthetic-debris test (registered with the real-trap design, before any fitting)

- Frames with contamination < 5 % (796 after common support); split 60/20/20 by leakage group (seed 20260927).
- Artifact: wtss.synthetic.draw_debris (textured yellow-green blob with bubbles), 48x48 box at 518 px; overlap with the lesion box $r \in \{0, 0.25, 0.5, 0.75, 1\}$; seeds 42/123/456; arms erm, mask, inpaint, balanced, dfr, leace + proposed (I2E, MtE, U-I2E, U-MtE).
- Claims: CS1 mask – ERM decreases with overlap ($r=1$ vs $r=0$ interaction < 0); CS2 mask – ERM < 0 at $r=1$; CS3 mask – ERM > 0 at $r=0$; CS4 U-MtE – mask > 0 at $r=1$. Small test split (137 frames): wide CIs expected.

R10.0.5 Results — DINOv2@518, real contamination, template arms (results/capsule/dino518_main)

Reversed AUROC deltas vs ERM [95 % CI]:

- **C1 supported:** mask – ERM (Trap B) = +0.465 [+0.440, +0.490].
- **C2 not supported:** mask – ERM (Trap A) = +0.097 (0.684 vs 0.587): masking helps, but a large residual shortcut remains (test_corr 0.975 vs test_rev 0.684).
- **C3 supported:** crossover = +0.368 [+0.340, +0.397].
- **C4 supported:** balanced +0.19, DFR +0.217 [+0.200, +0.235] (Trap A); DFR is the best arm (0.805).
- **Template MtE fails:** MtE – mask (Trap A) = –0.395 [–0.420, ...]; inpaint also hurts (–0.17 vs ERM). Interpretation (post hoc): erosions carry yellow-white fibrin that resembles debris; a subspace estimated from real-debris templates / debris-guided inpainting removes disease evidence. Boundary condition for erasure-based repair: the artifact subspace must be separable from the disease signal.

R10.0.6 Results — controlled synthetic debris, DINOv2@518 (results/synthetic/capsule/dino518_debris_c)

Reversed AUROC vs ERM [95 % CI]:

- **CS3 supported:** mask – ERM at $r=0$ = +0.103 [+0.060, +0.149].
- **CS2 supported:** mask – ERM at $r=1$ = –0.102 [–0.162, –0.040].
- **CS1 supported:** location interaction ($[r=0] - [r=1]$) = +0.205 [+0.150, +0.259].
- **CS4 supported:** U-MtE – mask at $r=1$ = +0.081 [+0.050, +0.112]; U-MtE_balanced – mask = +0.193 [+0.150, +0.235]; U-MtE_balanced – balanced = +0.047 [+0.004, +0.091]. U-MtE_balanced shows no location interaction (–0.000 [–0.023, +0.023]): its benefit does not depend on where the artifact sits.

R10.0.7 Replication — MedSigLIP-448 (results/capsule/medsiglip448_main, medsiglip448_universal)

Trap A reversed (clean): ERM 0.593, mask 0.702, template MtE 0.222, U-MtE 0.465, U-MtE_protect 0.667, U-MtE_protect_balanced 0.871 (0.904), U-MtE_balanced 0.854, balanced 0.810, DFR 0.801, mask+DFR 0.901 (0.908). Same pattern as DINOv2: masking helps less in Trap A than B; erasure fails on real debris unless protected; with artifact labels, masking + DFR is the most robust arm.

R10.0.8 Replication — ConvNeXt-Base (CNN), generic library (results/capsule/convnext384_universal)

Crossover +0.292 [+0.248, +0.337]; Trap A reversed: ERM 0.545, mask 0.639, U-MtE 0.559, U-MtE_protect 0.643, mte_aug 0.655, U-MtE_balanced 0.803, mask+DFR 0.864 (corr 0.878; most robust). Same pattern as DINOv2/MedSigLIP.

R10.0.9 Amendment (before fitting): controlled synthetic-debris replication with MedSigLIP-448 — same design and claims CS1-CS4.

Results — MedSigLIP replication of the controlled debris sweep (results/synthetic/capsule/medsiglip448_d

- CS3 supported: mask – ERM at $r=0$ = +0.335 [+0.257, +0.409].
- CS2 **not supported**: mask – ERM at $r=1$ = +0.140 [+0.095, +0.186] (benefit shrinks, no harm).
- CS1 supported: location interaction = +0.194 [+0.125, +0.264].
- CS4 **not supported**: U-MtE – mask at $r=1$ = -0.114 [-0.147, -0.082] (erasure removes lesion signal with this backbone); U-MtE_balanced – mask = +0.309 [+0.253, +0.369]; U-MtE_balanced – balanced = -0.026 [-0.080, +0.031].

R11 Pre-registration: ovarian ultrasound traps

Source: docs/PREREGISTRATION_OVARY_TRAPS.md

Committed before any model is fitted on this cohort. Protocol = thyroid traps (docs/PREREGISTRATION_THYROID_TRAPS.md).

- Data: MMOTU OTU_2d (Kaggle orville/mmotu-ovarian-ultrasound-images-dataset, CC BY 4.0), 1,469 images with tumour masks and 8 classes. Task: benign cystic (chocolate cyst, serous cystadenoma, simple cyst; $y=0$) vs lesions with solid components (teratoma, theca cell tumour, mucinous cystadenoma, high-grade serous carcinoma; $y=1$); normal ovary excluded → 1,202 images.
- Markers: the frozen thyroid detector, unchanged. Audit (8 per cell, scratch panel): Trap A detections are dotted measurement lines crossing the tumour (high precision); ~2/8 “marker-free” images show small marks (dilutes, does not bias); Trap B detections are mostly other on-screen overlays (dotted zoom-box arcs, focus markers).
- A0 = no detected marker pixel; A1 = ≥ 15 px; Trap A $r \geq 0.5$, Trap B $r < 0.1$.
- Cells (A0Y0/A0Y1/A1Y0/A1Y1): Trap A 170/178/341/276, Trap B 170/178/91/103; reversed positives 196 (gates pass).
- Groups: pHash ≤ 8 (1,059 groups, largest 13); no patient ids (limitation).
- Backbone DINOv2@518 (primary). Arms: as thyroid (template + generic library, incl. mte_protect, mte_aug, jtt, mask_dfr); then the controlled synthetic-caliper sweep on the marker-free images (user rule: synthetic after real).
- Claims: O1 mask – ERM > 0 (Trap B); O2 mask – ERM < 0 (Trap A); O3 crossover > 0 ; O4 U-MtE – mask > 0 (Trap A); O5 U-MtE – mte_aug > 0 (Trap A); O6 U-MtE_balanced best by min(rev, corr) among label-using arms (Trap A). Reported whichever way they go.

R11.0.1 Controlled synthetic-caliper sweep (registered with the real traps, before fitting)

Marker-free MMOTU images (343 after common support; test split 64 images — underpowered, wide CIs expected); draw_caliper, 65x19 box, $r \in \{0, 0.25, 0.5, 0.75, 1\}$; seeds 42/123/456; DINOv2@518; claims as TS1-TS4.

R11.0.2 Results — DINOv2@518, template arms (results/ovary/dino518_main)

- **O1 supported**: mask – ERM (Trap B) +0.30 (0.725 vs 0.423).
- **O2 not supported**: mask – ERM (Trap A) +0.13 (0.545 vs 0.414): masking helps less in-ROI, no harm.
- **O3 supported**: crossover +0.170 [+0.116, +0.224].
- Template MtE – mask (Trap A) +0.070 [+0.030, +0.106]; MtE_balanced reversed 0.729 / corr 0.790 (min 0.729) — best robust arm (balanced 0.707, DFR 0.689).

R11.0.3 Results — DINOv2@518, generic library (results/ovary/dino518_universal)

Trap A reversed / corr: mask 0.545 / 0.828; U-MtE 0.564 / 0.855; U-MtE_protect 0.588 / 0.832; mte_aug 0.529; JTT 0.428; balanced 0.707 / 0.732; mask_balanced 0.687; U-MtE_balanced 0.736 / 0.811; mask+DFR 0.752 / 0.757.

- **O4 not supported**: U-MtE – mask +0.019 [-0.007, +0.043]; protected U-MtE – mask **+0.043 [+0.025, +0.061]**.
- **O5 supported**: U-MtE – mte_aug +0.035 [+0.010, +0.059]. U-MtE – JTT +0.137 [+0.098, +0.179].
- **O6 not supported**: by min(rev, corr) mask+DFR (0.752) $>$ U-MtE_balanced (0.736) $>$ balanced (0.707).

R11.0.4 Results — controlled synthetic calipers, DINOv2@518 (results/synthetic/ovary/dino518_caliper_c343_images)

- TS3 not supported: mask – ERM at $r=0$ = +0.042 [−0.034, +0.119].
- **TS2 supported:** mask – ERM at $r=1$ = **−0.197 [−0.292, −0.100]** (masking harms when the caliper is inside).
- **TS1 supported:** location interaction +0.239 [+0.175, +0.306].
- **TS4 supported:** U-MtE – mask at $r=1$ = +0.141 [+0.091, +0.195]; protected +0.115 [+0.067, +0.165]; U-MtE_balanced – mask +0.253 [+0.187, +0.325]; U-MtE_balanced – balanced −0.031 [−0.116, +0.054].

R12 Breast ultrasound feasibility audit (not used)

Source: docs/BREAST_MARKER_AUDIT.md

Data: BUSI (Kaggle aryashah2k, 647 benign/malignant with tumour masks) + BUS-BRA (Kaggle orville, 1,875 images, 1,064 patients, biopsy-proven pathology, tumour masks). Both show sonographer calipers, often inside the tumour (BUSI dotted measurement lines cross the lesion — the in-ROI case).

Detector: wtss.data.us_markers.marker_mask (thyroid-frozen '+'/dotted-line rules) + an optional thick 'x' template for GE-style asterisk calipers (shapes=("plus", "x"); thyroid default unchanged).

- v1 ('+' only): BUS-BRA "marker-free" images contained visible 'x' calipers in 7/8 audited images → unusable A0.
- v2 two-detector design (A0 = nothing found by a sensitive detector, thr 0.50/30; A1 = strict detector 0.62/45): cells A0 386/190, Trap A 171/104, Trap B 470/184 (benign/malignant), but a fresh 10+10 audit found ~3/10 A0 images with faint grey '+' calipers missed, and text false positives among A1. Decision: labels too noisy (A0 purity $\approx 70\%$) for a confirmatory trap; not pre-registered, no model fitted. Revisit with manual caliper annotation (≈ 50 images, SLAS few-shot probe) if needed.

R12.0.1 Attempt 2 (2026-09-27): annotation-free caliper probe — PARKED (definitive)

A DINOv2 patch-token caliper probe (wtss.slas) trained on thyroid + ovary detector masks reaches patch AUROC 0.983 on held-out thyroid/ovary images (scripts/data/breast_probe.py). Applied to BUSI + BUS-BRA with thresholds fixed before any breast label was inspected (A0: no patch ≥ 0.30 ; A1: ≥ 2 patches ≥ 0.70), only 19 of 2,522 images qualify as caliper/overlay-free (BUS-BRA 6, BUSI 13): nearly every breast image carries on-screen markings, so no artifact-free comparison group exists and the trap design is infeasible. Breast ultrasound is not used.

R13 Pre-registration: universal insertion-erasure (U-MtE) and ablations U1-U14

Source: docs/PREREGISTRATION_UNIVERSAL_I2E.md

Committed before any U-I2E / U-MtE result exists. The insertion library is wtss.synthetic.draw_generic_artifact (version "generic_artifact_v1"): one procedural family — thin curvilinear strokes, thick bands, solid/translucent patches, blobs, dashed lines, small glyphs; random colour (half greyscale), opacity 0.35–1, width, 1–3 elements, half centred inside the ROI. It contains **no ruler with ticks and no real hair**, and is **identical for every artifact, dataset and modality** (no per-artifact tuning). Eraser: paired-difference subspace, energy 0.90, max rank 64 (unchanged from I2E).

- **U-I2E:** erase the generic-insertion subspace from the original-image features; head on erased features.
- **U-MtE:** ROI-mask the image, insert generic structure *then* mask, erase that subspace in the masked view; head trained and tested on masked images (no artifact localisation, no artifact labels).
- **_balanced** variants add image-level artifact labels (group-balanced head).

R13.0.1 Claims (same protocols and bootstraps as the corresponding experiments)

- **U1** synthetic ruler (fixed and variable appearance), 100 % overlap: U-I2E – ERM reversed AUROC > 0 (CI).
- **U2** synthetic ruler, 100 % overlap: U-MtE – mask > 0 (removes masking's in-ROI harm) and U-MtE – ERM > 0 .

- **U3** real hair Trap A (author protocol, DINOv2 @518 and DermLIP @224): U-MtE – mask > 0 (CI) and U-MtE – ERM ≥ 0; Trap B: U-MtE – ERM > 0.
- **U4** no gaming: clean AUROC loss ≤ 0.02 vs ERM and correlated ≥ reversed – 0.02 wherever U1-U3 are claimed.
- **U5** (labels available) U-I2E_balanced / U-MtE_balanced ≥ balanced on reversed AUROC with clean AUROC ≥ balanced's. Outcomes are reported either way.

R13.0.2 Results (appended after the runs)

Synthetic ruler, DINOv2 @518, 100 % overlap (reversed AUROC Δ vs ERM): U-I2E +0.173 [+0.133, +0.218] (U1 □; clean 0.822 vs ERM 0.792; corr 0.903 ≥ rev 0.688); U-MtE –0.007 [–0.053, +0.040] (U2 □); U-I2E_balanced +0.291 [+0.258, +0.325] vs balanced +0.281, clean 0.824 vs 0.806 (U5 □); U-MtE_balanced +0.243. Real hair, author protocol, DINOv2 @518: U-I2E +0.046 [+0.025, +0.067] (A), +0.032 [+0.020, +0.044] (B); U-MtE – mask +0.052 [+0.044, +0.062] (A), +0.025 [+0.009, +0.040] (B); U-MtE – ERM +0.015 [–0.004, +0.033] (A), +0.132 [+0.105, +0.160] (B) (U3 □); U-I2E_balanced +0.238 (A) vs balanced +0.225 with clean 0.791 vs 0.775 (U5 □ A), +0.207 vs +0.209 (B); U-MtE_balanced +0.254 (B, best). No arm inverts correlated/reversed except none (U4 □).

R13.0.3 Amendment (2026-09-27, before fitting) — erase vs augment ablation (reviewer baseline)

Closest generic baseline to U-MtE: the SAME generic overlays used as training augmentation (random-erasing / occlusion-augmentation style) instead of subspace erasure. Arms: insert_aug (ERM view + overlaid copies of a random 50 % of training images) and mte_aug (masked view + masked overlaid copies). Claim **U7**: U-MtE – mte_aug

0 (reversed AUROC, Trap A) on ISIC 2019 hair and thyroid markers (DINOv2@518). Reported either way.

U7 results (results/ablation_u7.json; reversed AUROC, Trap A)

- Thyroid: U-MtE – mte_aug = **+0.207 [+0.182, +0.235]** (mte_aug – mask only +0.015); Trap B –0.003 [–0.020, +0.014].
- ISIC 2019 hair: U-MtE – mte_aug = **+0.045 [+0.037, +0.054]** (mte_aug – mask +0.007); Trap B +0.015 [–0.000, +0.030]. **U7 supported in both cohorts**: the same generic overlays help only when used to *estimate and erase* a subspace, not as training augmentation.

R13.0.4 Amendment (2026-09-27, before fitting) — annotation-free balancing (U8)

Balanced heads need image-level artifact labels A. Label-free replacement: an “overlay detector” (logistic, C=1) trained on (original, generic-overlay) feature pairs of the training pool is applied to the real training images; Otsu's threshold on its logits gives pseudo-A (wtss.heads.pseudo_artifact_labels). Arms: pbal (balanced ERM head with pseudo-A) and umte_pbal (U-MtE + pseudo-A balancing). Claims **U8a** umte_pbal – U-MtE > 0 and **U8b** pbal – ERM > 0 (reversed AUROC, Trap A), thyroid + ISIC hair, DINOv2@518. Diagnostic: AUROC of the detector score vs true A. Reported either way.

R13.0.5 Amendment (2026-09-27, before fitting) — disease-protected erasure (U9)

Motivation: on capsule endoscopy, template MtE removed disease evidence (debris resembles erosion fibrin). Method: W = orthonormal basis of logistic label weights (C=0.1, 5 bootstrap fits) on the ARTIFACT-FREE training images (A=0) in the masked view; erased subspace U' = orth((I – W W^T) U) (insertion.protect). Guarantee: w^TP(x) = w^Tx for w ∈ span(W). Arms mte_protect / mte_protect_balanced, both with the template library (where it exists) and the generic library. Claims **U9a** mte_protect – mte > 0 (Trap A) on capsule; **U9b** mte_protect – mask > 0 (Trap A) on capsule, thyroid and ISIC hair; **U9c** mte_protect – mte ≥ –0.02 on thyroid and ISIC hair (protection costs little where erasure already works). Reported either way.

U8 results (annotation-free pseudo-group balancing)

- Thyroid: overlay-detector AUROC vs true marker presence 0.84 (A) / 0.79 (B); umte_pbal reversed 0.663 vs U-MtE 0.622 (Trap A); pbal 0.409 vs ERM 0.289.
- ISIC hair: detector AUROC 0.57 / 0.63 only; umte_pbal 0.510 vs U-MtE 0.506 (no gain); pbal 0.498 vs ERM 0.491. U8 helps only where the generic overlays resemble the real artifact in feature space (markers, not hair).

Cross-cohort summary of the generic (artifact-agnostic) arms, DINOv2@518, Trap A reversed AUROC

cohort	ERM	mask	balanced	DFR	U-MtE	U-MtE + balanced
thyroid markers	0.289	0.400	0.675	0.738	0.622	0.788
capsule debris	0.587	0.684	0.775	0.805	0.569	0.857
ISIC hair	0.491	0.454	0.716	0.743	0.506	0.704
ISIC hair, DermLIP	0.670	0.585	0.813	0.780	0.509	0.648

U-MtE + balanced is best where the shortcut is carried by the artifact pixels (markers, debris); where the shortcut is mostly carried by non-pixel correlates (hair: site/age/sex), label-based group methods win.

R13.0.6 Amendment (2026-09-27, before fitting) — erasure + DFR (U10)

Arms: mask_dfr (DFR head on the masked view) and mte_dfr (DFR head on the masked, generic-erased view; eraser unchanged). Claim **U10**: mte_dfr – dfr > 0 and mte_dfr – mask_dfr > 0 (reversed AUROC, Trap A) on thyroid, capsule and ISIC hair (DINOv2@518, generic library). Reported either way.

U9 results (paired_deltas.csv in each results dir; reversed AUROC, Trap A)

- **U9a supported** (capsule): mte_protect – mte = +0.383 [+0.357, +0.414] (template), +0.105 [+0.092, +0.117] (generic).
- **U9b**: capsule not supported (–0.012 [–0.028, +0.003] template; –0.010 [–0.020, –0.001] generic); thyroid supported +0.217 [+0.174, +0.254]; ISIC hair supported +0.041 [+0.030, +0.054].
- **U9c supported**: protection cost vs mte –0.005 [–0.036, +0.024] (thyroid), –0.012 [–0.020, –0.003] (ISIC).
- **Balanced**: protect_balanced – mte_balanced +0.125 (capsule template), –0.009 (capsule generic), +0.016 (thyroid), –0.020 (ISIC). Interpretation: protection is a safety net — it removes the catastrophic failure when the artifact resembles the disease (capsule template –0.395 → –0.012 vs mask) at ≤ 0.012 cost elsewhere; it does not by itself beat masking when the artifact subspace overlaps the disease subspace.

U10 results (reversed AUROC, Trap A; paired_deltas.csv)

cohort	mte_dfr – dfr	mte_dfr – mask_dfr	mask_dfr – dfr
thyroid	+0.101 [+0.083, +0.120]	+0.007 [+0.002, +0.011]	+0.095 [+0.075, +0.116]
capsule	+0.080 [+0.054, +0.107]	+0.000 [–0.015, +0.014]	+0.080 [+0.062, +0.099]
ISIC hair	–0.039 [–0.066, –0.009]	+0.014 [–0.004, +0.032]	–0.053 [–0.076, –0.030]

U10 (mte_dfr > dfr and > mask_dfr) **not supported as a whole**: once a group-robust head (DFR) is used, the generic erasure adds ≤ 0.014 over masking + DFR. The gain over DFR comes from *masking* in marker/debris cohorts (+0.08–0.10) and masking costs –0.05 on hair — again location/type dependent.

R13.0.7 Amendment (2026-09-27, before fitting) — backbone-type replication (CNN)

ConvNeXt-Base (timm convnext_base.fb_in22k_ft_in1k_384, global-average-pooled, frozen) on the thyroid and capsule traps with the generic library; same arms as the DINOv2 runs. Claims as T6 / C5 / U7 / U9 (U-MtE – mask > 0, U-MtE – mte_aug > 0 in Trap A). Purpose: show the methods are not specific to ViTs. Reported either way.

Replication — MedSigLIP-448, thyroid, generic library (results/thyroid/medsiglip448_universal)

Trap A reversed AUROC: U-MtE – mask = +0.204 [+0.174, +0.237]; U-MtE – mte_aug = +0.177 [+0.148, +0.211] (U7); U-MtE_protect – mask = +0.262 [+0.241, +0.282]; U-MtE_balanced – balanced = +0.096 [+0.083, +0.110]. U-MtE_balanced: rev 0.792 / corr 0.800 / clean 0.798 — most robust arm (min(rev,corr)).

R13.0.8 Amendment (2026-09-27, before fitting) — label-free group robustness baseline: JTT (U11)

JTT (Liu et al. 2021): ERM → training errors at the clean-validation threshold → retrain with errors upweighted by $\lambda \in \{5, 20, 50\}$ (λ , C by clean-validation AUROC). Needs no artifact/group labels. Arms: jtt (ERM view), mask_jtt, umte_jtt (masked + generic-erased view). Claims **U11a** umte_jtt – jtt > 0 and **U11b** umte_jtt – U-MtE ≥ 0 (reversed AUROC, Trap A) on ISIC hair, thyroid and capsule (DINOv2@518). Among label-free arms, the most robust (min(rev, corr)) is reported per cohort. Reported either way.

U11 results — label-free arms (Trap A reversed AUROC; paired_deltas.csv)

cohort	JTT	mask+JTT	U-MtE	U-MtE+JTT	U-MtE – JTT	U-MtE+JTT – JTT
thyroid	0.423	0.479	0.622	0.369	+0.198 [+0.167, +0.231]	-0.054 [-0.069, -0.040]
capsule	0.607	0.769	0.569	0.771	-0.038 [-0.065, -0.013]	+0.164 [+0.098, +0.223]
ISIC hair	0.564	0.509	0.506	0.541	-0.058 [-0.089, -0.031]	-0.023 [-0.061, +0.007]

U11a/U11b not supported uniformly. Without any labels, the best arm is cohort-dependent: U-MtE where the shortcut is in distinct overlay-like pixels (calipers), U-MtE+JTT or mask+JTT where erasure alone removes disease signal (debris), and JTT on the unmasked view where masking harms and the shortcut is carried by correlates (hair).

CNN replication — ConvNeXt-Base, thyroid, generic library (results/thyroid/convnext384_universal)

Crossover (mask) +0.284 [+0.253, +0.315]; Trap A mask – ERM +0.003 (masking useless in-ROI) vs Trap B +0.288. U-MtE – mask +0.128 [+0.112, +0.145]; U-MtE – mte_aug +0.120 [+0.102, +0.139] (U7); U-MtE_protect – mask +0.195 [+0.153, +0.238]; U-MtE_balanced – balanced +0.084 [+0.051, +0.111]; U-MtE_balanced min(rev,corr) 0.745 (best; mask+DFR 0.722, DFR 0.669). The methods are not ViT-specific.

U9 extension — DermLIP, ISIC hair (results/spec_e13/dermlip224_spec_protect; post-registration arm set, same method)

Trap A reversed: ERM 0.670, mask 0.585, U-MtE 0.509, U-MtE_protect 0.589, U-MtE_protect_balanced 0.693, U-MtE_balanced 0.648, JTT 0.670 (balanced alone 0.813). Protection removes the erasure failure ($\approx$ mask), as on capsule; on DermLIP hair masking itself harms and label-based balancing is best.

R13.0.9 Amendment (2026-09-27, before fitting) — protected U-MtE in the controlled sweeps

Arms umte_protect / umte_protect_balanced added to the synthetic driver (same protection as U9, using the artifact-free training images of the correlated training environment). Run on capsule (DINOv2, MedSigLIP) and thyroid (DINOv2, MedSigLIP) sweeps, tag "protect". Claim: umte_protect – mask > 0 at $r = 1$ where U-MtE – mask ≤ 0 .

Protected U-MtE in controlled sweeps — capsule, MedSigLIP (results/synthetic/capsule/medsiglip448_debris)

At $r = 1$: umte_protect – mask = **+0.061 [+0.032, +0.092]** (claim supported; unprotected U-MtE – mask was -0.114); umte_protect – umte = +0.175 [+0.137, +0.217]; umte_protect_balanced – balanced = -0.002 [-0.048, +0.046].

- Capsule, DINOv2 (.../dino518_debris_corr_protect): umte_protect – mask = +0.118 [+0.077, +0.167]; – umte = +0.037 [+0.007, +0.072].
- Thyroid, MedSigLIP (.../medsiglip448_caliper_corr_protect): umte_protect – mask = -0.041 [-0.089, -0.005] (claim **not supported**); – umte = +0.035 [-0.013, +0.085]; umte_protect_balanced at $r=1$: +0.486 vs ERM. Summary: protection turned 1 of 2 MedSigLIP controlled failures into a win and never cost > 0.012 anywhere; the balanced (protected or not) variant is the only one that beats masking in every controlled sweep.

R13.0.10 Amendment (2026-09-27, before fitting) — SPLINCE baseline (U12)

wtss.heads.SpliceProjection: task-preserving oblique projection ($\text{Cov}(\text{PX}, A) = 0$, $\text{Cov}(\text{PX}, Y)$ preserved), fitted on the training environment with image-level artifact labels; arms splice (ERM view) and mask_splice (masked view). Compared with U-MtE_protect (label-free w.r.t. A) and U-MtE_balanced (uses A) on thyroid, capsule, ISIC hair (DINOv2) and chest drains (RAD-DINO). Claims **U12a** U-MtE_balanced – mask_splice > 0 and **U12b** U-MtE_protect – mask_splice ≥ 0 (reversed AUROC, Trap A), with gaming flags reported. Reported either way.

R13.0.11 Amendment (2026-09-27, before fitting) — location-adaptive selection ("auto", U13)

Per (trap, seed, fold) the candidate with the best validation score is chosen and its test predictions are used. Candidates (none trains on val_groups): erm, mask, balanced, mask_balanced, U-MtE, U-MtE_protect, U-MtE_balanced, U-MtE_protect_balanced, jtt, mask_jtt, umte_jtt. Score on val_groups (image-level A needed on validation only): min(AUROC(Y1A0 vs Y0A1), AUROC(Y1A1 vs Y0A0)). References: DFR and mask+DFR (trained on val_groups, so not candidates). Cohorts: thyroid, capsule, ISIC hair (DINOv2), chest drains (RAD-DINO). Claims **U13a** auto – mask > 0 (Trap A, reversed) in all four; **U13b** auto is within 0.02 (reversed and min(rev,corr)) of the best fixed candidate in each cohort; **U13c** auto $\geq$ DFR – 0.02 in each cohort. Reported either way. (scripts/analysis/adaptive_select.py)

U12 results — SPLINCE (Trap A; rev / corr (clean); min(rev, corr))

cohort	mask_splice	splice	U-MtE_balanced	U-MtE_pro
thyroid (DINOv2)	0.878 / 0.646 (0.777) †; 0.646	0.808 / 0.622 (0.724) †; 0.622	0.788 / 0.818 (0.807); 0.788	0.617 / 0.7
capsule (DINOv2)	0.951 / 0.734 (0.854) †; 0.734	0.901 / 0.640 (0.784) †; 0.640	0.857 / 0.928 (0.893); 0.857	0.673 / 0.9
ISIC hair (DINOv2)	0.831 / 0.601 (0.720) †; 0.601	0.841 / 0.600 (0.724) †; 0.600	0.704 / 0.852 (0.779); 0.704	0.495 / 0.8
drains (RAD-DINO)	0.516 / 0.623 (0.682); 0.516	0.536 / 0.639 (0.726); 0.536	0.460 / 0.857 (0.826); 0.460	0.269 / 0.9

† gaming flag (correlated < reversed - 0.02). In the three cohorts where artifact and label are strongly correlated in training, SPLINCE flips the shortcut (reversed □ correlated) — the same failure as unpaired LEACE — because Cov(X, A) and Cov(X, Y) are nearly collinear. **U12a supported by min(rev, corr)** in thyroid, capsule and hair; not on drains (SPLINCE 0.516 vs 0.460, at a clean cost of -0.19). U12b not supported on raw reversed AUROC (SPLINCE's reversed values are inflated by flipping).

U13 results — location-adaptive selection (results/adaptive_select_summary.csv; Trap A)

cohort	auto min(rev,corr)	main choice	auto - mask	auto - DFR	be
thyroid (DINOv2)	0.795	U-MtE_balanced (14/25)	+0.395 [+0.379, +0.411]	+0.057 [+0.033, +0.083]	U
capsule (DINOv2)	0.852	U-MtE_balanced (11/25)	+0.168 [+0.152, +0.183]	+0.047 [+0.016, +0.077]	U
ovary (DINOv2)	0.733	U-MtE_balanced (19/25)	+0.188 [+0.151, +0.226]	+0.010 [-0.022, +0.044]	U
ISIC hair (DINOv2)	0.705	balanced (17/25)	+0.251 [+0.227, +0.274]	-0.038 [-0.056, -0.019]	ba
drains (RAD-DINO)	0.538	balanced (5/5)	+0.299 [+0.280, +0.320]	-0.144 [-0.180, -0.109]	ba

U13a supported (5/5); U13b supported (5/5: within 0.02 of the best candidate); U13c supported 3/5 — fails on hair and drains, the correlate-carried regime, where DFR (not a candidate: it trains on the validation set) is best. Follow-up registered below: split-validation selection that admits DFR as a candidate.

R13.0.12 Amendment (before fitting) — split-validation adaptive selection (U14)

val_groups is split per image (stable hash "val_half") into two halves. DFR-type candidates dfr_half and mask_dfr_half are trained on half 0 only; ALL candidates (U13 set + these two) are scored on half 1 only with the U13 score. Claims: **U14a** auto_split - DFR ≥ -0.02 in all five cohorts (Trap A, reversed); **U14b** auto_split within 0.02 of the best fixed candidate (min(rev, corr)) in all five. (Image-level split: near-duplicate images can fall in both halves — small optimistic bias, reported as a limitation.) Reported either way.

U14 results — split-validation selection (logs/auto2_select.log; results/adaptive_select_split_summary.csv)

Trap A min(rev, corr): auto_split vs [U13 auto] vs best single arm — thyroid 0.798 [0.795] vs 0.788 (U-MtE_bal); capsule 0.871 [0.852] vs 0.885 (mask+DFR); ovary 0.715 [0.733] vs 0.752 (mask+DFR); ISIC hair 0.688 [0.705] vs 0.741 (DFR); drains 0.656 [0.538] vs 0.683 (DFR). auto_split - DFR (reversed): thyroid +0.060 [+0.037, +0.082], capsule +0.066 [+0.043, +0.088], ovary -0.007 [-0.041, +0.031], hair -0.055 [-0.083, -0.028], drains -0.026 [-0.044, -0.009]. **U14a supported 3/5** (fails hair, drains by -0.055 / -0.026); **U14b supported 2/5** (thyroid, capsule). Admitting DFR fixes the drain regime (+0.118 over U13) but halving the selection set makes choices noisier (ovary, hair worse). Neither selector is uniformly best; both beat masking everywhere (+0.17 to +0.42).

Correction (2026-09-27) — SPLINCE implementation The U12 results above used a simplified SPLINCE (oblique projection with both constraints but in the un-whitened metric). Checked against the paper (Holstege, Ravfogel & Wouters, NeurIPS 2025, arXiv 2506.10703, Theorem 1), the method minimises distortion in the whitened metric. SpliceProjection now implements that (verified: both constraints hold exactly, P is idempotent, and a numerical search over all admissible ranges finds no lower distortion). The simplified variant remains available (euclidean=True). U12 is re-run as tag *_splice_v2 with the same arms and claims; both versions are reported.

U12 re-run with the faithful SPLINCE (tag *_splice_v2) — supersedes the simplified-variant numbers above Trap A reversed / correlated (clean); min(rev, corr): | cohort | splice | mask_splice | U-MtE_bal | best | |---|---|---|---| | thyroid | 0.563 / 0.545 (0.562) | 0.628 / 0.596 (0.616); 0.596 | 0.788 / 0.818 (0.807); **0.788** | U-MtE_bal | | capsule | 0.795 / 0.613 (0.706) | 0.868 / 0.683 (0.784); 0.683 | 0.857 / 0.928 (0.893); **0.857** | U-MtE_bal | | ISIC hair | 0.639 / 0.564 (0.606); 0.564 | 0.628 / 0.562 (0.592) | 0.704 / 0.852 (0.779); 0.704 | balanced 0.716 | | drains | 0.609 / 0.609 (0.717); **0.609** | 0.577 / 0.588 (0.700) | 0.460 / 0.857 (0.826) | DFR 0.683 | The faithful (whitened, minimal-distortion) SPLINCE no longer flips the shortcut in thyroid or on drains but removes much disease signal (clean AUROC 0.56-0.78 vs 0.69-0.87 for masking) and still flips it on capsule. **U12a (U-MtE_bal > SPLINCE by min(rev, corr)) supported in 3/4** (thyroid, capsule, hair); on drains SPLINCE is higher (0.609 vs 0.460), below

DFR (0.683). Name correction: the method is **SPLINCE** (Simultaneous Projection for LINear concept removal and Covariance prEservation; arXiv 2506.10703 abstract). Arms keep the internal names splice / mask_splice.

R14 Pre-registration: SLAS (few-shot patch suppression)

Source: docs/PREREGISTRATION_SLAS.md

Committed before any SLAS model is evaluated on a trap. Protocol, cohorts, seeds, folds, thresholds, bootstrap: unchanged from docs/REPLICATION_SPEC.md (E13, ISIC 2019 hair) and docs/PREREGISTRATION_THYROID_TRAPS.md.

R14.0.1 Motivation (evidence available before this registration)

Pixel masking cannot remove an artifact that lies inside the ROI (E13 Trap A: mask – ERM < 0). Pixel artifact detectors are poor on thin in-ROI artifacts (hair IoU 0.22). Localisation-only test (results/slas/ localisation_isic_hair.json; no trap outcome used): a linear probe on DINOv2@518 patch tokens trained on real Wegley hair/ruler masks localises hair on held-out images with patch AUROC 0.977 (0.969 inside the lesion); a probe trained on synthetic generic overlays only reaches 0.64 — so SLAS uses a few real annotated images.

R14.0.2 Method (frozen)

1. Probe: logistic regression (C=1, balanced) on DINOv2 ViT-B/14 @518 patch tokens; patch label = artifact coverage ≥ 0.2 (positive) / 0 (negative). Trained on $k = 50$ annotated images drawn (seed 20260927) from the donor pool ($0.1 \leq r < 0.5$; hair_frac ≥ 0.02), which is disjoint from both trap pools and shares no leakage group with them. The same procedure is used for every artifact type — only the k annotated images change.
2. Representation: L2-normalised mean of patch tokens. Patches with probe probability $\geq \tau = 0.5$ are dropped; ROI patches = $\geq 50\%$ lesion pixels.
3. Arms (heads, thresholds and validation exactly as E13): tok_erm (all patches) · tok_mask (ROI patches) · slas (probe-clean patches) · mts (ROI $\cap$ probe-clean; “mask-then-suppress”) · mts_oracle (ROI minus real-mask patches; ceiling, not a method) · tok_balanced · mts_balanced. Secondary: $k \in \{10, 25, 100\}$ (learning curve); thyroid markers (probe trained on 50 donor images with detector masks); DermLIP / MedSigLIP token versions if the primary is positive.

R14.0.3 Claims (reversed-test AUROC, 95 % hierarchical paired CI, seeds as clusters)

- **S0 (sanity, token space reproduces the thesis):** tok_mask – tok_erm < 0 in Trap A and > 0 in Trap B.
- **S1 (primary):** mts – tok_mask > 0 in Trap A.
- **S2:** slas – tok_erm > 0 in Trap A.
- **S3 (no harm out of ROI):** mts – tok_mask CI lower bound > –0.01 in Trap B.
- **S4 (no gaming):** clean-AUROC loss of mts vs tok_mask ≤ 0.02 in both traps.
- **S5:** mts_balanced – tok_balanced > 0 in Trap A.
- **S6:** mts reaches $\geq 50\%$ of the oracle gain (mts_oracle – tok_mask) in Trap A. All outcomes are reported whichever way they go. If S0 fails (token pooling does not reproduce the in-ROI harm), SLAS results are reported as exploratory only.

R14.0.4 Results — ISIC 2019 hair, DINOv2@518, $k = 50$ (results/slas/isic_k50/bootstrap.csv)

Reversed-test AUROC deltas [95 % CI]:

- **S0 not supported:** tok_mask – tok_erm = +0.030 [+0.019, +0.040] in Trap A (vs +0.138 [+0.124, +0.151] in Trap B). Token-level ROI pooling does not *harm* in Trap A; the location gap (B $\square$ A) persists. Per the registration, SLAS results on this cohort are therefore **exploratory**.
- S1 supported: mts – tok_mask = +0.022 [+0.012, +0.031] (Trap A).
- S2 not supported: slas – tok_erm = +0.008 [–0.002, +0.017].
- S3 supported: mts – tok_mask = +0.006 [+0.002, +0.011] in Trap B. S4 supported (clean +0.003).
- S5 supported: mts_balanced – tok_balanced = +0.051 [+0.028, +0.072]; mts_balanced has the best reversed (0.749) and clean (0.812) AUROC of all token arms.

- S6 supported, but the ceiling is tiny: removing EVERY real-hair patch (oracle) gains only +0.015 [+0.002, +0.024] over tok_mask. **Interpretation.** On real ISIC hair the residual shortcut after masking is mostly *not* carried by the hair patches themselves (oracle removal $\approx$ +0.015), consistent with the earlier diagnostic that $\sim$ 55 % of the hair shortcut is carried by metadata correlates (site / age / sex). No artifact-removal method can exceed that ceiling here; group re-weighting (+0.21) is what moves this trap.

R14.0.5 Results — thyroid ultrasound markers, DINOv2@518, k = 50 detector-annotated donors (results/slas/thyroid_k50)

- S0 **not supported**: tok_mask – tok_erm = +0.059 [+0.050, +0.069] in Trap A vs +0.232 [+0.212, +0.252] in Trap B (location gap persists; no harm) → exploratory, as registered.
- S1 supported: mts – tok_mask = +0.081 [+0.076, +0.087] (Trap A); larger than the detector-mask “oracle” (+0.047 [+0.045, +0.050]) — the probe also removes marker patches the rule-based detector misses.
- S2 not supported in size terms: slas – tok_erm = +0.007 [+0.006, +0.008].
- S3 supported (+0.014 [+0.007, +0.021] in Trap B); S4 supported (clean +0.025).
- S5 supported: mts_balanced – tok_balanced = +0.070 [+0.053, +0.089]; mts_balanced: reversed 0.758, clean 0.796 (best token arm). For reference the CLS-feature MtE_balanced (real-marker templates) reached 0.818 / 0.812.

R15 Pre-registration: end-to-end fine-tuning

Source: docs/PREREGISTRATION_FINETUNE.md

Committed before any fine-tuned model is trained. Purpose: test whether the frozen-backbone findings hold when the whole network is trained.

- Traps: the thyroid (and capsule) spec traps of docs/PREREGISTRATION_THYROID_TRAPS.md / _CAPSULE_TRAPS.md; seed 42, folds 0-4 (5 bootstrap clusters; image-level paired bootstrap within folds).
- Model: ResNet-50 (ImageNet, timm), 224 px, AdamW, one-cycle LR 1e-4 (head $\times$ 10), 8 epochs, flips/rot90 augmentation, fixed epochs (no early stopping); threshold on clean validation.
- Arms: erm; mask (train+test on ROI-masked images); balanced; mask_balanced; mte_ft (masked images + generic overlay inserted then masked; loss = BCE + $\|\text{normalise}(g(x)) - \text{normalise}(g(x^+))\|^2$ on the penultimate embedding — the end-to-end analogue of U-MtE; no artifact label or mask).
- Claims (reversed AUROC, 95 % CI): **F1** crossover — (mask – ERM)_B > (mask – ERM)_A (reported per trap); **F2** mte_ft – mask > 0 in Trap A; **F3** mask_balanced – balanced > 0 in Trap A. Reported either way.

R15.0.1 Results — thyroid, ResNet-50 fine-tuned end-to-end (results/finetune/thyroid/resnet50; 5 folds)

Trap A reversed / corr (clean): ERM 0.431 / 0.744 (0.587); mask 0.536 / 0.827 (0.691); balanced 0.599; mask_balanced 0.747 / 0.677 (0.721); **mte_ft 0.679 / 0.703 (0.698)**.

- **F1 not supported**: crossover (mask – ERM)_B – (mask – ERM)_A = +0.057 [–0.021, +0.130] (direction as predicted: +0.161 in B vs +0.105 in A; CI includes 0 with 5 folds).
- **F2 supported**: mte_ft – mask (Trap A) = +0.143 [+0.099, +0.188]; no gaming (corr 0.703 $\geq$ rev 0.679); clean +0.007 vs mask. Trap B: –0.034 [–0.062, –0.008].
- **F3 supported**: mask_balanced – balanced (Trap A) = +0.147 [+0.104, +0.192]. min(rev, corr), Trap A: mte_ft 0.679 $\approx$ mask_balanced 0.677 > mask 0.536 > balanced 0.599 — the label-free end-to-end U-MtE matches the best label-using arm.

R15.0.2 Results — capsule, ResNet-50 fine-tuned (results/finetune/capsule/resnet50; 5 folds)

Trap A reversed / corr (clean): ERM 0.495 / 0.900 (0.719); mask **0.211** / 0.915 (0.596); balanced 0.670; mask_balanced 0.687 / 0.780 (0.754); mte_ft 0.190 / 0.904 (0.567).

- **F1 supported (strongly)**: crossover = +0.584 [+0.469, +0.691]; masking HARMS in-ROI end-to-end (–0.284 [–0.347, –0.226]) and helps out-of-ROI (+0.301 [+0.225, +0.380]).
- **F2 not supported**: mte_ft – mask (A) –0.021 [–0.039, –0.003]; mte_ft also lowers clean AUROC (debris $\approx$ fibrin regime, as in the frozen analysis).
- F3 not supported: mask_balanced – balanced (A) +0.017 [–0.070, +0.105].

R15.0.3 Amendment (before training): ovarian ultrasound (MMOTU) fine-tuning with the same design and claims F1-F3.

R15.0.4 Results — ovary, ResNet-50 fine-tuned (results/finetune/ovary/resnet50; 5 folds; small cohort → wide CIs)

Trap A reversed / corr (clean): ERM 0.457 / 0.671 (0.576); mask 0.556 / 0.570 (0.578); balanced 0.602; mask_balanced 0.593; **mte_ft 0.649 / 0.743 (0.687)** — best reversed, correlated and clean AUROC. F1 not supported (crossover +0.136 [−0.085, +0.329]); F2 not supported at 95 % (mte_ft − mask +0.094 [−0.025, +0.217]; mte_ft − ERM +0.192 [+0.103, +0.278]); F3 not supported (−0.009). Underpowered (1,202 images).

R15.0.5 Amendment (before training): ViT-S/16 (timm vit_small_patch16_224.augreg_in21k_ft_in1k), lr 3e-5, thyroid, same

design and claims F1-F3 (transformer replication of the ResNet-50 check).

R15.0.6 Results — thyroid, ViT-S/16 fine-tuned (results/finetune/thyroid/vit_small_patch16_224.augreg_in21k_ft_in1k)

Trap A reversed / corr (clean): ERM 0.377 / 0.860 (0.640); mask 0.465 / 0.868 (0.688); balanced 0.693; mask_balanced 0.710 / 0.788 (0.760); mte_ft 0.475 / 0.878 (0.695).

- **F1 supported:** crossover +0.168 [+0.074, +0.265] (mask − ERM: +0.088 in-ROI vs +0.256 out-of-ROI).
- **F2 not supported:** mte_ft − mask (A) +0.009 [−0.021, +0.040] (ResNet-50: +0.143) — the end-to-end invariance analogue does not transfer to ViT-S at this learning rate / epoch budget.
- F3 not supported: mask_balanced − balanced (A) +0.017 [−0.023, +0.055].

R16 Pre-registration: natural (unaltered) test distributions

Source: docs/PREREGISTRATION_NATURAL.md

Committed before any model is fitted for this analysis.

- Thyroid (TN3K + TNCD): train on the official trainval split with its natural association between in-ROI calipers and the label (in-ROI calipers: 37.6 % of benign vs 24.1 % of malignant images); test on the untouched official test split (614 images). 5 seeds of a group-safe 80/20 train/validation split = bootstrap clusters.
- A (for group-using arms) = in-ROI caliper (≥ 15 px, $r \geq 0.5$).
- Metrics on the natural test set: AUROC; cross-group AUROC (min over the two counter-/pro-shortcut pairings of groups; the natural analogue of min(reversed, correlated)); AUROC within the in-ROI-caliper subgroup.
- Arms: erm, mask, balanced, mask_balanced, U-MtE, U-MtE_protect, U-MtE_balanced, U-MtE_protect_balanced, DFR, mask+DFR, JTT (DINOv2@518, generic library).
- Claims: **N1** cross-group AUROC(U-MtE_protect) > cross-group AUROC(mask) and **N2** overall AUROC(U-MtE_protect) $\geq$ AUROC(mask) − 0.01 (no cost on the natural distribution). Reported whichever way they go.

R16.0.1 Results — thyroid, DINOv2@518 (results/natural/thyroid_dino518; 5 seeds; official TN3K test, 614 images)

arm	AUROC	cross-group AUROC	AUROC within in-ROI-caliper images
ERM	0.736	0.679	0.781
mask	0.731	0.576	0.738
U-MtE	0.742	0.627	0.760
U-MtE_protect	0.715	0.518	0.723
balanced	0.716	0.713	0.768
U-MtE_balanced	0.731	0.664	0.755
DFR	0.697	0.625	0.749

- Under the natural distribution, masking lowers the cross-group AUROC by 0.10 (0.679 → 0.576): the in-ROI caliper is retained while other evidence is removed — the thesis effect without any resampling.
- **N1 not supported** (U-MtE_protect 0.518 < mask 0.576); unprotected U-MtE recovers half the loss (0.627).

- **N2 not supported** (U-MtE_protect AUROC – mask = $-0.017 [-0.027, -0.007]$); unprotected U-MtE has the best overall AUROC (0.742). Group balancing gives the best cross-group AUROC (0.713). Subset bootstraps (results/natural/thyroid_dino518/subset_boot.json; post hoc, secondary):
- Hard pairs (malignant WITH in-ROI caliper vs benign WITHOUT): mask – ERM = **$-0.102 [-0.139, -0.066]$** ; U-MtE – mask = $+0.051 [+0.033, +0.072]$; balanced – mask = $+0.142 [+0.107, +0.178]$.
- Easy pairs (malignant without vs benign with): mask – ERM = $+0.048 [+0.019, +0.079]$. Masking helps where the natural shortcut agrees with the label and harms where it disagrees — the in-ROI shortcut is amplified by masking in unaltered data.

R16.0.2 Amendment (2026-09-27, before fitting) — more natural-distribution tests

Same design and arms as the thyroid analysis (scripts/run_natural.py):

- **ISIC 2019 leave-one-hospital-out** (isic_BCN, isic_HAM, isic_MSK): train on the other two sources (natural association: in-lesion hair 31 % of melanomas vs 20 % of benign lesions, similar in every source), test on the held-out source; A = in-lesion hair (> 30 native px, $r \geq 0.5$); all QC-passed images.
- **Capsule**: SEE-AI, per seed a group-safe 80/20 split of frame-block groups; A = debris ≥ 10 % with $r \geq 0.5$ (natural association: 11.6 % of erosions vs 20.9 % of polyp-like lesions). The “hard” (shortcut-conflicting) pairing is determined from the test-set association direction. Claims: **N3** mask – ERM < 0 on hard pairs (thesis effect in natural data) and **N4** U-MtE – mask > 0 on hard pairs, per cohort. Reported either way.

R16.0.3 Results — amendment cohorts (DINOv2@518, 5 seeds; results/natural/*_dino518/natural_boot.js)

Hard = shortcut-conflicting pairs (direction from the test-set association). Δ reversed-free AUROC on hard pairs:
| cohort | P(A|Y=1) / P(A|Y=0) | mask – ERM (N3: < 0) | U-MtE – mask (N4: > 0) | U-MtE_bal – mask | best cross-group AUROC | |---|---|---|---| | thyroid (official split) | 0.241 / 0.376 | **$-0.102 [-0.139, -0.066]$** | **$+0.051 [+0.033, +0.072]$** | — | balanced 0.713 | | ISIC, HAM held out | 0.274 / 0.184 | $+0.053 [+0.037, +0.068]$ | **$+0.023 [+0.017, +0.032]$** | $+0.065 [+0.056, +0.075]$ | **U-MtE_bal 0.786** | | ISIC, BCN held out | 0.362 / 0.247 | **$-0.016 [-0.027, -0.005]$** | $+0.003 [+0.000, +0.006]$ | $+0.034 [+0.029, +0.039]$ | balanced 0.735 $\approx$ U-MtE_bal 0.734 | | ISIC, MSK held out | 0.114 / 0.076 | **$-0.056 [-0.084, -0.029]$** | $-0.008 [-0.015, -0.002]$ | $+0.041 [+0.029, +0.053]$ | balanced 0.734 | | capsule | 0.101 / 0.232 | $+0.021 [+0.000, +0.044]$ | $+0.000 [-0.010, +0.009]$ | $+0.014 [+0.003, +0.025]$ | **U-MtE_bal 0.947** | **N3 supported in 3/5** (thyroid, BCN, MSK): on unaltered data masking harms the shortcut-conflicting cases whenever it does. **N4 supported in 2/5** (thyroid, HAM). U-MtE_balanced improves on masking for hard pairs in every cohort where measured (4/4) and gives the best or tied-best cross-group AUROC in 3/5; plain balancing is best in 2/5.

R16.0.4 Integrity note and clinical metrics (2026-09-27, afternoon)

- results/natural/thyroid_dino518/predictions.csv.gz failed a gzip CRC check (the only one of 146 result archives; probably two writers at once). It was regenerated with the same deterministic run: every per-seed AUROC and every bootstrap CI above is identical. New from the regenerated run: hard pairs U-MtE_bal – mask = $+0.088 [+0.071, +0.108]$.
- Clinical secondary metrics (AUROC, AUPRC, Brier, ECE, sensitivity/specificity at the validation-fixed operating point; scripts/analysis/clinical_metrics.py → results/natural/clinical_metrics.csv, paper/tables/natural_clinical.tex). Thyroid (official split): masking lowers sensitivity from 0.734 to 0.347 while specificity rises from 0.619 to 0.888 (balanced accuracy 0.677 → 0.618); U-MtE recovers part of it (0.425 / 0.854) with the best AUPRC (0.638) and Brier score (0.203). ISIC BCN: masking worsens Brier (0.239 → 0.286) and ECE (0.279 → 0.332). HAM, MSK and capsule: masking improves every metric, matching the AUROC results.

R17 Sensitivity to artifact-label definitions

Source: docs/SENSITIVITY_ARTIFACT_LABELS.md

Motivation: marker masks come from an audited rule-based detector (no clinician audit available). If conclusions depended on borderline detections they would be fragile. Variants (thyroid and ovary, DINOv2@518, generic library):

- **main**: A1 ≥ 15 detected px; Trap A $r \geq 0.5$; Trap B $r < 0.1$.
- **large-markers**: A1 ≥ 50 px (large detections are almost certainly real calipers).

- **strict-location:** Trap A $r \geq 0.7$; Trap B $r < 0.05$. (A 5-px variant changes ≤ 1 image per cell and is omitted.) Arms: erm, mask, balanced, U-MtE, U-MtE_protect, U-MtE_balanced. Checked claims: P1 crossover > 0 ; P2 U-MtE_protect – mask > 0 ; P4 U-MtE_balanced – balanced > 0 (Trap A). Robust = same sign and CI excluding 0 in every variant. Command: `scripts/run_thyroid_traps.py --cohort {thyroid,ovary} --generic --min_px/--rA/--rB --tag sens_*`.

R17.0.1 Capsule Trap B definition (found during figure review, 2026-09-27)

$r = |\text{debris} \cap \text{ROI}| / |\text{debris}|$ can be small for a large debris blob that still covers part of the lesion: 26.9 % of capsule Trap B images have debris covering > 20 % of the lesion box (median 10.5 %; Trap A median 41 %). This makes the capsule crossover conservative. Variant **capsule-strict-B:** Trap B additionally requires lesion coverage < 5 % (`--max_cover_B 0.05`). Registered before running; same arms and claims.

R17.0.2 Results (DINOv2@518, generic library; Trap A reversed AUROC [95 % CI])

cohort	variant	crossover	U-MtE_protect – mask	U-MtE_bal – balanced
thyroid	main	+0.230 [+0.204, +0.255]	+0.217 [+0.174, +0.257]	+0.112 [+0.090, +0.134]
thyroid	large markers (≥ 50 px)	+0.262 [+0.228, +0.294]	+0.282 [+0.243, +0.318]	+0.132 [+0.114, +0.148]
thyroid	strict location (0.7 / 0.05)	+0.215 [+0.184, +0.244]	+0.252 [+0.221, +0.279]	+0.136 [+0.105, +0.165]
ovary	main	+0.170 [+0.116, +0.224]	+0.043 [+0.025, +0.061]	+0.028 [–0.001, +0.058]
ovary	large markers	+0.186 [+0.135, +0.238]	+0.067 [+0.049, +0.082]	+0.025 [–0.011, +0.060]
ovary	strict location	+0.195 [+0.143, +0.246]	+0.046 [+0.023, +0.071]	–0.012 [–0.041, +0.017]

Robust: the crossover and U-MtE_protect – mask hold (CI excludes 0) in every variant and grow under stricter definitions, as expected if detector errors dilute rather than create the effects. The ovary balanced claim is not significant in any variant (as in the main analysis). | capsule | main | +0.368 [+0.340, +0.397] | –0.010 [–0.020, –0.001] | +0.082 [+0.064, +0.100] | | capsule | strict Trap B (lesion coverage < 5 %) | +0.259 [+0.213, +0.302] | –0.010 [–0.020, –0.001] | +0.082 [+0.064, +0.100] | Capsule strict-B: the crossover remains strongly positive but is smaller than in the main analysis (Trap B masking gain +0.355 vs +0.465; the strict Trap B has far fewer benign images, 32 vs 232) — so the expectation that the main definition was conservative was **not** confirmed; the claim itself is robust. (Trap A arms are unchanged by the Trap B definition.)

R18 Artifact-mask visual audit

Source: `docs/ARTIFACT_MASK_AUDIT.md`

Judging whether an on-screen caliper/overlay is present is a non-clinical visual task. Panels of 40 random images per cell (seed 101; `scripts/audit/make_audit_panels.py`, ids saved next to the panels) were inspected at 200 px; very faint marks may be missed at that size (limitation). An earlier 20-image audit is in `THYROID_MARKER_AUDIT.md`.

cohort	cell	criterion	result
Thyroid (TN3K)	detected (A1)	detection lies on a real caliper / measurement mark	38/40 (95 %); 2 border fals
Thyroid	marker-free (A0)	no visible caliper	38–39/40 (95–97 %); depth
Ovary (MMOTU)	detected (A1)	an on-screen overlay is present	40/40; caliper specifically :
Ovary	marker-free (A0)	no visible caliper	37/40 (92 %)
Ovary	marker-free (A0)	no overlay of any kind (arrows, boxes)	~29/40 (72 %); text labels
Capsule (SEE-AI)	probe debris masks	vs expert masks (540 held-out frames)	pixel accuracy 0.92, IoU 0.
ISIC 2019	Wegley hair/ruler masks	published semi-automatic masks	not re-audited

Implications: label noise on A dilutes the constructed correlation (biasing effects toward zero), it does not create the location effect. The sensitivity analysis (`docs/SENSITIVITY_ARTIFACT_LABELS.md`) re-runs the thyroid and ovary claims with large-marker-only and strict-location definitions. For the ovary cohort, “marker-free” means caliper-free, not overlay-free — stated as a limitation in the paper.

R19 Pre-registration: LaMa inpainting comparison

Source: `docs/PREREGISTRATION_INPAINT_LAMA.md`

Committed before any model is fitted on LaMa-inpainted images.

- Method: LaMa (big-lama; Suvorov et al., WACV 2022) inpaints the artifact pixels (artifact mask dilated 3 px) of every image with a non-empty mask (thyroid/ovary: detected calipers; capsule: probe debris masks; ISIC: Wegley hair/ruler masks) — an ORACLE w.r.t. the artifact masks, like the classical Telea arm. Cache: scripts/data/precompute_lama.py.
- Arms: lama (ERM head on LaMa-inpainted images), mask_lama (inpaint, then ROI-mask), compared with Telea inpaint, mask, U-MtE_protect and U-MtE_balanced (existing runs; same deterministic environments → paired).
- Claims (reversed AUROC, Trap A): **L1** lama – mask; **L2** U-MtE_protect – mask_lama; reported with CIs, no directional prediction (LaMa uses oracle masks; U-MtE uses none). Reported either way.

R19.0.1 Results — DINOv2@518, Trap A reversed AUROC (results/{thyroid,ovary}/dino518_lama; results/lama_comparison.json)

	thyroid	ovary
LaMa – ERM	+0.136 [+0.130, +0.142]	+0.158 [+0.141, +0.176]
LaMa – mask	+0.025 [+0.009, +0.040]	+0.027 [−0.017, +0.069]
mask•LaMa – mask	+0.226 [+0.213, +0.238]	+0.123 [+0.089, +0.155]
L2 U-MtE_protect – mask•LaMa	−0.009 [−0.045, +0.023]	−0.080 [−0.108, −0.053]
U-MtE_bal – mask•LaMa	+0.162 [+0.152, +0.173]	+0.067 [+0.037, +0.095]

Modern inpainting of the (detected) caliper pixels followed by masking is a strong baseline: it needs artifact masks, which U-MtE does not. U-MtE_protect matches it on thyroid (n.s. difference) and is below it on ovary; with image-level labels, U-MtE_bal beats it on both. Reversed / corr (clean) of mask•LaMa: thyroid 0.626 / 0.825 (0.735), ovary 0.668 / 0.812 (0.741).

ISIC hair (results/spec_e13/dino518_spec_lama) LaMa – mask +0.145 [+0.116, +0.177]; mask•LaMa – mask +0.127 [+0.116, +0.138]; U-MtE_protect – mask•LaMa −0.086 [−0.098, −0.074]; U-MtE_bal – mask•LaMa +0.123 [+0.105, +0.140]. With the (semi-automatic, published) hair masks, LaMa inpainting is clearly better than mask-free U-MtE on hair; with image-level labels U-MtE_bal is best.

R20 Pre-registration: text-prompted erasure and zero-shot reliance

Source: docs/PREREGISTRATION_TEXT_PROMPT.md

Committed before any model is fitted or any zero-shot score is computed for this analysis.

R20.0.1 Motivation

MedSigLIP and DermLIP embed images and text in one space. A text description of the artifact (“a thyroid ultrasound image with measurement calipers”) therefore gives an artifact direction with **no image, mask or label of the artifact** — a modern annotation-free alternative (cf. biased-prompt debiasing, Chuang et al. 2023) to U-MtE's synthetic overlays.

R20.0.2 Method (frozen; wtss.text_directions)

- For each cohort: 4 neutral templates × 4 artifact phrases; direction set = {emb(template + phrase) − emb(template)}; SVD, rank k = smallest with 90 % energy (max 8). Prompts are fixed in PROMPTS (not tuned on any result).
- Erasure on the cached joint-space image features: $x \mapsto x - U U^T (x - \mu)$, μ = mean of training features.
- Arms: text_erase (unmasked), mask_text_erase, mask_text_erase_balanced, mask_text_erase_protect (disease protection as in U9). Compared with mask, U-MtE (mte), U-MtE_protect, U-MtE_balanced on the same deterministic environments.
- Backbones/cohorts: MedSigLIP — thyroid, capsule, ovary; DermLIP — ISIC 2019 hair (spec traps).

R20.0.3 Zero-shot reliance (no training)

Zero-shot score = $\cos(x, t_{\text{pos}}) - \cos(x, t_{\text{neg}})$ with the class prompts in `PROMPTS[cohort]["classes"]`, on the unmasked and masked views of the trap test environments. Question: does a foundation model *without any task training* already rely on in-ROI artifacts (correlated vs reversed AUROC gap), and does masking change that?

R20.0.4 Claims (reversed AUROC, Trap A, 95 % CI)

- **V1** mask_text_erase – mask > 0 (text-prompted erasure removes in-ROI shortcut).
- **V2** mask_text_erase – U-MtE: reported with CI, no directional prediction.
- **V3** mask_text_erase_balanced – balanced > 0.
- **Z1** zero-shot: report correlated – reversed AUROC gap (Trap A) per cohort (descriptive). Reported whichever way they go.

R20.0.5 Results (Trap A reversed AUROC [95 % CI]; results/*/medsiglip448_text, results/spec_e13/dermlip)

cohort (backbone)	V1 mask_text_erase – mask	U-MtE – mask	V2 mask_text_erase – U-MtE	V3 text_bal
thyroid (MedSigLIP)	+0.005 [+0.003, +0.008]	+0.204 [+0.174, +0.237]	–0.199 [–0.230, –0.170]	+0.056 [+0.026, +0.086]
ovary (MedSigLIP)	+0.008 [–0.001, +0.021]	+0.215 [+0.182, +0.246]	–0.207 [–0.235, –0.178]	–0.010 [–0.040, +0.020]
capsule (MedSigLIP)	+0.001 [–0.000, +0.003]	–0.236 [–0.252, –0.221]	+0.238 [+0.222, +0.253]	+0.066 [+0.036, +0.096]
ISIC hair (DermLIP)	+0.004 [–0.002, +0.011]	–0.076 [–0.094, –0.060]	+0.080 [+0.064, +0.097]	–0.112 [–0.142, –0.082]

- **V1 essentially not supported:** text-prompted erasure is a near no-op ($\leq +0.008$; significant only on thyroid). The text-space artifact direction does not coincide with the image-space direction along which the artifact moves the representation (modality gap), so erasing it leaves the shortcut intact.
- V2: U-MtE is far better where the artifact is a distinct overlay (thyroid, ovary); text erasure “wins” on capsule and hair only because it does nothing while U-MtE (unprotected) removes disease signal there.
- V3: supported 2/4. New replication: U-MtE – mask on ovary with MedSigLIP = +0.215.

R20.0.6 Zero-shot reliance (Z1; results/zero_shot/)

No task training. Correlated – reversed AUROC gap (unmasked, Trap A): thyroid +0.066, capsule –0.398, ovary –0.126, hair +0.081 — the foundation models' zero-shot scores are already swayed by artifacts (capsule debris by 0.40 AUROC). Masking effect on zero-shot reversed AUROC, in-ROI vs out-of-ROI (post hoc crossover test): thyroid +0.013 vs +0.116 (crossover +0.103 [+0.079, +0.127]); capsule +0.007 vs +0.056 (+0.049 [+0.024, +0.076]); ovary –0.235 vs –0.054 (+0.181 [+0.135, +0.228]); hair (DermLIP) –0.078 vs –0.027 (+0.051 [+0.031, +0.071]). **The location law holds even without any training** (4/4 crossovers > 0). Caveat: MedSigLIP zero-shot thyroid classification is at chance (clean AUROC 0.45), so that row reflects artifact effects on an uninformative score.

R21 Statistical plan and multiplicity

Source: docs/STATISTICAL_PLAN.md

R21.0.1 Estimation

Every comparison is a hierarchical paired bootstrap (10,000 replicates): clusters = training seeds (or folds for the chest-drain and fine-tuning analyses), paired test images resampled within clusters, AUROC differences computed within clusters and averaged. Two-sided bootstrap p-values (percentile method, floored at 1/10,000) are reported alongside 95 % CIs (`wtss.stats.boot_p`).

R21.0.2 Primary family (16 tests; results/PRIMARY_CLAIMS.md)

Each test below was individually pre-registered before its data were analysed (docs/PREREGISTRATION_*.md). The grouping into one *primary family* for multiplicity control was defined on 2026-09-27 after results were available, to answer the question a reviewer will ask (“are the headline claims robust to multiple testing?”); we therefore apply the conservative Holm correction over the whole family and report BH for reference.

- P1 location crossover, [mask – ERM]_out-of-ROI – [mask – ERM]_in-ROI > 0: ISIC hair, thyroid, capsule, ovary.
- P2 protected U-MtE – mask > 0 (in-ROI): ISIC hair, thyroid, capsule, ovary.
- P3 erase – augment (same generic overlays) > 0 (in-ROI): ISIC hair, thyroid, capsule, ovary.
- P4 U-MtE + balanced – balanced > 0 (in-ROI): ISIC hair, thyroid, capsule, ovary. All on DINOv2 ViT-B/14 @518 (the backbone with every cohort); other backbones are replications.

R21.0.3 Secondary / exploratory

Everything else (other backbones, controlled sweeps, JTT, SPLINCE, DFR combinations, adaptive selection, SLAS, fine-tuning, chest drains) is secondary: reported with unadjusted 95 % CIs and interpreted as replication or exploration, not as confirmatory evidence.

R21.0.4 Robustness summary

Reversed-test AUROC is always read together with correlated-test and clean AUROC; min(reversed, correlated) is the (post hoc) robustness summary, and arms with correlated < reversed – 0.02 are flagged as shortcut-flipping.

R22 Theory (linear-Gaussian account)

Source: docs/THEORY.md

Numerically verified in scripts/analysis/theory_sim.py (max |simulation – theory| = 0.009 AUROC over 29 settings; results/theory/theory_sim.csv; figure paper/figures/theory.pdf).

R22.0.1 Setup

A frozen encoder maps an image to $z = (z_s, z_a, \varepsilon)$: a disease block whose class-mean difference has signal-to-noise ratio S , an artifact block with SNR A per unit of artifact presence, and independent noise. Training uses $P(A=1 | Y=1) = p_1 = 0.9$, $P(A=1 | Y=0) = p_0 = 0.1$; the reversed test swaps them. A linear head fitted to the training distribution (LDA-optimal; logistic regression approximates it) weights each block by its training SNR. The artifact's *effective* training SNR is

$$\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1) A),$$

and the reversed-environment AUROC is

$$\text{AUROC}_{\text{rev}}(S, \tilde{A}) = \Phi((S - \tilde{A}) / \sqrt{2(S + \tilde{A})}),$$

increasing in S and decreasing in $\tilde{A}$. (Without the artifact, $\tilde{A} = 0$ and $\text{AUROC} = \Phi(\sqrt{S/2})$.)

R22.0.2 Consequences (each matches an empirical result)

1. **Out-of-ROI artifacts.** Masking removes the artifact ($\tilde{A} \rightarrow 0$) and changes the disease SNR to S_m : $\text{AUROC}_{\text{rev}} = \Phi(\sqrt{S_m/2})$. It helps whenever $S_m > 2 \cdot [\Phi^{-1}(\text{AUROC}_{\text{rev}}^{\text{ERM}})]^2$, i.e. unless the mask destroys most of the disease signal. → masking is the strongest single fix out of the ROI (all cohorts, Trap B).
2. **In-ROI artifacts.** Masking keeps the artifact ($\tilde{A}$ unchanged, or larger when the artifact becomes a larger share of what remains) and changes S to S_m . Because $\text{AUROC}_{\text{rev}}$ is increasing in S , masking **harms iff it removes disease-informative context** ($S_m < S$) and helps only through denoising ($S_m > S$). → harm in dermoscopy (lesion surroundings are informative) and in the controlled sweeps (the synthetic artifact is fully retained), small gains in real thyroid/ovary/capsule (masking removes clutter).
3. **The crossover is always positive.** For the same S_m , $\Phi(\sqrt{S_m/2}) - \Phi((S_m - \tilde{A})/\sqrt{2(S_m + \tilde{A})}) > 0$ for every $\tilde{A} > 0$: masking is always worth more for an out-of-ROI artifact than for an in-ROI one. → P1 supported in all four cohorts.
4. **Erasure.** Projecting out a subspace that contains the artifact direction sets $\tilde{A} \rightarrow 0$ but scales the disease SNR by $(1 - \rho^2)$, ρ = overlap between the erased subspace and the disease direction: $\text{AUROC}_{\text{rev}} = \Phi(\sqrt{(S_m(1 - \rho^2))/2})$. Erasure beats in-ROI masking unless ρ is large. → U-MtE works for calipers (ρ small) and fails for debris that resembles erosion fibrin (ρ large); **disease protection** forces $\rho \rightarrow 0$ along the protected label directions.
5. **Group balancing** removes the training association between artifact and label (effective $p_1 = p_0 = \tilde{A} = 0$) for the artifact *and everything correlated with it*: $\text{AUROC}_{\text{rev}} = \Phi(\sqrt{S/2})$, independent of where the artifact sits. → balanced / U-MtE_balanced are location-invariant in every controlled sweep.

6. **Correlate-carried shortcuts.** If part of the shortcut is carried by features that neither masking nor pixel-based erasure touch (hair $\leftrightarrow$ body site/age/sex; drain $\leftrightarrow$ treated lung), a residual $\tilde{A}_c$ remains after masking/erasure and only label-based reweighting removes it. $\rightarrow$ DFR/balancing win on hair and drains.
7. **Augmentation vs erasure.** Overlay augmentation only decorrelates the label from the *generated* overlay direction; the real artifact direction u_a keeps its association unless u_a lies in that direction. Erasure removes the whole subspace spanned by overlay-induced feature changes, which contains u_a whenever overlays and the real artifact move the representation along shared "foreign-overlay" directions. $\rightarrow$ erase $\square$ augment (P3).

R22.0.3 Empirical test on real encoders (docs/PREREGISTRATION_THEORY_PREDICTION.md)

Generalised to any test composition $q_y = P(A=1|Y=y)$: $\Delta = S + (p_1 - p_0)(q_1 - q_0)A/(1 + vA)$, $\sigma_y^2 = S + w_a^2(1 + A q_y(1 - q_y))$, $AUROC = \Phi(\Delta/\sqrt{(\sigma_1^2 + \sigma_0^2)})$. Fitting (S, A) per cell to the observed clean and correlated AUROC and predicting the held-out reversed AUROC gives MAE 0.039 ($r = 0.92$, 148 ERM/mask cells); the resulting location crossovers agree with the observed ones across 14 cohort $\times$ backbone pairs ($r = 0.997$, MAE 0.014 (13/14 signs; the miss is a near-zero chest-radiograph cell), better than simple reference predictors (scripts/analysis/theory_predict.py; paper/figures/theory_predict.pdf). By the pre-registered rule it remains a *qualitative account* (one sign miss). It fails for chest drains, whose shortcut is carried by correlates that the clean/correlated environments do not reveal (§6).

R22.0.4 Limits of the model

Linear heads on fixed features; Gaussian class-conditional features; the mask is modelled only through $(S_m, \tilde{A}_m)$. Per-cell parameters must be estimated from the clean and correlated environments; it cannot see correlate-carried shortcuts that those environments do not expose.

R23 Pre-registration: theory versus real encoders

Source: docs/PREREGISTRATION_THEORY_PREDICTION.md

Committed before any prediction is computed. Motivation: a reviewer noted that the theory (docs/THEORY.md) was only checked against simulations drawn from its own assumptions. This test uses it as a **zero-free-parameter predictor of held-out environments** on the real experiments already run.

R23.0.1 Model (docs/THEORY.md, generalised to any test composition)

Training $P(A=1|Y=1)=p_1=0.9$, $P(A=1|Y=0)=p_0=0.1$ (the design value of every trap and sweep). With disease SNR S , artifact SNR A , $v = [p_1(1-p_1)+p_0(1-p_0)]/2$ and an LDA-optimal head ($w_s = \sqrt{S}$, $w_a = \sqrt{A(p_1-p_0)/(1+vA)}$), a test environment with $P(A=1|Y=y) = q_y$ has $\Delta = S + (p_1 - p_0)(q_1 - q_0)A/(1 + vA)$, $\sigma_y^2 = S + w_a^2(1 + A q_y(1 - q_y))$, $AUROC = \Phi(\Delta / \sqrt{(\sigma_1^2 + \sigma_0^2)})$. (Reduces to $AUROC_{rev} = \Phi((S - \tilde{A})/\sqrt{2(S + \tilde{A})})$ for $q = 0.1/0.9$.)

R23.0.2 Procedure (scripts/analysis/theory_predict.py)

For every cell (result directory $\times$ trap or overlap $\times$ arm):

1. Observed AUROC per environment = mean over training seeds/folds; q_y per environment read from the saved test predictions (artifact_present by label).
2. Fit ($S \geq 0$, $A \geq 0$) so that the model reproduces the **clean** and **correlated** AUROC (least squares in probit space; two equations, two unknowns).
3. **Predict the reversed AUROC**, which is never used in the fit. No parameter is shared across cells or tuned.

Cells (fixed now; one canonical run per cohort $\times$ backbone):

- Real location traps (Trap A in-ROI, Trap B out-of-ROI): ISIC hair (spec_e13/dino518_spec, dermlip224_spec); thyroid (dino518_main, medsiglip448_main, convnext384_universal); capsule (dino518_main, medsiglip448_main, convnext384_universal); ovary (dino518_main, medsiglip448_text); chest-radiograph traps (cxr_traps/raddino518 $\times$ Atelectasis, Consolidation, Effusion, Infiltration).
- Chest drains (in-ROI only): cxr_drain/raddino518_universal.
- Controlled overlap sweeps (5 overlaps each): synthetic/isic2018 {dino518, dermlip224, dino224} ruler_fixed_corr_main; nih_ptx {raddino518, dino518} tube_corr_main; capsule {dino518, medsiglip448} debris_corr_main; thyroid {dino518, medsiglip448} caliper_corr_main; ovary dino518_caliper_corr_main.
- Secondary (outside the model's assumptions — end-to-end networks, not linear heads on frozen features): finetune/{thyroid, capsule, ovary}/resnet50, finetune/thyroid/ViT-S.

Arms: **primary** = ERM and mask (the two arms that define the location law). **Secondary** = every other arm whose head is plain ERM on the 0.9/0.1 training set (erasure/inpainting/projection arms). Arms that reweight or change the training distribution (balanced, DFR, JTT, GroupDRO, prevcal, augmentation, selectors) are excluded because the model's p1/p0 no longer describe their training set.

R23.0.3 Endpoints

- **T1** reversed AUROC, predicted vs observed, over all primary cells: mean absolute error (MAE), Pearson r, share within ± 0.05 . Compared with two reference predictors that use the same inputs: (a) rev = clean ("no shortcut harm"), (b) rev = 2·clean – corr (linear symmetry).
- **T2** location crossover, predicted vs observed, per real trap cohort $\times$ backbone: [rev(mask) – rev(ERM)]_TrapB – [rev(mask) – rev(ERM)]_TrapA. Sign agreement, Pearson r, MAE.
- **T3** controlled sweeps: predicted vs observed mask – ERM reversed-AUROC gain at every overlap. Sign agreement, Pearson r, MAE.
- **T4** (secondary) in-ROI mask effect: does sign(S_mask – S_ERM) (fitted from clean/correlated only) predict the sign of the observed Trap-A mask gain? Accuracy.

R23.0.4 Decision rule (fixed now)

We call the theory **quantitatively predictive** if T2 signs agree in all cells and Pearson $r \geq 0.7$ for both T1 and T2, and the theory's T1 MAE is smaller than reference (a). Otherwise the paper describes it as a **qualitative account** ("reproduces signs and orderings") and does not use the word "predicts" for magnitudes. Reported whichever way it goes.

R23.0.5 Results (results/theory/prediction_summary.json, prediction_cells.csv, prediction_crossover.csv; Fig. theory_predict)

Deviation: the DINOv2 chest-tube sweep (synthetic/nih_ptx/dino518_tube_corr_main) has no saved predictions and was skipped; everything else as listed.

endpoint	theory	ref. (a) rev = clean
T1 reversed AUROC, ERM & mask (148 cells)	MAE 0.039 , $r = 0.917$, 76 % within ± 0.05	MAE 0.234, $r = 0.592$
T1 all ERM-head arms (570 cells)	MAE 0.031, $r = 0.945$	MAE 0.163
T2 crossover (14 cohort $\times$ backbone)	$r = 0.997$, MAE 0.014, signs 13/14	$r = 0.930$, MAE 0.108,
T3 sweeps, mask gain (45 overlap points)	$r = 0.965$, MAE 0.043, signs 42/45	$r = 0.670$, 31/45
T4 in-ROI sign from S_mask – S_ERM	11/14	
End-to-end fine-tuning (outside the model)	crossover signs 4/4; rev AUROC MAE 0.018 (16 cells)	

- **Verdict by the pre-registered rule: "qualitative account"**, because one of 14 crossover signs disagrees: chest-radiograph Consolidation, observed -0.002 vs predicted $+0.010$ (both ≈ 0). All other criteria are met with wide margins (T1 $r = 0.92 \geq 0.7$; T2 $r = 0.997 \geq 0.7$; T1 MAE $0.039 < 0.234$).
- Fitted only to clean and correlated performance, the closed form's out-of-sample estimates agree with the held-out reversed-test AUROC to within 0.04 on average and with the location crossover to within 0.014, better than both reference predictors on every endpoint; they also agree for end-to-end fine-tuned networks (4/4 crossover signs), which violate its assumptions. Per the decision rule, the paper does not say the theory "predicts" magnitudes; it reports these agreements and calls the theory a qualitative account.
- The clear failure is the chest-drain cohort (predicted 0.70–0.76, observed 0.19–0.24): the clean and correlated environments do not reveal a shortcut carried by correlates of the drain (the treated lung), which is consistent with the correlate-carried account (THEORY.md §6) and marks the model's boundary.

R24 Pre-registration: embedding-based leakage groups

Source: docs/PREREGISTRATION_EMBEDDING_GROUPS.md

Committed before any model is refitted with these groups. Motivation: TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). A reviewer asked for a stricter grouping. ISIC (lesion IDs) and chest drains (patient IDs) are unaffected.

R24.0.1 Rule (fixed now; scripts/data/embedding_groups.py)

- Embedding = cached DINOv2 ViT-B/14 @518 CLS feature of the unmasked image, L2-normalised.
- Edge between two images if cosine similarity $\geq \tau$; groups = connected components of these edges **united with the existing pHash / frame-block groups** (so the new grouping is never looser than the old one).
- τ = the smallest value on the grid 0.99, 0.98, ..., 0.80 for which the largest group holds ≤ 5 % of the cohort (the most aggressive threshold that still allows 5-fold grouped splits). Chosen without labels or results.
- Images without a cached embedding keep their existing group.

Amendment (before any refit, after inspecting only the similarity distribution): raw CLS cosine is dominated by the shared ultrasound appearance (ovary: median nearest-neighbour similarity 0.99; at $\tau = 0.99$ the largest group already held 26 % of the cohort, so no grid value met the 5 % cap). Embeddings are therefore **mean-centred per cohort before L2 normalisation**, for all three cohorts; the rest of the rule is unchanged.

R24.0.2 Re-analysis

Same traps, same seeds, same arms (erm, mask, balanced, U-MtE, U-MtE_balanced, U-MtE_protect, overlay augmentation; generic overlays), only the grouping changes: `run_thyroid_traps.py --cohort {thyroid,ovary,capsule} --generic --groups_csv ... --tag emb_groups`. Reported for each cohort: P1 crossover and the P2-P4 contrasts, next to the main estimates. Claim: conclusions are robust if every estimate keeps its sign and its CI still excludes 0 wherever it did before. Reported whichever way it goes.

R24.0.3 Results (results/leakage/embedding_groups.json, embedding_groups_compare.csv)

Groups: thyroid 3,455 $\rightarrow$ 3,081 ($\tau = 0.80$, largest group 4.6 %); ovary 1,059 $\rightarrow$ 969 ($\tau = 0.80$, 2.1 %); capsule 137 $\rightarrow$ 135 ($\tau = 0.93$, 3.7 %). Trap cell counts are unchanged (only the split assignment changes).

cohort	test	main [95 % CI]	embedding groups [95 % CI]	robust
Thyroid	P1 crossover	+0.230 [+0.204, +0.255]	+0.235 [+0.211, +0.259]	yes
Thyroid	P2 U-MtE_protect – mask	+0.217 [+0.174, +0.257]	+0.251 [+0.219, +0.279]	yes
Thyroid	P3 erase – augment	+0.207 [+0.182, +0.235]	+0.192 [+0.162, +0.218]	yes
Thyroid	P4 U-MtE_bal – balanced	+0.112 [+0.090, +0.134]	+0.152 [+0.138, +0.167]	yes
Ovary	P1 crossover	+0.170 [+0.116, +0.224]	+0.157 [+0.080, +0.235]	yes
Ovary	P2	+0.043 [+0.026, +0.061]	+0.047 [+0.020, +0.072]	yes
Ovary	P3	+0.035 [+0.010, +0.059]	+0.020 [–0.011, +0.053]	no
Ovary	P4	+0.028 [+0.000, +0.057]	+0.039 [+0.015, +0.064]	yes
Capsule	P1 crossover	+0.368 [+0.340, +0.397]	+0.394 [+0.359, +0.430]	yes
Capsule	P2	–0.010 [–0.021, –0.001]	–0.007 [–0.013, –0.001]	yes
Capsule	P3	–0.130 [–0.147, –0.110]	–0.136 [–0.153, –0.122]	yes
Capsule	P4	+0.082 [+0.064, +0.100]	+0.068 [+0.050, +0.085]	yes

11/12 robust. The location law (P1) is unchanged in all three cohorts without patient IDs. The one change: ovary erase – augment (P3) loses significance (+0.020 [–0.011, +0.053]), so “erase beats augment” is robust in thyroid and ISIC hair (lesion IDs) but fragile in ovary. Embedding grouping is still a proxy for patient identity; it removes visually near-identical images (same patient/session) but cannot rule out different-looking images of one patient.

R25 Pre-registration: backbone scale

Source: docs/PREREGISTRATION_SCALE.md

Committed before any feature is extracted. Question: do the location law and U-MtE depend on the size of the frozen foundation model? (DINOv3 would be the newest family but its weights are gated and access was refused, 403.)

R25.0.1 Design (frozen)

- Backbones: DINOv2 ViT-S/14 (21 M parameters, 384-d) and ViT-L/14 (304 M, 1024-d) at 518 px, alongside the main ViT-B/14 (86 M). Same preprocessing, CLS token, linear heads, seeds, folds and traps.
- Cohorts: thyroid, ovary, capsule (the three cohorts run by `run_thyroid_traps.py`; ISIC is too large for the time budget). Arms: erm, mask, balanced, U-MtE (mte), U-MtE_balanced, U-MtE_protect, overlay augmentation (mte_aug); generic overlays. `run_thyroid_traps.py --backbone {dinos518,dinoL518} --cohort ... --generic --tag scale`.

R25.0.2 Claims (reversed AUROC, 95 % CI)

- **S1** location crossover > 0 for both backbones in all three cohorts (6 tests).
- **S2** U-MtE_{protect} – mask > 0 (Trap A) in thyroid and ovary (distinct overlays), both backbones.
- **S3** descriptive: does the ERM shortcut gap (correlated – reversed, Trap A) or the crossover grow or shrink with scale (S $\rightarrow$ B $\rightarrow$ L)? No directional prediction. Reported whichever way they go.

R25.0.3 Results (results/scale/scale_compare.csv; ViT-B = main runs)

cohort	size	crossover [95 % CI]	U-MtE _{protect} – mask (Trap A)	U-MtE – mask	ERM gap corr – rev (Trap A)
thyroid	S	+0.244 [+0.209, +0.275]	+0.181 [+0.146, +0.218]	+0.182	0.547
thyroid	B	+0.230 [+0.204, +0.255]	+0.217 [+0.174, +0.257]	+0.222	0.590
thyroid	L	+0.271 [+0.234, +0.308]	+0.222 [+0.195, +0.256]	+0.213	0.621
ovary	S	+0.298 [+0.229, +0.365]	+0.103 [+0.078, +0.128]	+0.120	0.319
ovary	B	+0.170 [+0.116, +0.224]	+0.043 [+0.026, +0.061]	+0.019	0.409
ovary	L	+0.202 [+0.153, +0.250]	+0.060 [+0.043, +0.077]	+0.094	0.570
capsule	S	+0.370 [+0.341, +0.401]	–0.008 [–0.014, –0.003]	–0.153	0.353
capsule	B	+0.368 [+0.340, +0.397]	–0.010 [–0.021, –0.001]	–0.115	0.354
capsule	L	+0.408 [+0.378, +0.438]	–0.040 [–0.046, –0.035]	–0.093	0.325

- **S1 supported 6/6** (and 9/9 with ViT-B): the location law does not depend on model size.
- **S2 supported 4/4**: protected U-MtE beats masking on thyroid and ovary with ViT-S and ViT-L.
- **S3**: larger models exploit the in-ROI caliper shortcut *more* (ERM gap thyroid 0.547 $\rightarrow$ 0.590 $\rightarrow$ 0.621; ovary 0.319 $\rightarrow$ 0.409 $\rightarrow$ 0.570); capsule is flat. Scale does not remove the shortcut. On capsule, disease protection is less complete with ViT-L (–0.040 vs masking).

R26 Pre-registration: U-MtE rank ablation

Source: docs/PREREGISTRATION_UMTE_ABLATION.md

Committed before running. The main U-MtE fixes the rank rule a priori (smallest k whose held-out paired-difference energy falls below 10 %, i.e. energy 0.90, max 64). Reviewers will ask how sensitive the results are to that choice.

R26.0.1 Design

Cached DINOv2 ViT-B/14 @518 features, generic overlays, Trap A (in-ROI) of thyroid, ovary and capsule; same seeds, folds and heads as the main runs. Settings: energy $\in \{0.50, 0.70, 0.80, 0.90 \text{ (main)}, 0.95, 0.99\}$ and fixed k $\in \{1, 4, 16, 64\}$. Arms: mask, U-MtE, U-MtE_{protect} (scripts/analysis/umte_ablation.py). The energy-0.90 setting must reproduce the main U-MtE numbers exactly (built-in check).

Amendment (before any ovary/capsule result): to fit the time budget the grid was trimmed to energy $\in \{0.50, 0.90, 0.99\}$ and k $\in \{1, 4, 16, 64\}$; thyroid e0.70/e0.80/e0.95 (already run) are kept in the results folder.

R26.0.2 Reported

Reversed, correlated and clean AUROC per setting. Descriptive; the question is whether U-MtE – mask keeps its sign over a wide range of ranks (robust) or only near the chosen one (fragile). No setting will replace the main one.

R26.0.3 Results (results/ablation_umte/summary.csv; Trap A reversed AUROC, mean over seeds)

setting	thyroid U-MtE – mask	ovary U-MtE – mask	ovary protect – mask	capsule U-MtE – mask	capsule
energy 0.50	+0.211	–0.002	+0.033	–0.054	–0.008
energy 0.90 (main)	+0.222	+0.019	+0.043	–0.115	–0.010
energy 0.99	+0.222	+0.019	+0.043	–0.115	–0.010
k = 1	+0.071	+0.012	+0.007	+0.001	+0.001
k = 4	+0.063	–0.041	+0.015	–0.008	–0.002
k = 16	+0.180	–0.010	+0.020	–0.050	–0.009
k = 64	+0.222	+0.019	+0.043	–0.115	–0.010

- Built-in check passed: energy 0.90 reproduces the main U-MtE numbers exactly. In these cohorts the 90 % rule reaches the rank cap (identical to $k = 64$).
- **Robust where U-MtE works:** thyroid +0.18 to +0.22 for every setting that erases ≥ 16 directions (or energy ≥ 0.5); one to four directions recover only a third of it (+0.06/+0.07) — the overlay shift is multi-dimensional.
- The protected variant never harms capsule by more than 0.010 and helps ovary in every setting (+0.007 to +0.043). Unprotected erasure harms capsule more the more directions it removes (debris $\approx$ fibrin), as the theory's ρ predicts.

R27 Pre-registration: fine-tune, then erase

Source: *docs/PREREGISTRATION_FT_ERASE.md*

Committed before running. Motivation (external review): the end-to-end evidence for U-MtE is weaker than the frozen-feature evidence (an invariance-penalty analogue works on thyroid with ResNet-50 but not with ViT-S or on capsule). U-MtE is a representation-level method; the natural end-to-end use is to apply it to the representation of a network that was fine-tuned on masked images.

R27.0.1 Design (scripts/run_finetime_spec.py --arms mte_post --traps trapA)

Same traps, fold seeds, ResNet-50, 8 epochs and training as the existing mask arm (results/finetime/*/resnet50). After fine-tuning: penultimate features of masked training images with and without a generic overlay $\rightarrow$ difference subspace (90 % held-out energy, as in the frozen setting) $\rightarrow$ erase $\rightarrow$ logistic head refitted on the training set with C chosen on clean validation (mte_post). Control mask_post: identical body and head fitting without erasure. Cohorts: thyroid first (the distinct-overlay regime), then ovary and capsule if time allows; Trap A (in-ROI) only.

R27.0.2 Claims (reversed AUROC, Trap A, 95 % CI over 5 folds)

- **F1** (primary) mte_post – mask_post > 0 on thyroid (the erasure step helps after fine-tuning).
- **F2** mte_post – mask > 0 on thyroid (vs the fine-tuned masked network's own head).
- Ovary / capsule: reported with CIs; capsule is expected to fail without disease protection (debris $\approx$ fibrin). Reported whichever way it goes.

R27.0.3 Results (results/finetime/thyroid/resnet50/paired_deltas.csv; thyroid, Trap A, 5 folds)

Deviation: the first launch failed because half-precision penultimate features overflowed (NaN); features are now extracted in fp32 (no other change). Ovary and capsule were not run (time budget).

contrast	reversed AUROC Δ [95 % CI]
F1 mte_post – mask_post (erasure step alone)	+0.004 [+0.000, +0.012]
F2 mte_post – mask (vs the network's own head)	–0.027 [–0.053, –0.001]

F1 essentially null, F2 not supported. Applied after fine-tuning, the generic-overlay subspace no longer contains the direction along which the fine-tuned network encodes the caliper: once the network has been trained on the correlated data, the shortcut feature is task-specific, and overlays that resemble no particular artifact do not move the representation along it. U-MtE is therefore a method for frozen (foundation-model) representations; the end-to-end route that works is the invariance penalty during training (thyroid ResNet-50 +0.143), not post-hoc erasure. This supports the framing of U-MtE as a representation-level intervention for frozen encoders.

R28 Pre-registration: erase-while-fine-tuning

Source: *docs/PREREGISTRATION_FT_UMTE.md*

Committed before running. Motivation: post-hoc U-MtE on a fine-tuned ResNet-50 was null (+0.004; PREREGISTRATION_FT_ERASE.md) because unconstrained fine-tuning on the correlated data builds a task-specific caliper feature that generic overlays do not move. A remedy for fully fine-tuned networks must act *during* training.

R28.0.1 Method (umte_ft, src/wtss/experiments/finetune.py)

Masked training image x and the same image with a generic overlay x^+ (same library, half inside the ROI) in every batch. Let g be the penultimate features of the network being trained.

1. Running overlay-shift subspace: $C \leftarrow 0.98 C + 0.02 E[(g(x^+) - g(x))(g(x^+) - g(x))^T]$ (detached); every 20 steps $U = \text{top eigenvectors of } C \text{ covering } 90\% \text{ of its energy } (\leq 64)$, $\mu = \text{running mean of } g(x)$.
2. Head sees the erased features: $\text{logit} = h(g(x) - U U^T (g(x) - \mu))$; the final U is used at test time.
3. Plus the invariance penalty of mte_ft ($\lambda = 1$): $\| \hat{g}(x) - \hat{g}(x^+) \|^2$, $\hat{g} = \text{L2-normalised features}$. No artifact example, artifact mask or artifact label; only the ROI mask; architecture-agnostic. Secondary arm cons_ft: prediction consistency, $\text{BCE} + \lambda (\text{logit}(x) - \text{logit}(x^+))^2$.

R28.0.2 Design

Thyroid Trap A, 5 folds, same training as the existing arms: ResNet-50 (lr 1e-4) and ViT-S/16 (lr 3e-5), 8 epochs.

```
run_finetune_spec.py --cohort thyroid --arms umte_ft --traps trapA [--arch vit_small_patch16_224.augreg_in21k_ft_in1k --lr 3e-5].
```

R28.0.3 Claims (reversed AUROC, Trap A, 95 % CI over folds)

- **G1** umte_ft – mask > 0 for ResNet-50 and for ViT-S (success = both).
- **G2** umte_ft – mte_ft reported (does erasure add to invariance?); cons_ft – mask reported if run. Clean and correlated AUROC reported alongside (a gain that only flips the shortcut is not a success). Reported whichever way it goes.

R28.0.4 Amendment (before running it) — combined arm

Observed so far: umte_ft works for ResNet-50 (+0.160 [+0.114, +0.207]) but is not significant for ViT-S (+0.061 [−0.005, +0.141]); cons_ft is null for ResNet-50 (+0.007 [−0.014, +0.029]) but significant for ViT-S (+0.062 [+0.031, +0.087]). Pre-registered combined arm **umte_cons_ft** = projection + feature invariance + prediction consistency ($\lambda = 1$ each; no tuning). **G3**: umte_cons_ft – mask > 0 for both ResNet-50 and ViT-S (thyroid, Trap A). Also running: umte_ft on ovary (ResNet-50), reported with CI.

R28.0.5 Amendment 2 (before running) — other cohorts

umte_cons_ft on ovary and capsule (ResNet-50, Trap A, 5 folds), reported with CIs. Capsule debris resembles erosion fibrin; without disease protection the frozen U-MtE failed there, so failure is the expected outcome.

R28.0.6 Results (thyroid, Trap A, 5 folds; results/finetune/thyroid/*/paired_deltas.csv)

arm – mask (reversed AUROC)	ResNet-50	ViT-S/16
mte_ft (invariance only; earlier)	+0.143 [+0.099, +0.188]	+0.009 [−0.021, +0.040]
umte_ft (projection + invariance)	+0.160 [+0.114, +0.207]	+0.061 [−0.005, +0.141]
cons_ft (prediction consistency)	+0.007 [−0.014, +0.029]	+0.062 [+0.031, +0.087]
umte_cons_ft (all three)	+0.157 [+0.114, +0.205]	+0.143 [+0.103, +0.187]

AUROC (clean / correlated / reversed): ResNet-50 mask 0.691 / 0.827 / 0.536; umte_cons_ft 0.688 / 0.661 / 0.693. ViT-S mask 0.688 / 0.868 / 0.465; umte_cons_ft 0.712 / 0.798 / 0.608.

- **G1 partial** (ResNet-50 yes, ViT-S not significant). **G3 supported**: the combined arm improves on masking for both architectures with no artifact annotation, and each component alone works for only one architecture.
- On ResNet-50 the correlated AUROC falls below the reversed one (0.661 vs 0.693): mild over-correction (flag), its min(rev, corr) 0.661 is close to the label-using mask_balanced (0.677). On ViT-S there is no flip (0.798 vs 0.608) and clean AUROC improves (0.712 vs 0.688), but the shortcut is only partly removed.
- Interpretation: post-hoc erasure fails because unconstrained fine-tuning builds a task-specific shortcut feature; estimating the overlay subspace *during* training on the network's own features, and requiring overlay-invariant features and predictions, stops that feature from forming.

R28.0.7 Status of Amendment 2

First attempt not run: at 15:28 UTC the GPU became unavailable inside the container (nvidia-smi: Input/output error, host-side driver problem; both jobs failed at CUDA initialisation before training). Commands to run when a GPU is available: `python scripts/run_finetune_spec.py --cohort ovary --arms umte_cons_ft --traps trapA` and the same with `--cohort capsule`. `umte_ft` on ovary (ResNet-50, run before the failure): +0.107 [−0.003, +0.229].

Recovery (15:37 UTC): the kernel module and `/dev/nvidia*` were intact; only the host-mounted `libcuda.so.610.43.02` was unreadable. The identical userspace libraries were extracted from NVIDIA's official 610.43.02 package into `/root/data/cache/nvidia_driver/lib` and the runs were relaunched with `LD_LIBRARY_PATH` pointing there (same driver version, same GPU; no other change). Results below.

R28.0.8 Results of Amendment 2 (ResNet-50, Trap A, 5 folds)

cohort	umte_cons_ft − mask	AUROC clean / corr / rev (umte_cons_ft)	mask
ovary	+0.094 [−0.011, +0.210]	0.687 / 0.726 / 0.650	0.578 / 0.570 / 0.556
capsule	−0.022 [−0.046, +0.001]	0.555 / 0.879 / 0.190	0.596 / 0.915 / 0.211

- **Ovary: positive, not significant** (the 1,202-image cohort is underpowered for fine-tuning: every ovary contrast has a CI of about ± 0.11). It improves every environment over masking and has the best min(rev, corr) of the annotation-free arms together with `umte_ft` (0.650 / 0.663 vs mask 0.556 and the label-using `mask_balanced` 0.551).
- **Capsule: fails, as pre-registered.** Fine-tuned masking collapses on in-ROI debris (reversed 0.211 vs ERM 0.495); debris resembles erosion fibrin, so an overlay-driven method cannot separate it from disease (theory: large p). Only label-based balancing helps there (`mask_balanced` reversed 0.687).
- Overall: erase-while-fine-tuning works across architectures (ResNet-50, ViT-S) for distinct-overlay artifacts (thyroid significant, ovary positive), and — like frozen U-MtE without protection — not for pathology-like artifacts.

R29 Related work and novelty assessment

Source: `docs/RELATED_WORK_NOVELTY.md`

Honest map of what is and is not new, for the paper's positioning. Each row: closest prior work → our delta.

Our component

Location-dependent failure of ROI masking (in-ROI vs out-of-ROI artifacts), pre-registered traps, 3 modalities (dermoscopy hair, Generic procedural overlay library (artifact-agnostic)
Paired-difference subspace erasure (I2E)
Mask-then-erase ordering (MtE / U-MtE)
Disease-protected erasure (U9, `insertion.protect`)
Pseudo-group balancing from overlay detector (U8)
SLAS token-level suppression

R29.0.1 Baselines still to add for the paper

- SPLINCE (needs A labels) vs `mte_protect`.
- NoiseCrop-style test-time masking (ours $\approx$ mask arm; verify equivalence or add).
- JTT (two-stage error-set upweighting) as a label-free group baseline vs `pbal`.

Sources: arXiv 2302.00070; arXiv 2506.10703 (SPLINCE); arXiv 2310.11991; arXiv 2306.12673; arXiv 2208.09756; CVPRW 2020 Bissoto; arXiv 2412.04030; arXiv 2504.18983.

R30 Findings overview (living document)

Source: `docs/FINDINGS_OVERVIEW.md`

Every number below is from a pre-registered analysis (`docs/PREREGISTRATION_*.md`) with 95 % hierarchical paired bootstrap CIs; full tables in `results/CROSS_COHORT.md` and `results/SUMMARY.md`. "Hard test" = shortcut-reversed test set (reversed AUROC). Trap A = artifact inside the ROI, Trap B = outside.

R30.0.1 1. Replication of the original pilot (the PDF)

73 of 88 PDF numbers MATCH (same sign, same CI verdict, $|\Delta| \leq 0.02$), 12 agree in sign and significance, 3 mismatch (E4 occlusion borderline CI, E12 contrast trap not null, one HAM-only stratum), 0 pending.

R30.0.2 2. The phenomenon: does ROI masking depend on where the artifact sits?

cohort	artifact	masking gain, out-of-ROI	masking gain, in-ROI	crossov
ISIC 2019 dermoscopy (real)	hair	+	-0.037 [-0.057, -0.019] (harm)	+0.145
Thyroid US (real, DINOv2)	calipers	+0.341	+0.111	+0.230
Thyroid US (real, MedSigLIP)	calipers	+0.228	+0.064	+0.163
Capsule endoscopy (real)	debris	+0.465	+0.097	+0.368
ISIC 2018 (controlled, DINOv2 / DermLIP)	ruler	+ (r=0)	harm (r $\geq$ 0.25)	sign rev
Thyroid US (real, ConvNeXt CNN)	calipers	+0.288	+0.003	+0.284
Capsule endoscopy (real, MedSigLIP)	debris	+0.441	+0.109	+0.332
Capsule endoscopy (real, ConvNeXt CNN)	debris	+	+0.094	+0.292
Capsule (controlled)	debris	+0.103 (r=0)	-0.102 (r=1)	+0.205
Thyroid (controlled)	calipers	+0.402 (r=0)	-0.030 (r=1, n.s.)	+0.432
Chest X-ray (controlled, RAD-DINO)	tube	-0.002 (not used by ERM)	+0.110	-0.113

Robust claim: masking leaves in-ROI shortcuts largely intact (it removes most of an out-of-ROI shortcut); it is actively harmful in dermoscopy and in the controlled capsule sweep, not in real thyroid/capsule or in CXR.

R30.0.3 3. The solution: artifact-agnostic mask-then-erase (U-MtE)

No artifact example, mask or label; procedural overlays $\rightarrow$ insertion pairs on masked images $\rightarrow$ erase subspace. | cohort (Trap A / r=1) | U-MtE - mask | erase - augment (U7) | |---|---|---| | Thyroid DINOv2 | +0.222 [+0.197, +0.251] | +0.207 [+0.182, +0.235] | | Thyroid MedSigLIP | +0.204 [+0.174, +0.237] | +0.177 [+0.148, +0.211] | | ISIC hair DINOv2 | +0.052 [+0.044, +0.062] | +0.045 [+0.037, +0.054] | | Thyroid ConvNeXt (CNN) | +0.128 [+0.112, +0.145] | +0.120 [+0.102, +0.139] | | Thyroid (controlled, r=1) | +0.278 [+0.232, +0.325] | - | | Capsule (controlled) | +0.081 [+0.050, +0.112] | - | | Chest X-ray (controlled) | +0.134 [+0.120, +0.151] | - | | Capsule (real) | -0.115 (fails: debris $\approx$ fibrin) $\rightarrow$ protected -0.010 | - | With image-level artifact labels, U-MtE + group balancing is the most robust model (min of reversed/correlated AUROC) in thyroid (0.788 DINOv2, 0.792 MedSigLIP); in real capsule masking + DFR is higher (0.885 vs 0.857); on dermoscopic hair plain DFR is best (0.741) — the hair shortcut is mostly carried by site/age/sex correlates (oracle removal of every hair patch gains only +0.015).

R30.0.4 4. Other solution attempts (reported whichever way)

- SLAS (few-shot patch-token artifact suppression): localises hair from 5 annotated images (patch AUROC 0.93), but downstream gains are small (+0.02 hair, +0.08 thyroid). Kept as a localisation tool.
- Disease-protected erasure (U9): safety net — removes the capsule failure, costs $\leq$ 0.012 elsewhere.
- Annotation-free pseudo-group balancing (U8): +0.04 on thyroid, none on hair.
- Erasure + DFR (U10): no gain over masking + DFR.
- JTT label-free baseline (U11): U-MtE beats JTT on thyroid (+0.198); U-MtE+JTT best on capsule (0.771); plain JTT best on hair (0.564). The winner follows what carries the shortcut.

R30.0.5 5. Datasets evaluated and not used (with reasons)

CANDID-PTX (no access), CXR CLiP real devices (masking helps both locations), CXR cardiomegaly (too few labelled), TCGA/GrandQC pathology (unreliable pen masks), ISIC ink (70 in-lesion images), breast US BUSI/BUS-BRA (marker labels $\approx$ 70 % pure), EAD endoscopy (no disease labels), BKAI-IGH NeoPolyp colonoscopy (specular highlights: 0 specular-free images, 41 in-polyp).

R30.0.6 6. Tools

python -m wtss.umte (U-MtE) and python -m wtss.slas (SLAS): any backbone (ViT or CNN), two commands.

R30.0.7 7. Latest (2026-09-27, 03:00-05:30 UTC)

- Real chest drains (NIH pneumothorax): extreme shortcut (ERM reversed 0.19 RAD-DINO, 0.28 DINOv2); lung masking +0.05 / +0.01; U-MtE +0.03 over masking with RAD-DINO, none with DINOv2; DFR / mask+DFR best. Drain = treated lung → correlate-carried regime.
- MedSigLIP controlled sweeps (capsule, thyroid): location interaction replicates (+0.19, +0.47); masking does not harm at full overlap; unbalanced U-MtE loses to masking (−0.11, −0.08); U-MtE_balanced beats masking (+0.31, +0.46) and is location-invariant.
- Disease protection in sweeps: fixes MedSigLIP capsule (+0.061 over masking), helps DINOv2 capsule (+0.118), does not fix MedSigLIP calipers (−0.041).
- Colonoscopy (BKAI NeoPolyp, specular): infeasible (no specular-free images).

R30.0.8 8. Recommendation for practitioners (paper's decision rule)

1. Measure where artifacts sit relative to the ROI. Out-of-ROI → mask.
2. In-ROI, no artifact labels → protected U-MtE (DINOv2/CNN backbones), or JTT when the shortcut is carried by correlates (check with the oracle-removal ceiling on a few annotated images, e.g. with SLAS).
3. In-ROI, image-level artifact labels available → U-MtE + balanced (overlay-like artifacts), masking + DFR (artifacts resembling the disease), DFR / balancing (correlate-carried shortcuts).

R30.0.9 9. Session 2 additions (2026-09-27, 06:00-09:30 UTC)

- **4th disease — ovarian tumour ultrasound (MMOTU, calipers)**: real crossover +0.170 [+0.116, +0.224]; controlled sweep: masking HARMS at full overlap (−0.197 [−0.292, −0.100]), U-MtE +0.141 over masking. Generic real U-MtE +0.019 (n.s.), protected +0.043 [+0.025, +0.061].
- **SPLINCE baseline** (closest published erasure): flips the shortcut in thyroid, capsule and hair (reversed □ correlated); by min(rev, corr) U-MtE_balanced wins in all three; on drains SPLINCE 0.516 vs 0.460 at −0.19 clean.
- **Location-adaptive selection (auto)**: beats masking in 5/5 cohorts (+0.17 to +0.40), within 0.02 of the best candidate in 5/5, beats DFR in 3/5 (fails on hair and drains = correlate-carried regime). Split-validation variant that admits DFR as a candidate: running (U14).
- **Fine-tuning (ResNet-50, end-to-end)**: thyroid — end-to-end U-MtE +0.143 [+0.099, +0.188] over masking and joint-best robust model without labels; capsule — masking HARMS in-ROI (−0.284) with a crossover of +0.584 [+0.469, +0.691], end-to-end U-MtE fails there (debris ≈ disease).
- Git: 76 commits had silently failed to push (branch name mismatch); fixed and verified.

R30.0.10 10. Session 3 additions (2026-09-27, 09:30-11:30 UTC) — reviewer-proofing

- **Statistics**: Holm-corrected primary family — 12/16 supported (P1 location crossover 4/4; protected U-MtE > mask 3/4; erase > augment 3/4; U-MtE+bal > balanced 2/4). docs/STATISTICAL_PLAN.md, results/PRIMARY_CLAIMS.md.
- **Natural distribution (thyroid, official split, no resampling)**: masking lowers AUROC on shortcut-conflicting cases by −0.102 [−0.139, −0.066]; U-MtE recovers +0.051; balancing +0.142. Protected U-MtE did not help (N1/N2 not supported).
- **Artifact-label robustness**: crossover and protected-U-MtE gain hold under large-marker and strict-location definitions (thyroid, ovary) and a strict capsule Trap B (+0.259). Visual audit: thyroid detections 95 % real.
- **Theory**: $AUROC_{rev} = \Phi((S - \tilde{A})/\sqrt{2(S + \tilde{A})})$; simulation agreement 0.009; explains all regimes (docs/THEORY.md).
- **LaMa inpainting** (oracle masks): strong; protected U-MtE matches it on thyroid, below on ovary; U-MtE+bal beats it.
- **Split-validation selector**: fixes drains (0.656), mixed elsewhere; neither selector uniformly best.
- **Paper**: compiles cleanly (14 pages), new figures (dose-response, forest, examples, theory), two review passes.

R30.0.11 Additions after external review (2026-09-27)

A colleague's review raised seven points; what changed:

- **Wording**. U-MtE "needs no artifact example, artifact mask or artifact label" — it does use the ROI mask (now said everywhere). Table 1's "I2E" rows renamed "Insert-then-erase (artifact templates)" and explained as U-MtE's artifact-specific precursor. GroupDRO is stated to appear only in the controlled dermoscopy sweeps. Chest radiography is named as a boundary case. U-MtE is framed as a representation-level intervention for frozen foundation-model features; end-to-end results are secondary. The "clean" environment is now described correctly (real traps: artifact in half of each class; sweeps: absent).

- **Theory vs real data** (pre-registered, PREREGISTRATION_THEORY_PREDICTION.md): fitted per model to clean and correlated AUROC only, the closed form matches held-out reversed AUROC (MAE 0.039, $r = 0.92$, 148 models) and the location crossover ($r = 0.997$, MAE 0.014, 13/14 signs); verdict by the pre-registered rule: qualitative account (one near-zero sign miss). Fails for chest drains (correlate-carried shortcut).
- **Leakage** (PREREGISTRATION_EMBEDDING_GROUPS.md): stricter DINOv2 near-duplicate groups; 11/12 primary contrasts unchanged; ovary erase > augment loses significance.
- **Clinical metrics** (PREREGISTRATION_NATURAL.md): AUPRC, Brier, ECE, sensitivity/specificity; masking halves thyroid sensitivity at the validation-fixed operating point.
- **Related work**: six verified references added (Pewton 2024; Wang 2024; Germani 2026 MedIA; Lin 2024 MICCAI; Zech 2018; Brown 2023).
- **Integrity**: one archived prediction file (thyroid natural) failed a gzip CRC check; regenerated, identical numbers.
- **Model scale** (PREREGISTRATION_SCALE.md): DINOv2 ViT-S/B/L — location law 9/9; protected U-MtE > mask on thyroid and ovary at every size; larger models exploit in-ROI calipers more (ERM gap thyroid $0.547 \rightarrow 0.621$).
- **U-MtE rank ablation** (PREREGISTRATION_UMTE_ABLATION.md): thyroid gain +0.18 to +0.22 for ≥ 16 erased directions; protected variant safe on capsule at every rank.
- **Fine-tune, then erase** (PREREGISTRATION_FT_ERASE.md): null (+0.004) — U-MtE is for frozen representations.
- **Mechanism figure** (paper/figures/mechanism.pdf): masking increases counterfactual reliance on an in-ROI artifact in every controlled sweep; U-MtE and balancing remove it.